



Mansoura University
Faculty of Computers and Information
Fall 2023-2024



DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

Ibrahim El-Hasnony

COURSE DETAILS

Course Syllabus

Chapter 1: Introduction to DSS

Chapter 2: Decision Making Concepts

Chapter 3: Decision Making Technologies and Environment

Chapter 4: Model Management

Chapter 5: Data Management System and Dialogue Management

Chapter 6: Designing and Developing Decision Support Systems

Chapter 7: Executive Support Systems

Chapter 8: Automated Decision Systems and Expert Systems

Chapter 9: Knowledge Management and Collaborative Systems

Chapter 10: DSS Intelligent Systems & Tools and Techniques

Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

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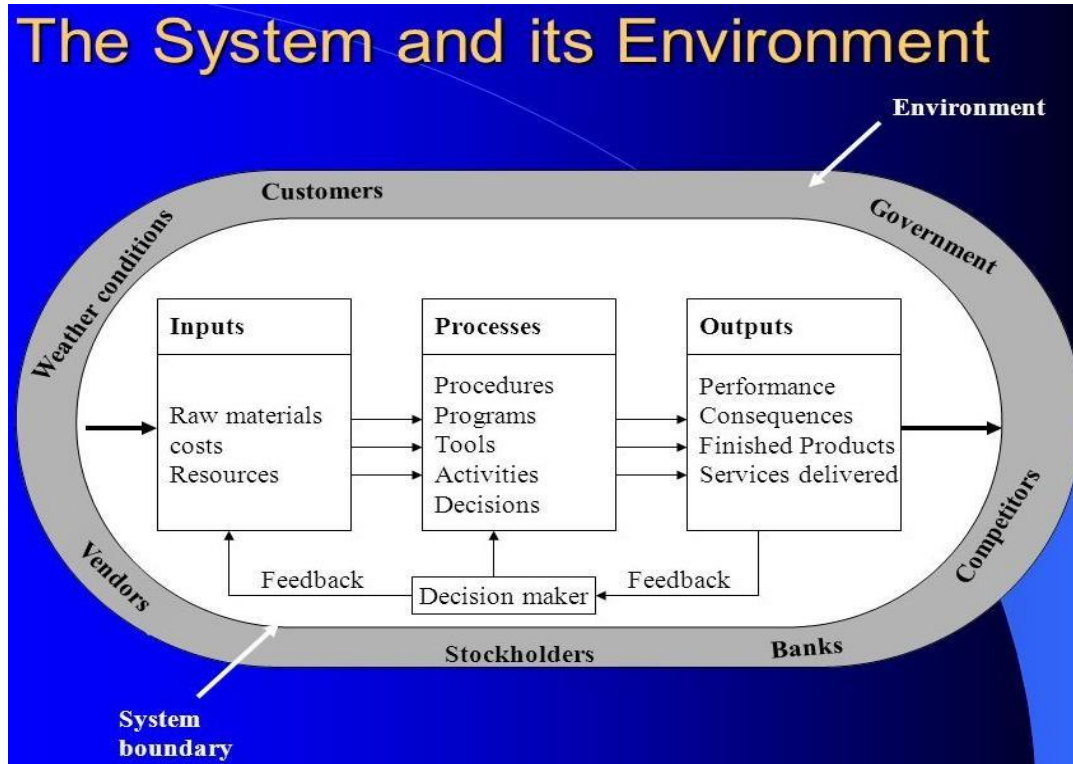
Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

CHAPTER I CONTENTS

- Systems
- Information
- Information systems (IS)
- Management information systems (MIS)
- Efficient and effective systems
- Components of IS
- Types of IS
- DSS evolution

SYSTEMS

A System: an interrelated set of components including people, activities, technology, and procedures that are designed or intended to achieve a predefined purpose.



MANAGERIAL DECISION MAKING AND INFORMATION SYSTEMS

- **Management** is a process by which organizational goals are achieved through the use of resources



- **Resources:** *Inputs*
- **Goal** Attainment: *Output*
- Measuring **Success**:

$$\text{Productivity} = \text{Outputs} / \text{Inputs}$$

Management is decision making

The **manager** is a decision maker

Now fast changing, complex environment

Trial-and-error: not a great approach

INFORMATION SYSTEMS AND ITS CONCEPTS (I OF 2)

- **Decision** is a choice made between alternative courses of action in a situation of uncertainty.
- **Knowledge** awareness or familiarity gained by experience of a fact or situation. Knowledge comprises things that are held to be true in each context and that drive us to action if there were no impediments.
- **Information**, defined as processed data is an increment in knowledge: it contributes to the general framework of concepts and facts that we know.
- **Data** are only the raw facts, the material for obtaining information. Information systems use data stored in computer databases to provide needed information.

INFORMATION SYSTEMS AND ITS CONCEPTS (2 OF 2)

- **A database** is an organized collection of interrelated data reflecting a major aspect of a firm's activities.
- **Downsizing** In information systems, transferring some or all the organization computing from centralized processing on mainframes or minicomputers to systems built around networked microcomputers (often in a client/server configuration).

INFORMATION AND ITS ROLE IN BUSINESS

Information systems **capture data** from the organization (internal data) and its environment (external data).

They **store** the database items over an extensive period.

When specific information is needed, the appropriate data items are **manipulated** as necessary, and the user receives the resulting information.

Depending on the type of information system, the information **output** may take the form a **query response, decision outcome, expert-system advice, transaction document**, or a report

QUALITY INFORMATION NEEDS TO POSSESS SEVERAL ATTRIBUTES

- **Timely:** Available when needed and not outdated when made available
 - **Complete:** Includes all the user needs to know about the situation where the information will be used
 - **Concise:** Does not include elements unneeded by the user
 - **Relevant:** Has direct bearing on the situation
 - **Precise:** Offers quantitative information with a degree of exactness appropriate to the underlying data
 - **Form:** The level of detail, tabular versus graphic display, and quantitative versus qualitative form are selected in accordance with the situation
- Internal and External Information

EFFECTIVENESS AND EFFICIENCY

Effectiveness

measures the extent to which the system meets its objectives.

Efficiency

is a measure of resources consumed to produce given outputs.

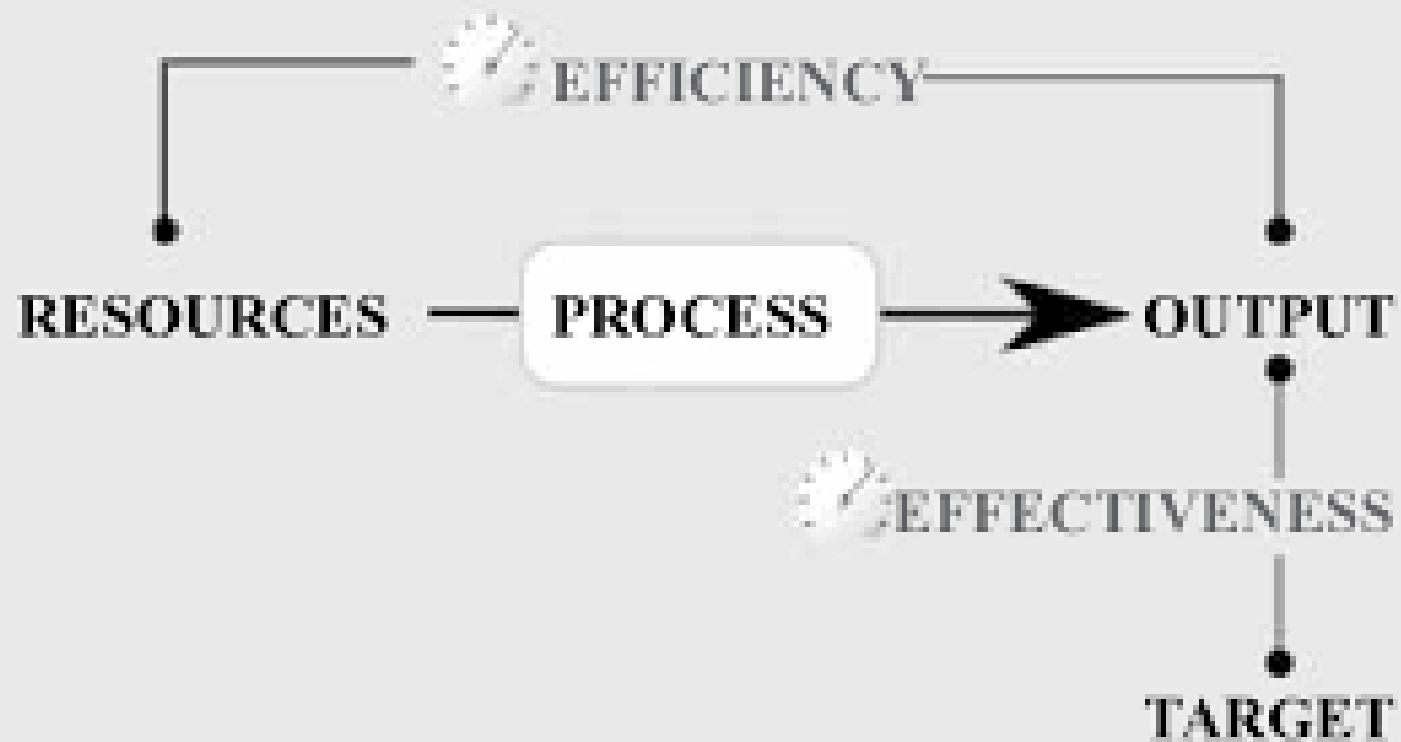
The fewer resources a system consumes in producing given outputs, the more efficient it is.

EFFECTIVENESS AND EFFICIENCY

Efficiency and Effectiveness Matrix



EFFECTIVENESS AND EFFICIENCY



COMPONENTS OF INFORMATION SYSTEMS

- **Information systems consist of the following general components:**
 - Hardware
 - Software
 - Databases
 - Human resources
 - Procedures

TYPES OF INFORMATION SYSTEMS

- Organizations employ several types of information systems. These include:

- 1) Transaction Processing Systems (TPS)
- 2) Management Reporting Systems (MRS)
- 3) Decision Support Systems (DSS)
- 4) Executive Information Systems (EIS)
- 5) Office Information Systems (OIS)
- 6) Professional Support Systems
- 7) Expert systems (ES)

TRANSACTION PROCESSING SYSTEMS

- **A Transaction** is an elementary **activity** conducted during business operations.
- **TPS** may work either in **batch mode**, processing accumulated transactions at a single time later on, or in **on-line** mode, processing incoming transactions immediately.
- Transaction processing systems **today** generally work in **on-line mode** by immediately processing a firm's business transactions.

MANAGEMENT REPORTING SYSTEMS

- The objective of management reporting systems is to provide routine information to managers.
- Managers receive performance **reports** within their specific areas of responsibility.
- Generally, these reports provide internal information rather than spanning corporate boundaries.
- They report on the **past** and the **present**, rather than projecting the future.

DECISION SUPPORT SYSTEMS

- Decision support systems directly support a **decision-making session**.
- These systems facilitate a **dialog between the user, who is considering alternative problem solutions**, and the system that provides **built-in models** and access to databases.
- The DSS databases are often extracted from the general databases of the enterprise or from external databases.

EXECUTIVE INFORMATION SYSTEMS

- Executive information systems support **top managers** with conveniently displayed **summarized information**, customized for them. They make a variety of internal and external information readily available in a highly summarized and convenient form. **EIS are used to:**
 - Monitor the performance of the organization.
 - Assess the business environment.
 - Develop strategic directions for the company's future.

OFFICE INFORMATION SYSTEMS

- The main objective of OIS is to facilitate **communication** between the members of an organization and between the organization and its environment. **OIS are used to:**
 - Help manage documents represented in an electronic format.
 - Handle messages, such as electronic mail, facsimile, and voice mail
 - Facilitate teleconferencing and electronic meetings.
 - Facilitate the use of the Internet for communication and access to information.
 - Facilitate the use of task-oriented teams using groupware.

EXPERT SYSTEMS IN INFORMATION SYSTEMS

- **Expert systems** are system that employs **knowledge** about its application domain and uses an **inferencing** (reason) procedure to solve problems that would otherwise require human competence or expertise

EVOLUTION OF DECISION SUPPORT SYSTEMS

- **mid-1950s** Transaction Processing Systems (TPS)
- **1960s** MIS
- **1970s** Office Automation Systems
DSS
- **1980s** DSS Expanded
Commercial applications of expert systems
Executive Information Systems
- **1990s** Group Support Systems
Neural Computing
Integrated, hybrid computer systems

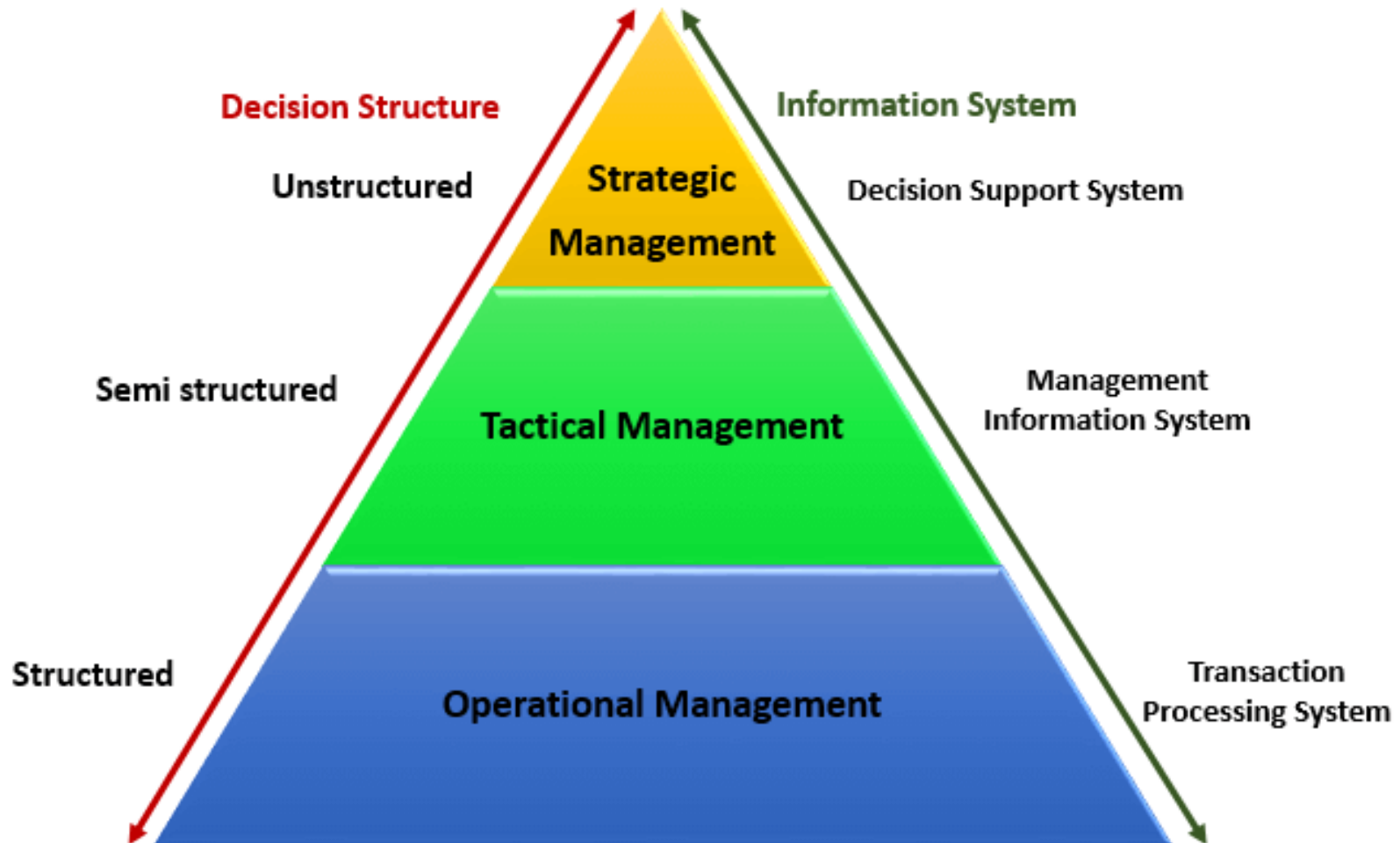
MANAGERS AND DECISION MAKING: WHY COMPUTERIZED SUPPORT?

- Speedy computations
- Overcome cognitive limits in processing and storage
- Cognitive limits may restrict an individual's problem-solving capability
- Cost reduction
- Technical support
- Quality support
- Competitive edge
- The *MANAGERS* are always responsible for decision making

MANAGERS AS INFORMATION SYSTEM USERS

- Managers exist at various **managerial levels** and within various **business areas** of the firm
- The 3 primary management levels are:
 1. **Strategic** planning level
 2. **Management** control level
 3. **Operational** control level

DSS AND MANAGEMENT



THE FUNDAMENTAL FUNCTIONS OF MANAGEMENT INCLUDE:

- **Planning:** establishing goals and selecting the actions needed to achieve them over a specific period of time.
- **Organizing:** This process of establishing worker relationships allows workers to work together to achieve their organizational goals.
- **Leadership** including the people in the organization to contribute to its goals.
- **Staffing:** Recruiting and selecting employees for positions within the company (within teams and departments).
- **Controlling:** measuring performance against the planned objectives and initiating corrective action, if needed.

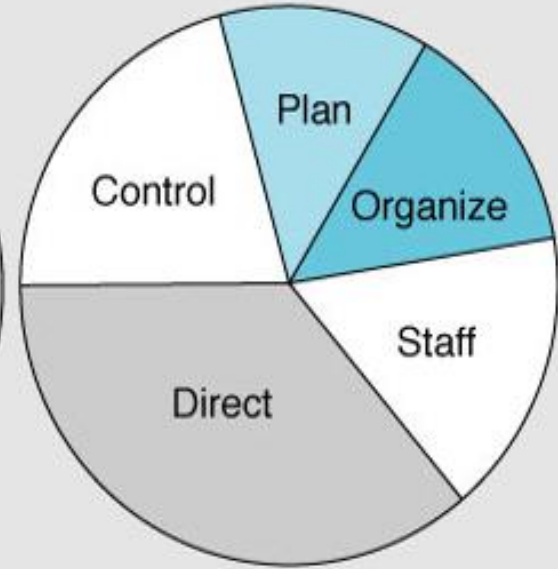
Strategic planning level



Management control level



Operational control level



THE PROCESS OF DECISION MAKING

- **Defining the problem** (a decision situation that may deal with some difficulty or with an opportunity)
- **Constructing a model** that describes the real-world problem
- Identifying **possible solutions** to the modeled problem and evaluating the solutions
- **Comparing, choosing, and recommending** a potential solution to the problem

DECISION MAKING

Process of choosing amongst alternative courses of action for the purpose of attaining a goal or goals.

- **The four phases of the decision process are:**
 - Intelligence
 - Design
 - Choice
 - Implementation



Thanks
Any questions



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CHAPTER 2 CONTENTS

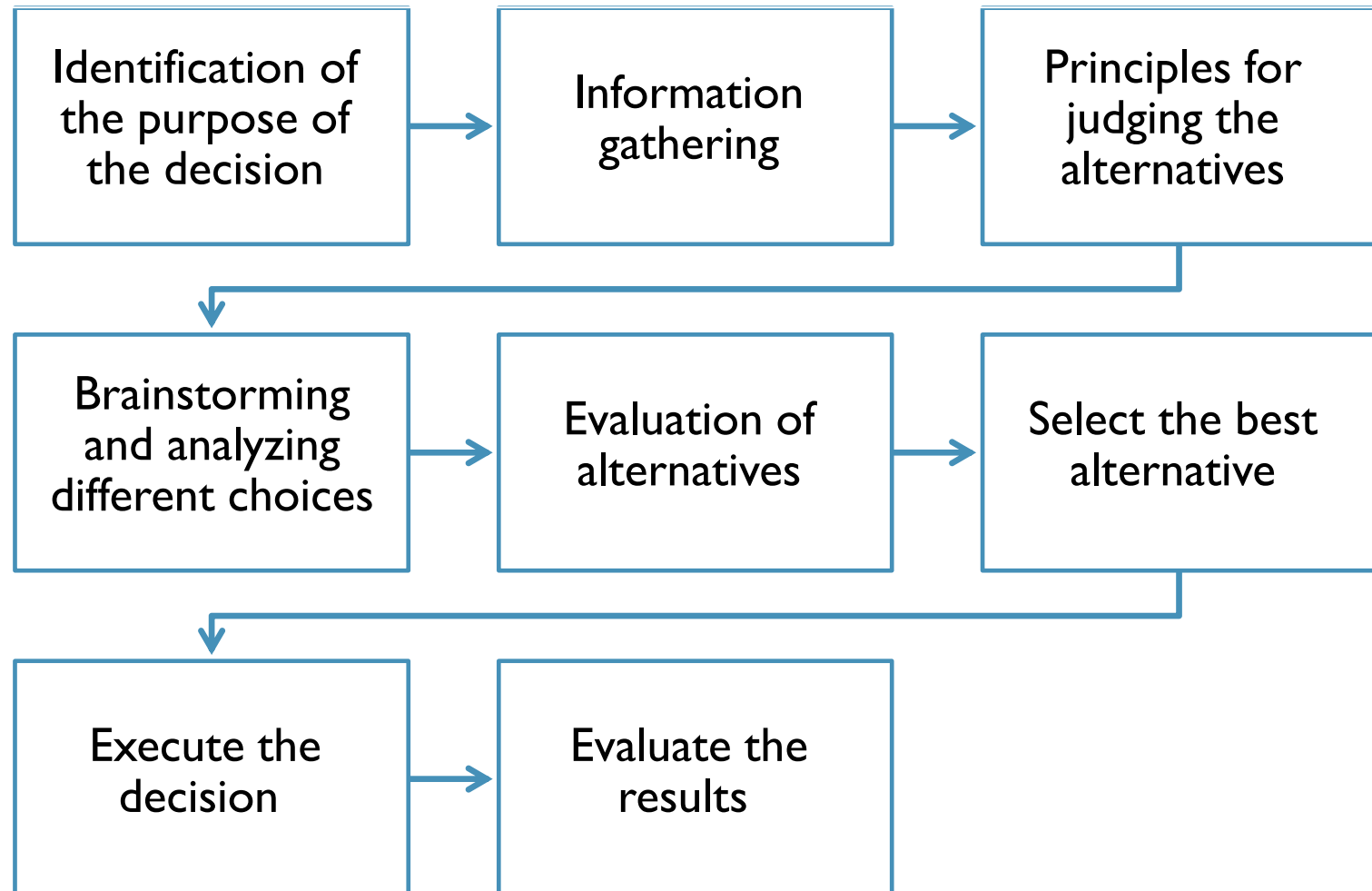
- Decision making process
- Decision support systems
- DSS characteristics
- DSS components
- DSS advantages and disadvantages
- DSS benefits

DECISION MAKING

Decision-making is a cognitive process that results in the selection of a course of action among several alternative scenarios



DECISION MAKING PROCESS



STEP 1: IDENTIFICATION OF THE PURPOSE OF THE DECISION

- What exactly is the problem?
- Why the problem should be solved?
- Who are the affected parties of the problem?
- Does the problem have a deadline or a specific timeline?

STEP 2: INFORMATION GATHERING

- In the process of solving the problem, you will have to gather as much as information related to the factors and stakeholders involved in the problem.
- For the process of information gathering, tools such as 'Check Sheets' can be effectively used.

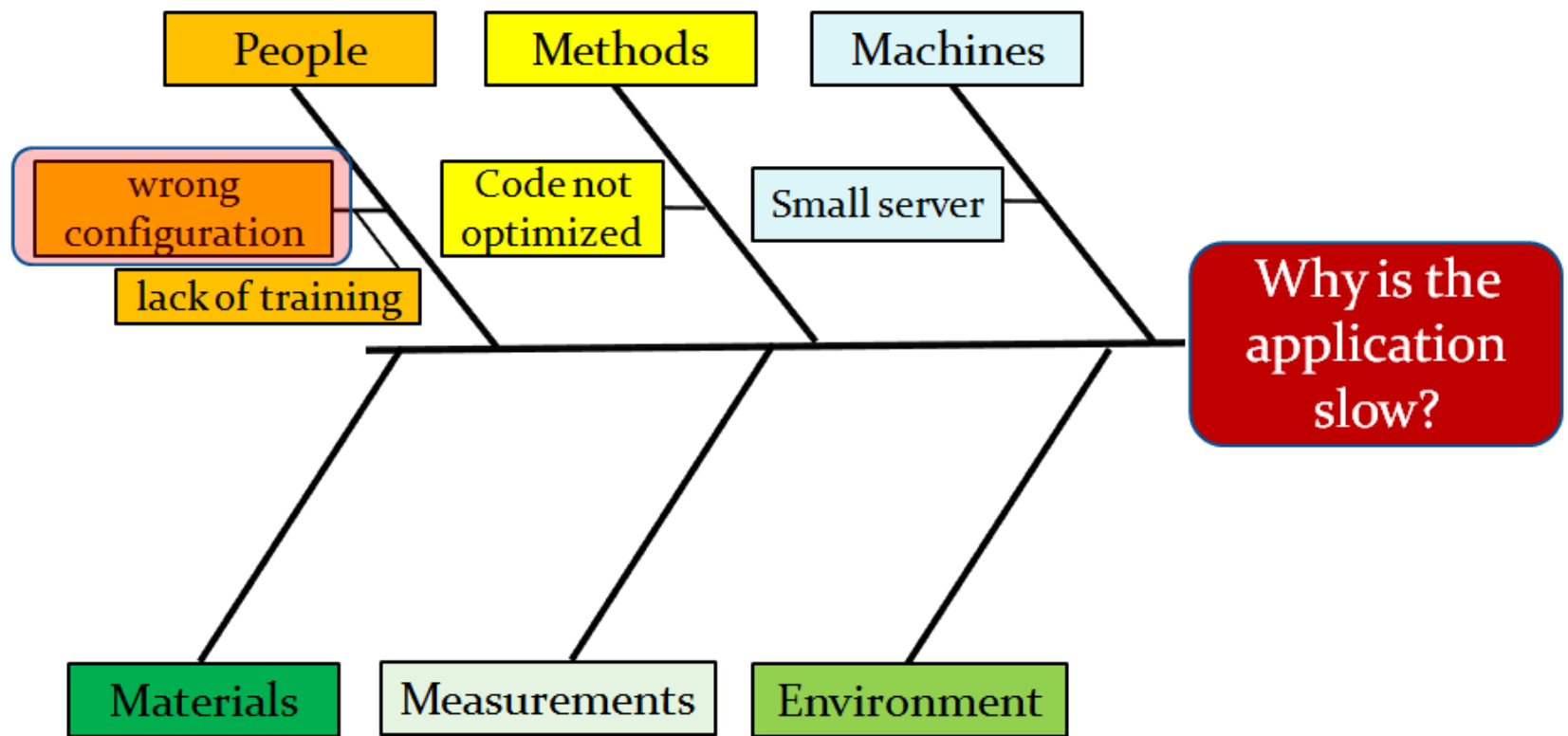
STEP 3: PRINCIPLES FOR JUDGING THE ALTERNATIVES

- The baseline **criteria** for judging the alternatives should be set up.
- When it comes to defining the criteria, organizational goals as well as the corporate culture should be taken into consideration.
- As an example, profit is one of the main concerns in every decision-making process.
- Companies usually do not make decisions that reduce profits unless it is an exceptional case

STEP 4: BRAINSTORM AND ANALYSE THE CHOICES

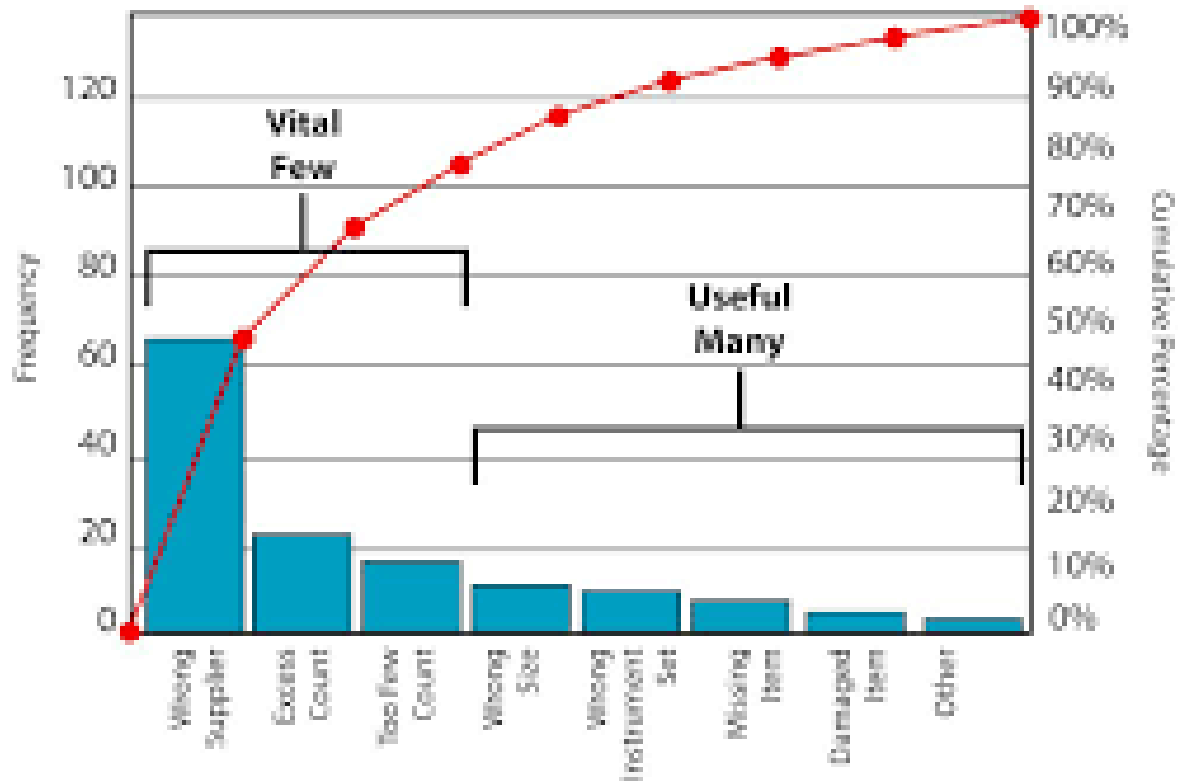
- You can make use of Cause-and-Effect diagrams and Pareto Chart tool.
- **Cause-and-Effect diagram** helps you to identify all possible causes of the problem
- **Pareto chart** helps you to prioritize and identify the causes with the highest effect.
- Then, you can move on generating all possible solutions (alternatives) for the problem in hand.

CAUSE-AND-EFFECT DIAGRAM



PARETO CHART

Pareto Chart: Types of Errors Discovered During Surgical Set-up



STEP 5: EVALUATION OF ALTERNATIVES

- Use your judgment principles and **decision-making criteria** to evaluate each alternative.
- In this step, **experience and effectiveness** of the judgment principles come into play.
- You need to compare each alternative for their positives and negatives.

STEP 6: SELECT THE BEST ALTERNATIVE

- The selection of the best alternative is an informed decision since you have already followed a methodology to derive and select the best alternative.

STEP 7: EXECUTE THE DECISION

- Convert your decision into a plan or a sequence of activities.
- Execute your plan by yourself or with the help of subordinates.

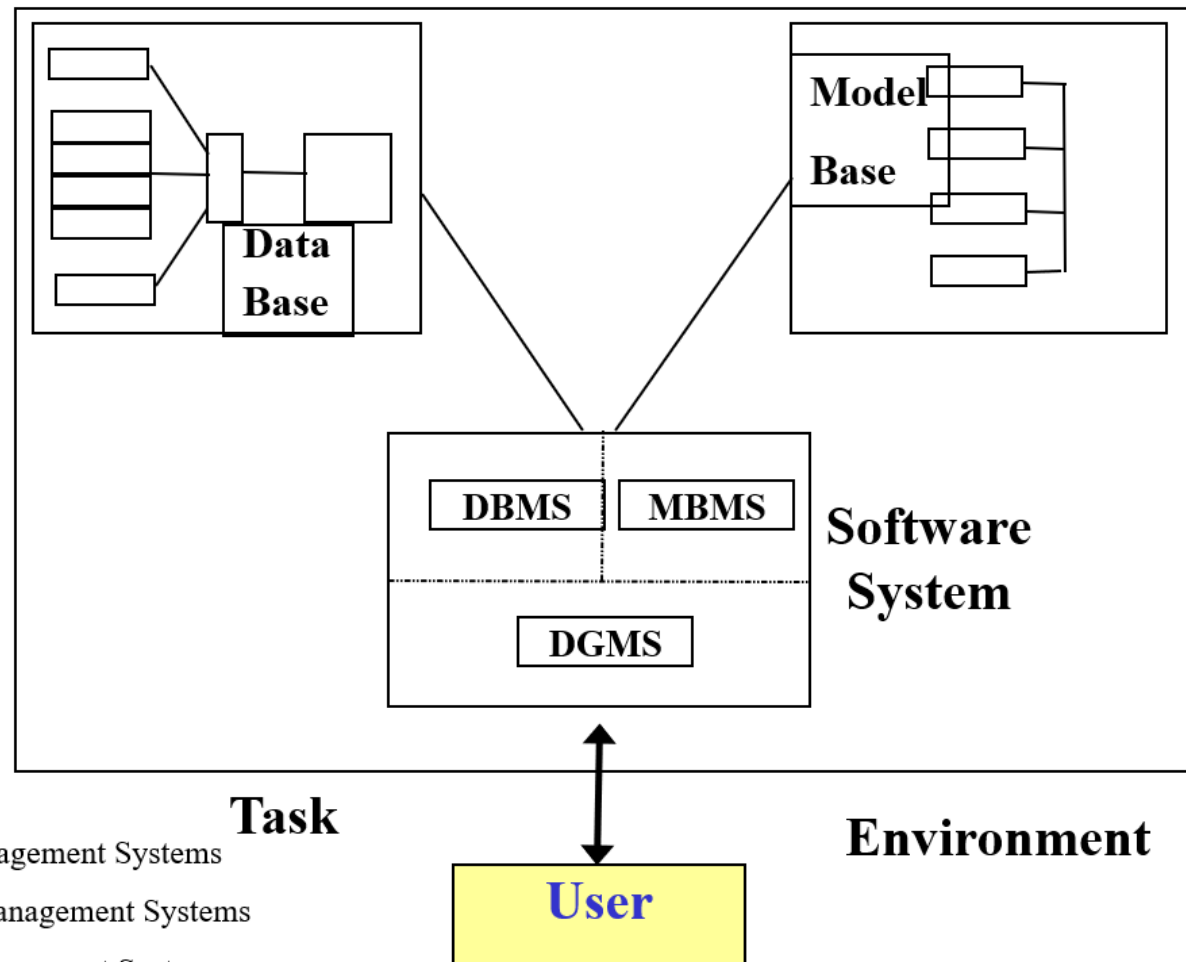
STEP 8: EVALUATE THE RESULTS

- Evaluate the outcome of your decision.
- See whether there is anything you should learn and then correct in future decision making.

DECISION SUPPORT SYSTEM (I)

- DSS is an information system application that helps in decision-making.
- DSS is used in planning and analyzing alternatives.
- DSS differs from most traditional information system in that each DSS is distinct from the other information system and is specifically made for managers.

DSS COMPONENTS



DBMS: DataBase Management Systems

MBMS: ModelBase Management Systems

DGMS: DialoGue Management Systems

DSS COMPONENTS

- **DSS Database:** It contains data from various sources, including internal data from the organization, the data generated by different applications, and the external data mined from the Internet, etc.
- **DSS Software System:** It consists of various mathematical and analytical models that are used to analyze complex data, thereby producing the required information.
- **DSS User Interface:** It is an interactive graphical interface which makes the interaction easier between the DSS and its users. It displays the results (output) of the analysis in various forms, such as text, table, charts or graphics.

DSS MODELS EXAMPLES

- **Statistical Models:** They contain a wide range of statistical functions, such as mean, median, mode, deviations etc. These models are used to establish relationships between the occurrences of an event and various factors related to that event.
- **Sensitivity Analysis Models:** These are used to provide answers to **what-if situations** occurring frequently in an organization
- **Optimization Analysis Models:** They are used to find optimum value for a target variable under given circumstances
- **Forecasting Models:** They use various forecasting tools and techniques
- **Backward Analysis Sensitivity Models:** Also known as **goal seeking** analysis, the technique followed in these models is just opposite to the technique applied in sensitivity analysis models.

DSS CHARACTERISTICS (I)

- 1) Should aid the decision-maker in decision-making.
- 2) Should be able to address **semi/un-structured** decision-making situations.
- 3) Should support decision-makers particularly at **tactical/strategic** levels.
- 4) Should be able to create general-purpose models, simulation capabilities and other analytical tools available to decision-maker.
- 5) Should enable users to use DSS without assistance from MIS/technical professionals.
- 6) Should be readily adapted to meet information requirement for any decision environment.

DSS CHARACTERISTICS (2)

- 7) Should provide mechanism to enable rapid response to a decision-makers request for information.
- 8) Should have the capability to interface with corporate database.
- 9) Should be flexible to accommodate variety of management styles.
- 10) Should facilitate communication between/among various levels of decision-making.
- 11) Should have in-built flexibility and ability to evolve as user-sophistication grows.
- 12) Using of interactive methods are better advised.

DSS BENEFITS

- 1) Improves personal efficiency.
- 2) Speed up the process of decision making.
- 3) Increases organizational control.
- 4) Encourages exploration and discovery on the part of the decision maker.
- 5) Speeds up problem solving in an organization.
- 6) Facilitates interpersonal communication.
- 7) Promotes learning or training.
- 8) Generates new evidence in support of a decision.
- 9) Creates a competitive advantage over competition.
- 10) Reveals new approaches to thinking about the problem space.
- 11) Helps automate managerial processes.

DSS ADVANTAGES AND DISADVANTAGES

Advantages

- Time savings.
- Enhance effectiveness
- Improve interpersonal communication
- Competitive advantage
- Cost reduction.
- Increase decision maker satisfaction
- Promote learning
- Increase organizational control

Disadvantages

- Monetary cost
- Overemphasize decision making
- Assumption of relevance
- Transfer of power
- Unanticipated effects.
- Status reduction.
- Information overload.
- False belief in objectivity



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QUIZ I

- 1)is a process by which organizational goals are achieved using resources
(Decision, Modelling, System ,Management)
- 2)is a choice made between alternative courses of action in a situation of uncertainty.
(Decision, Decision making, Decision support systems, Information system)
- 3)measures the extent to which the system meets its objectives.
(reliability, effectiveness, efficiency, completeness)
- 4) TPS may work inmode, processing accumulated transactions at a single time later
(batch, online, real time, immediate)
- 5) DSS support all organization levels except.....level.
(strategic, operational, management, tactical)

QUIZ I

- 1) Management reporting systems report on the past and the present, rather than projecting the future ()
- 2) Executive information systems support top managers with conveniently displayed summarized information, customized for them ()
- 3) The core management activates for planning in the management level of organization ()
- 4) Decision is a cognitive process that results in the selection of a course of action among several alternative scenarios ()
- 5) Pareto chart helps you to identify all possible causes of the problem ()
- 6) Forward Analysis Sensitivity Models known as goal seeking analysis ()

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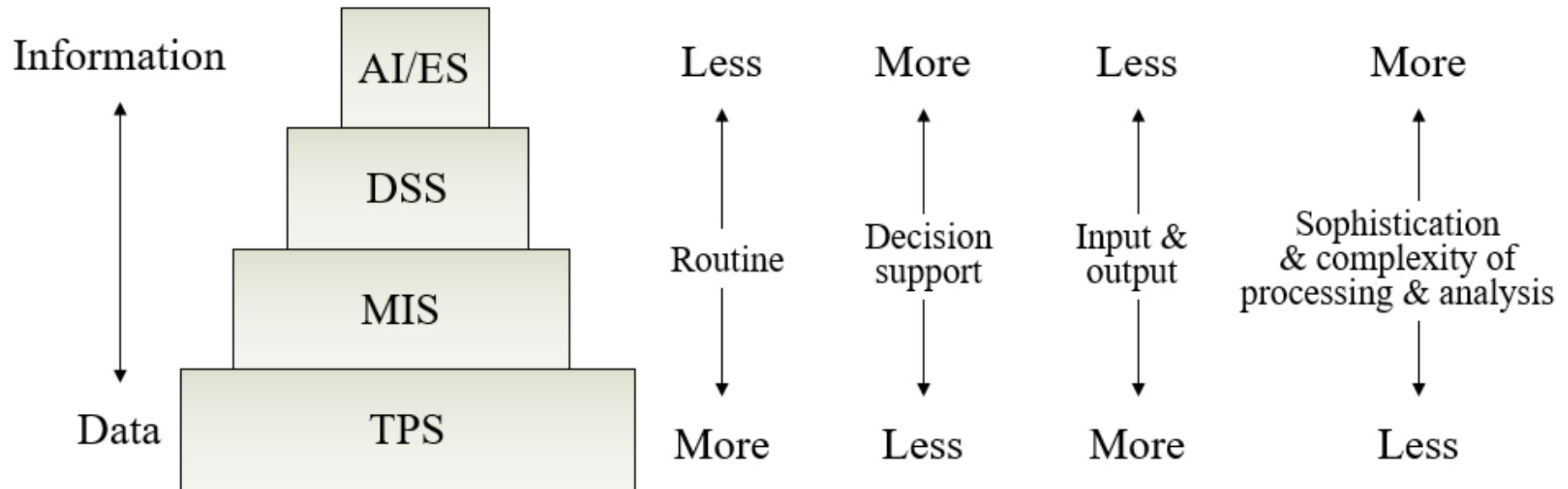
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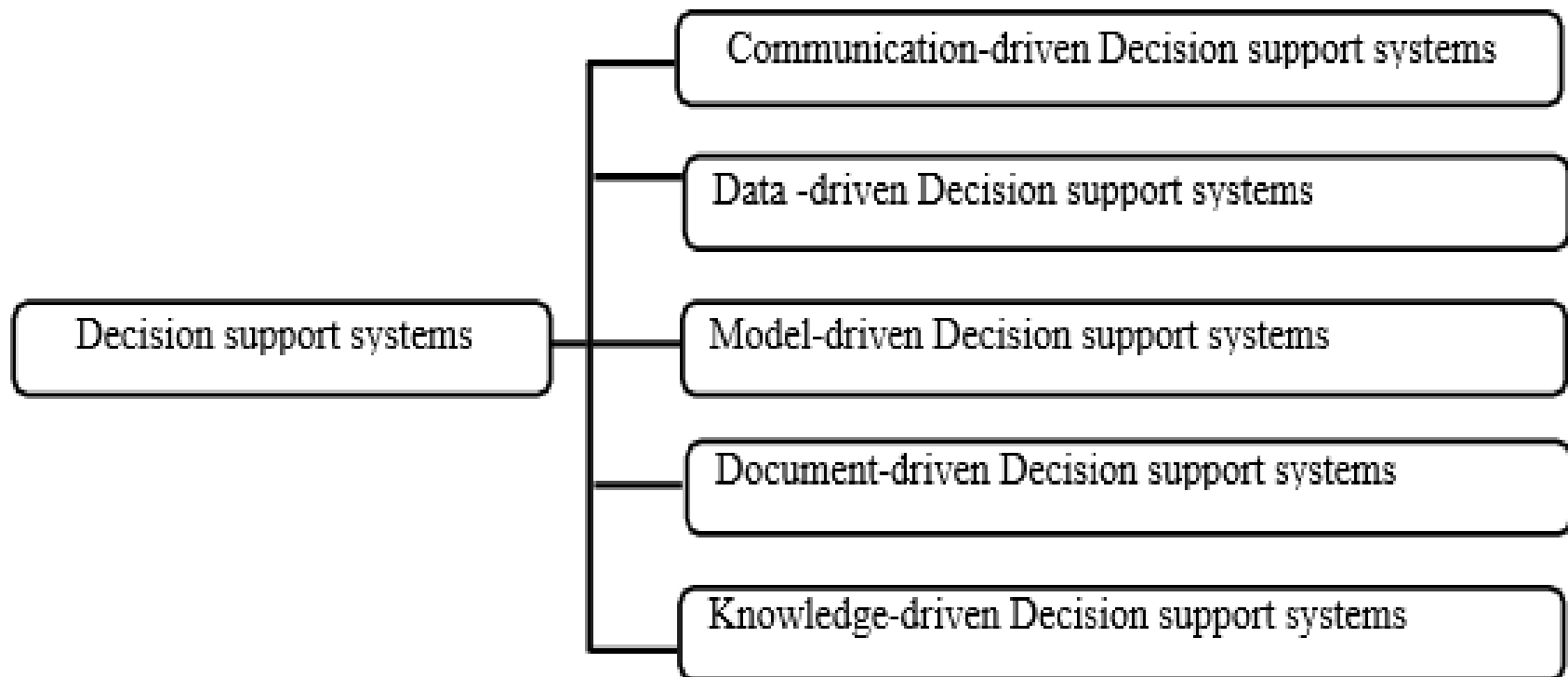
CHAPTER 3 CONTENTS

- Framework of Decisions Support Systems (DSS Classifications)
- DSS with Other Systems
- Decision-Making Levels
- Types of decisions
- Systems for Decision Support
- Four steps in decision making
- Certainty, Risk, and Uncertainty
- Decision Annalysis

REMEMBER



FRAMEWORK OF DECISIONS SUPPORT SYSTEMS (DSS CLASSIFICATIONS)



COMMUNICATION-DRIVEN DSS

- Most communications driven DSSs are targeted at **internal teams**, including partners
- Help **conduct a meeting**, or for users to **collaborate**.
- Uses network and communications technologies to facilitate collaboration and communication, and thus makes decision making faster and more productive.
- The most common technology used to deploy the DSS is a web or client server.
- Examples: chats and instant messaging software, online collaboration, and net-meeting systems.

DATA-DRIVEN DSS

- Most data driven DSSs are targeted at managers, staff and product/service suppliers.
- Supports decision making by analyzing given time-series of data and returning new information gained by those analysis
- It is used to query a database or data warehouse to seek specific answers for specific purposes.
- It is deployed via a main frame system, client/server link, or via the web.
- Examples: computer-based databases that have a query system to check (including the incorporation of data to add value to existing databases.

DOCUMENT-DRIVEN DSS

- Document-driven DSSs are more common, targeted at a broad base of user groups.
- The purpose of such a DSS is to search web pages and find documents on a specific set of keywords or search terms.
- The usual technology used to set up such DSSs are via the web or a client/server system.

KNOWLEDGE-DRIVEN DSS

- It is essentially used to provide management **advice or to choose products/services.**
- The typical deployment technology used to set up such systems could be client/server systems, the web, or software running on stand-alone PCs.

MODEL-DRIVEN DSS

- Model-driven DSSs are complex systems that help analyze decisions or choose between different options.
- These are used by managers and staff members of a business, or people who interact with the organization, for several purposes depending on how the model is set up - scheduling, decision analyses etc.
- These DSSs can be deployed via software/hardware in stand-alone PCs, client/server systems, or the web.

DECISION SUPPORT SYSTEM :TYPES (I)

1. **Traditional DSS**

Components: data management + model management
+ dialogue management + user interface

Output: decisions based on quantitative data

2. **Knowledge-based DSS (KBDSS)**

Components: DSS + Knowledge management

Output: Decisions based on quantitative &
qualitative data

DECISION SUPPORT SYSTEM :TYPES (2)

3. **Intelligent DSS based on online analytical processing (OLAP)**

Components: KBDSS + OLAP

Output: provide new hidden or
unexpected trends & patterns

4. **Group decision support systems (GDSS)**

Components: DSS + group thinking +
varying time/place facilities

Output: improve the DM process. Such
process is interactive executed by
a group of decision makers

DECISION SUPPORT SYSTEM :TYPES (3)

5. **Organizational DSS (ODSS)**

Output: improving communication and coordination among people with organizations to improve organization performance and solve problems

6. **Web-based decision support system**

Output: extend the DSS capabilities on a global basis

DSS WITH OTHER SYSTEMS (TASK REPORT)

- Transaction Processing Systems (TPS)
- Management Reporting Systems (MRS)
- Executive Information Systems (EIS)
- Office Information Systems (OIS)
- Professional Support Systems
- Expert Support systems (ESS)

BUSINESS INTELLIGENCE AND DECISION SUPPORT SYSTEMS (I)

- Because of the importance of high-quality decision making, firms are investing heavily in business intelligence systems, which consist of technologies and applications designed to help users make better business decisions.
- People's ability to combine learned knowledge with new information and change their behavior in such a way that they succeed at their task or adapt to a new situation.

BUSINESS INTELLIGENCE AND DECISION SUPPORT SYSTEMS (2)

Business intelligence (BI) is a technology-driven process for analyzing data and delivering actionable information that helps executives, managers and workers make informed business decisions.

- As part of the BI process, organizations collect data from internal IT systems and external sources, prepare it for analysis, run queries against the data and create data visualizations, BI dashboards and reports to make the analytics results available to business users for operational decision-making and strategic planning.

BUSINESS INTELLIGENCE AND DECISION SUPPORT SYSTEMS (2)



Systems and Technologies for Business Intelligence

DECISION-MAKING LEVELS (I)



DECISION-MAKING LEVELS (2)

Management support system	Needs to support	Length of time for decision-making	The nature of data & Information	Environment	type	decision / Management
IDSS / ES	+ ↑	+ ↑	- ↑	Unstructured	Strategic	Top
DSS / EIS / OR MIS				Semi Structured	Tactical	Middle
TPS OR MIS				Structured	Operational	Bottom

DECISION TYPES (I)

Unstructured

Each of these decisions is novel, important, and nonroutine, and there is no well-understood or agreed-on procedure for making them.

Structured

Repetitive and routine, and decision makers can follow a definite procedure for handling them to be efficient.

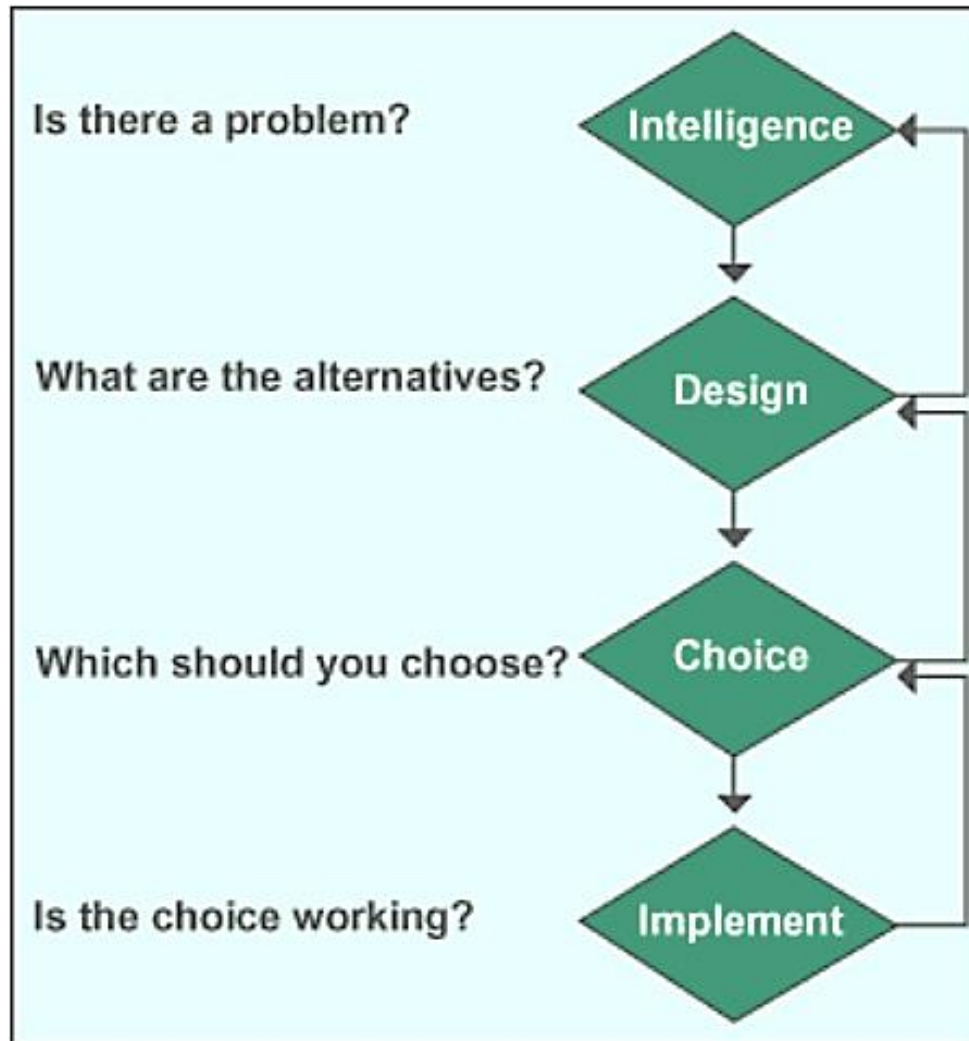
Semi-Structured

Only part of the problem has a clear-cut answer provided by an accepted procedure

DECISION TYPES (2)

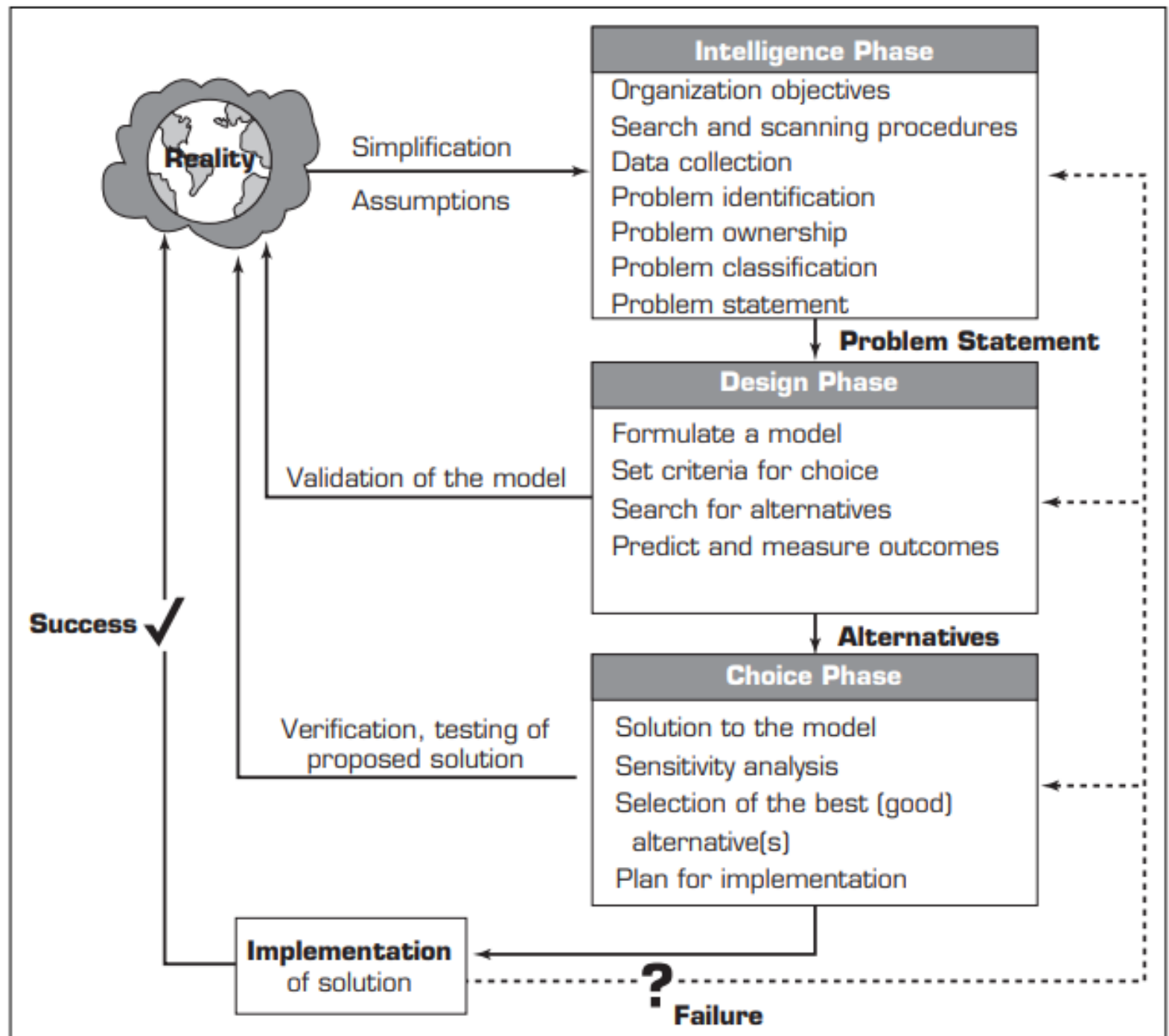
Type of Decision	Type of Control		
	Operational Control	Managerial Control	Strategic Planning
Structured	1 Accounts receivable Accounts payable Order entry	2 Budget analysis Short-term forecasting Personnel reports Make-or-buy	3 Financial management Investment portfolio Warehouse location Distribution systems
Semistructured	4 Production scheduling Inventory control	5 Credit evaluation Budget preparation Plant layout Project scheduling Reward system design Inventory categorization	6 Building a new plant Mergers & acquisitions New product planning Compensation planning Quality assurance HR policies Inventory planning
Unstructured	7 Buying software Approving loans Operating a help desk Selecting a cover for a magazine	8 Negotiating Recruiting an executive Buying hardware Lobbying	9 R & D planning New tech development Social responsibility planning

FOUR STEPS IN DECISION MAKING (I)

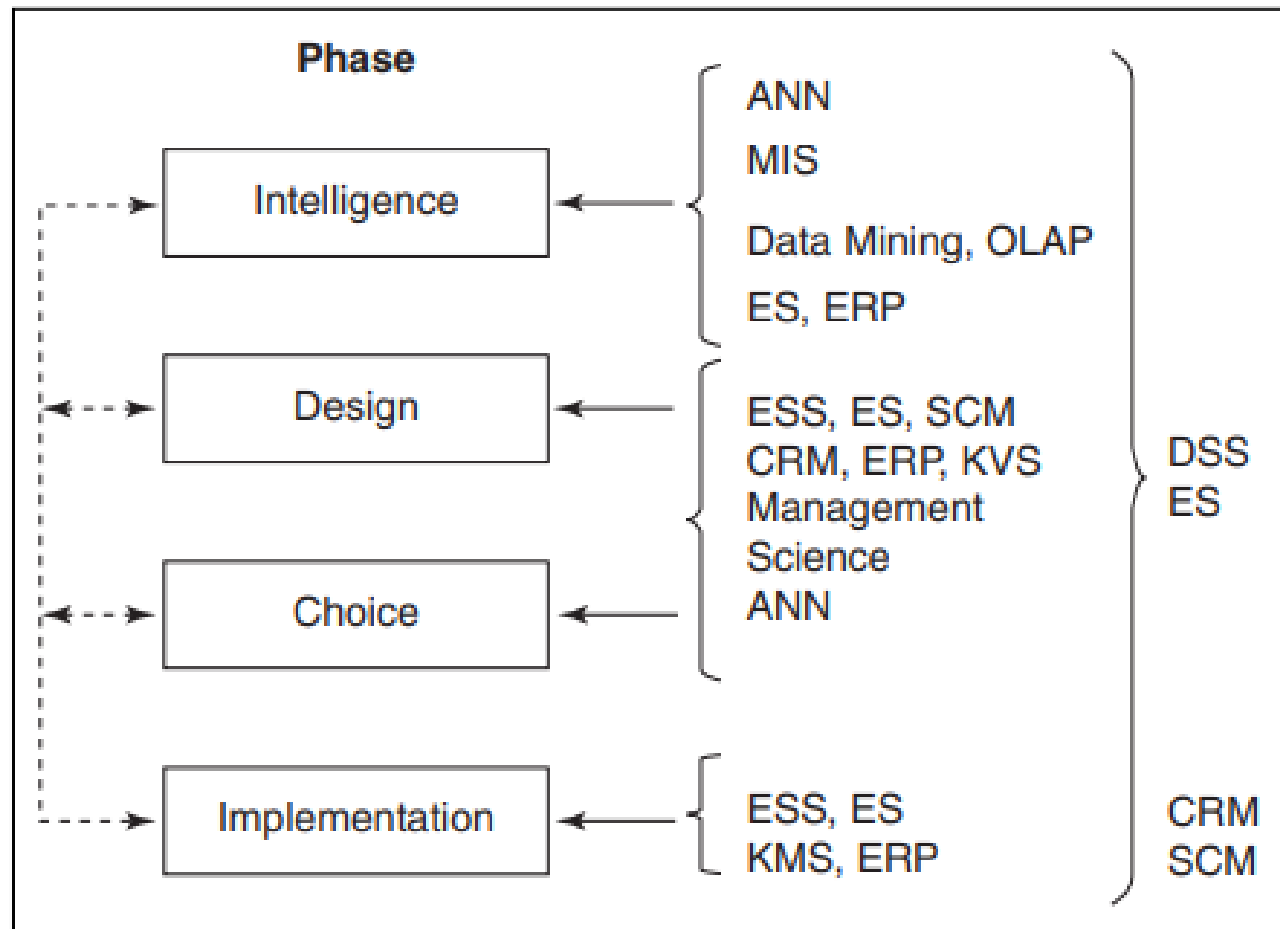


FOUR STEPS IN DECISION MAKING (2)

- **Intelligence** consists of discovering, identifying, and understanding the problems occurring in the organization—why is there a problem, where, and what effects is it having on the firm
- **Design** involves identifying and exploring various solutions to the problem.
- **Choice** consists of choosing among solution alternatives
- **Implementation** involves making the chosen alternative work and continuing to **monitor** how well the solution is working



DSS SUPPORT



DSS Support.

CERTAINTY, RISK, AND UNCERTAINTY

- Decision situations are often classified based on **what the decision-maker knows** (or believes) about the forecasted results. We classify this knowledge into three categories ranging from complete knowledge to total ignorance. These categories are:
 - Certainty
 - Risk
 - Uncertainty

DECISION-MAKING UNDER CERTAINTY

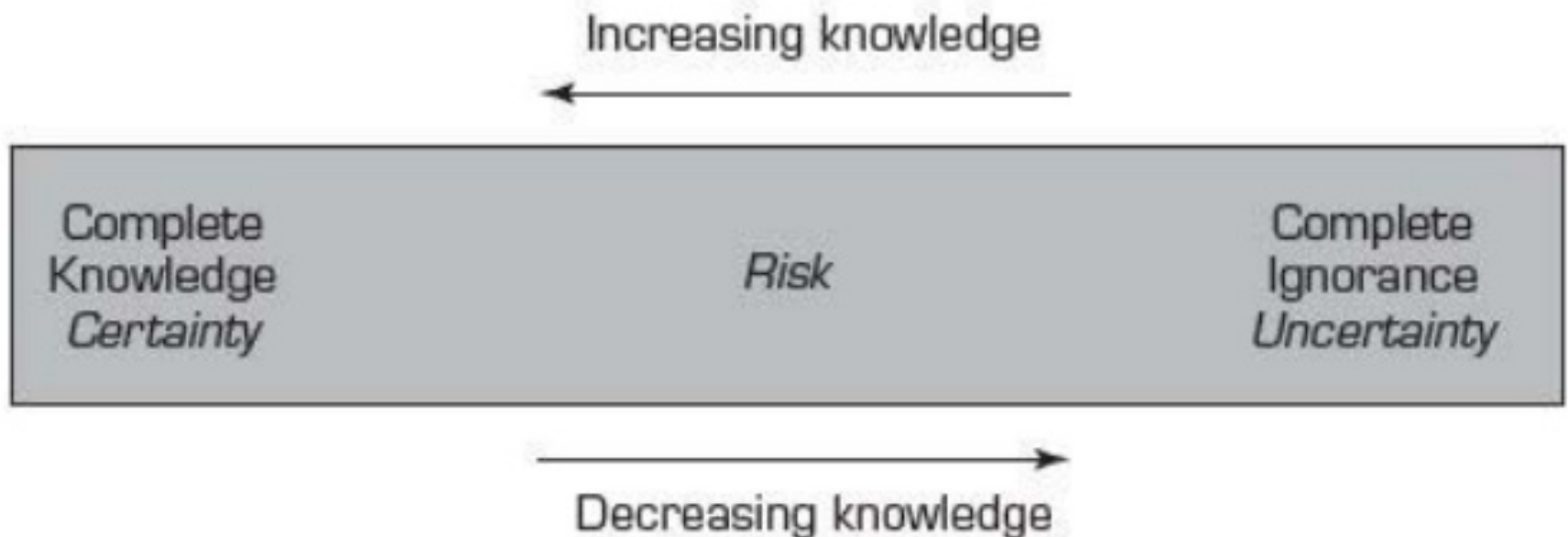
- It is assumed that **complete knowledge** is available so that the decision-maker knows exactly what the outcome of each course of action will be (as in a **deterministic** environment).
- The decision-maker is viewed as a **perfect predictor** of the future because it is assumed that there is only one outcome for each alternative
- Certainty models are relatively easy to develop and solve and can yield optimal solutions. Many financial models are constructed under assumed certainty

DECISION-MAKING UNDER UNCERTAINTY

- The decision-maker considers situations in which several outcomes are possible for each course of action.
- In this case the decision-maker does not know, or cannot estimate, the probability of occurrence of the possible outcomes.
- Decision-making under uncertainty is more difficult because of insufficient information.

DECISION-MAKING UNDER RISK

- The decision-maker must consider several possible outcomes for each alternative, each with a **given probability** of occurrence.



DECISION ANALYSIS

- **Problem Formulation**

- A decision problem is characterized by **decision alternatives**, **states of nature**, and resulting **payoffs**.
- The **decision alternatives** are the different possible strategies the decision maker can employ.
- The **states of nature** refer to future events, not under the control of the decision maker, which will ultimately affect decision results.

DECISION ANALYSIS MODELS

- **Decision theory problems are represented as one of the following:**
 - Payoff Table
 - Decision Tree

PAYOFF TABLES (I)

- The consequence resulting from a specific combination of a decision alternative and a state of nature is a **payoff**.
- **In a Payoff Table:**
 - ❖ The **rows** correspond to the possible decision **alternatives**.
 - ❖ The **columns** correspond to the possible future events (**States of Nature**).
 - ❖ The **body** of the table contains the **payoffs**.

PAYOFF TABLES (2)

THIS IS A PROFIT PAYOFF TABLE

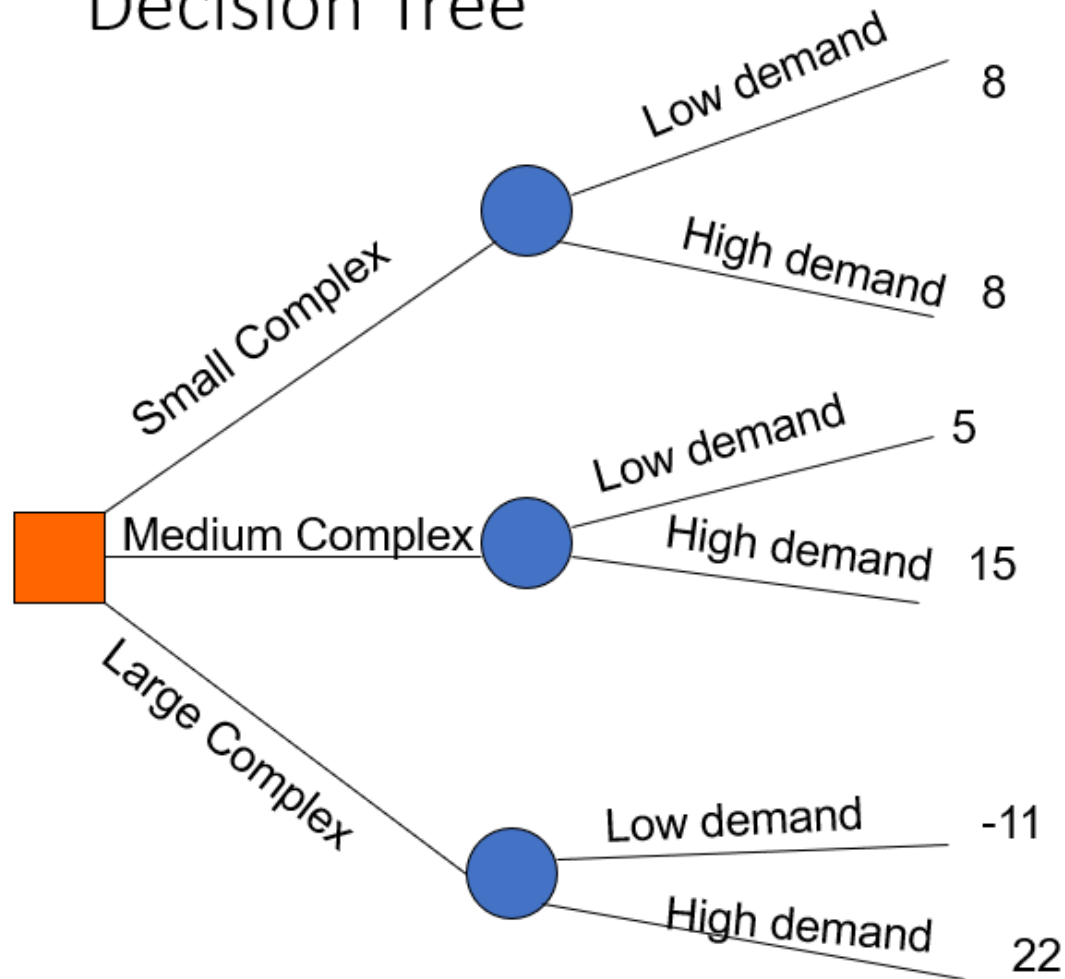
<u>Alternatives</u>	<u>States of Nature</u>	
	Low	High
Small	8	8
Medium	5	15
Large	-11	22

DECISION TREES (I)

- A **decision tree** is a graphical representation of the decision problem.
- Each decision tree has two types of nodes; **round nodes** correspond to the states of nature while **square nodes** correspond to the decision alternatives.
- The branches leaving each round node represent the different states of nature while the branches leaving each square node represent the different **decision alternatives**.
- At the end of each limb of a tree are the **payoffs** attained from the series of branches making up that limb.

DECISION TREES (2)

Decision Tree



TYPES OF DECISION-MAKING ENVIRONMENTS

- **Type 1: Decision making under certainty.**
 - Decision maker *knows* with certainty the consequences of every alternative or decision choice.
- **Type 2: Decision making under uncertainty.**
 - The decision maker *does not know* the probabilities of the various outcomes.
- **Type 3: Decision making under risk.**
 - The decision maker *knows the probabilities* of the various outcomes.

DECISION MAKING UNDER UNCERTAINTY

- **Probabilities can NOT be assigned to the occurrence of states of nature in the future. There are several methods such as:**
 - Maximax (optimistic)
 - Maximin (pessimistic)
 - Minimax regret

DECISION MAKING UNDER UNCERTAINTY

- **Maximax** (optimistic approach)
 - Choose decision with the maximum of the maximum payoffs.
- **Maximin** (conservative approach)
 - Choose decision with the maximum of the minimum payoffs.
- **Minimax regret** (regret approach)
 - Choose decision with the minimum of the maximum regrets for each alternative.

OPTIMISTIC APPROACH

- The optimistic approach would be used by an optimistic decision maker.
- This approach would be suitable for an ***optimist, or 'risk-seeking' investor***, who seeks to achieve the best results if the best happens.
- The decision with the best possible payoff is chosen.
- If the payoff table was in terms of costs, the decision with the lowest cost would be chosen.
- If the payoff table was in terms of profits, the decision with the highest profit would be chosen.

OPTIMISTIC APPROACH (EX.)

ALTERNATIVE	STATE OF NATURE		MAXIMUM IN A ROW (\$)
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)	
Construct a large plant	200,000	-180,000	200,000
Construct a small plant	100,000	-20,000	100,000
Do nothing	0	0	0

Maximax

CONSERVATIVE APPROACH

- The **conservative approach** would be used by a conservative decision maker.
- This approach would be appropriate for a ***pessimist who seeks to achieve the best results if the worst happens.***
- **If the payoff was in terms of costs**, the maximum costs would be determined for each decision and then the decision corresponding to the minimum of these maximum costs is selected. (Hence, the maximum possible cost is minimized.)
- **If the payoff was in terms of profits**, the minimum profits would be determined for each decision and then the decision corresponding to the maximum of these minimum profits is selected. (Hence, the minimum possible profit is maximized.)

CONSERVATIVE APPROACH (EX.)

ALTERNATIVE	STATE OF NATURE		MINIMUM IN A ROW (\$)
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)	
Construct a large plant	200,000	−180,000	−180,000
Construct a small plant	100,000	−20,000	−20,000
Do nothing	0	0	0

Maximin

MINIMAX REGRET APPROACH

- The minimax regret strategy is the one that **minimizes the maximum regret. It is useful for a risk-neutral decision-maker.** Essentially, this is the technique for a 'sore loser' who does not wish to make the wrong decision.
- 'Regret' in this context is defined as the **opportunity loss or cost of having made the wrong decision**
- The minimax regret approach requires the construction of a regret table or an opportunity loss table. This is done by calculating for each state of nature the difference between each payoff and the best payoff for that state of nature.
- Then, using this regret table, the maximum regret for each possible decision is listed.
- The decision chosen is the one corresponding to the minimum of the maximum regrets.

MINIMAX REGRET APPROACH (EX. I)

■ Opportunity Loss Tables

STATE OF NATURE	
FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
200,000 – 200,000	0 – (–180,000)
200,000 – 100,000	0 – (–20,000)
200,000 – 0	0 – 0

ALTERNATIVE	STATE OF NATURE	
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
Construct a large plant	0	180,000
Construct a small plant	100,000	20,000
Do nothing	200,000	0

MINIMAX REGRET APPROACH (EX. 2)

■ Opportunity

	STATE OF NATURE	
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
ALTERNATIVE	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
Construct a large plant	0	180,000
Construct a small plant	100,000	20,000
Do nothing	200,000	0

ALTERNATIVE	STATE OF NATURE		MAXIMUM IN A ROW (\$)
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)	
Construct a large plant	0	180,000	180,000
Construct a small plant	100,000	20,000	100,000
Do nothing	200,000	0	200,000

Minimax

DECISION MAKING UNDER RISK (I)

- Decision making when there are several possible states of nature, and we know the probabilities associated with each possible state.
- **Expected Value Approach**
 - If probabilistic information regarding the states of nature is available, one may use the expected value (EV) approach.
 - Here the expected return for each decision is calculated by summing the products of the payoff under each state of nature and the probability of the respective state of nature occurring.
- The decision yielding the best expected return is chosen

DECISION MAKING UNDER RISK (2)

- The expected value of a decision alternative is the sum of weighted payoffs for the decision alternative.
- The expected value (EV) of decision alternative d_i is defined as:

$$EV(d_i) = \sum_{j=1}^N P(s_j) V_{ij}$$

where: N = the number of states of nature

$P(s_j)$ = the probability of state of nature s_j

V_{ij} = the payoff corresponding to decision alternative d_i and state of nature s_j

EXAMPLE: BURGER PRINCE

Burger Prince Restaurant is contemplating opening a new restaurant on Main Street. It has three different models, each with a different seating capacity. Burger Prince estimates that the average number of customers per hour will be 0.4, 0.2, or 0.4. The payoff table (profits) for the three models is on the next slide.

DECISION MAKING UNDER RISK (EX. I)

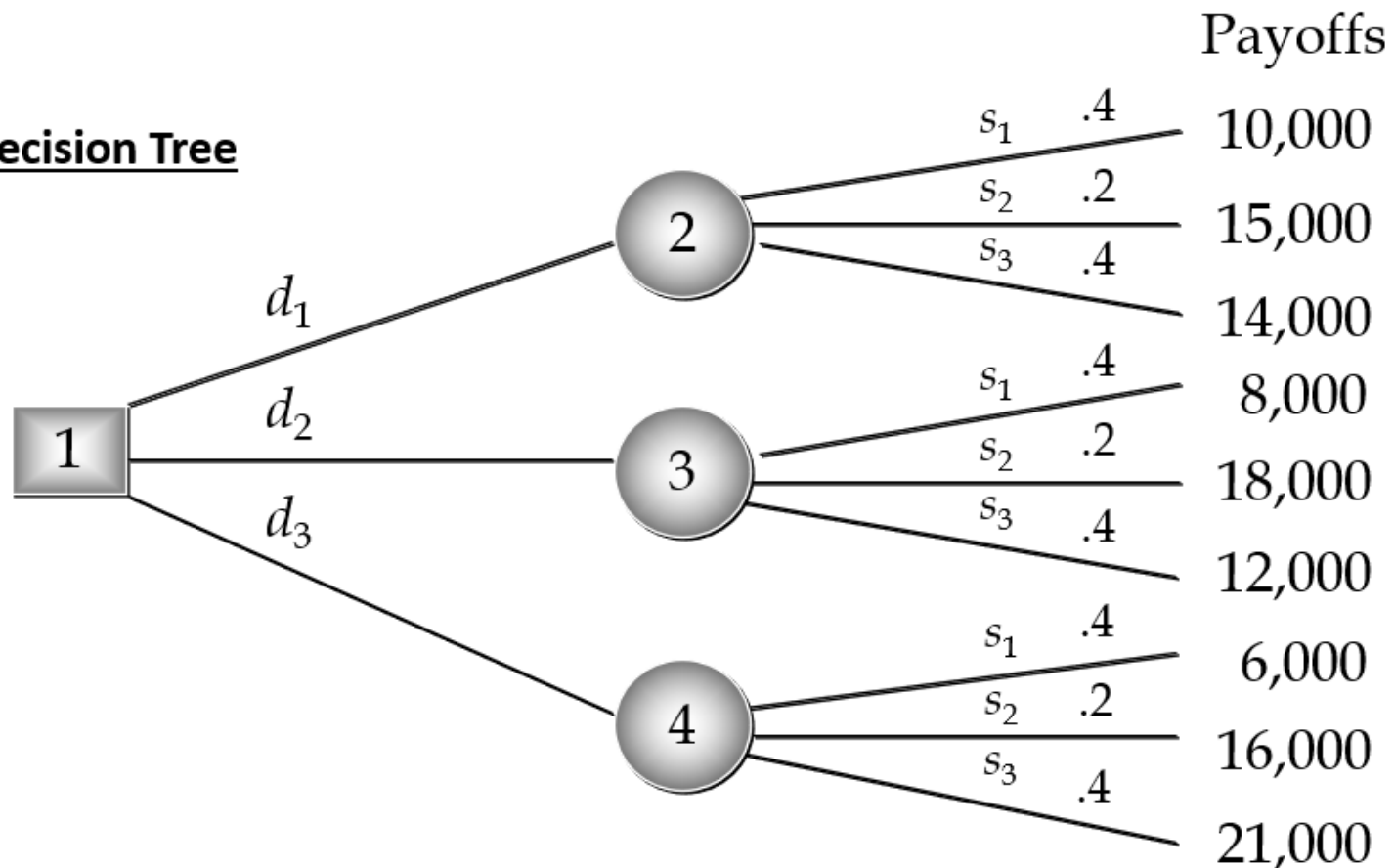
Average Number of Customers Per Hour

	$s_1 = 0.4$	$s_2 = 0.2$	$s_3 = 0.4$
Model A	\$10,000	\$15,000	\$14,000
Model B	\$ 8,000	\$18,000	\$12,000
Model C	\$ 6,000	\$16,000	\$21,000

DECISION MAKING UNDER RISK (EX. 2)

EXPECTED VALUE APPROACH

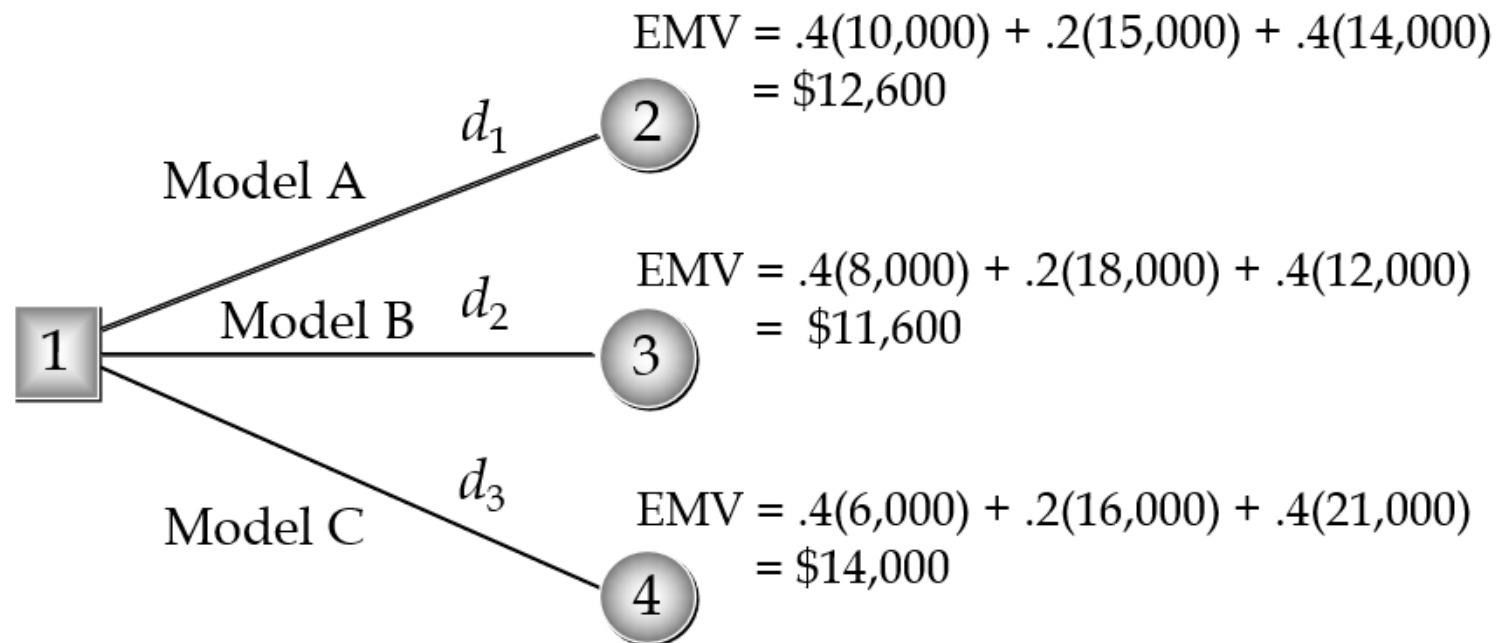
- Decision Tree



DECISION MAKING UNDER RISK (EX. 3)

EXPECTED VALUE APPROACH

⊕ Expected Value For Each Decision



Choose the model with largest EV, Model C.

DECISION MAKING UNDER RISK

EXPECTED VALUE APPROACH

■ Expected Value of Perfect Information

- Frequently information is available which can improve the probability estimates for the states of nature.
- The expected value of perfect information (EVPI) is the increase in the expected profit that would result if one knew with certainty which state of nature would occur.
- The EVPI provides an upper bound on the expected value of any sample or survey information.

DECISION MAKING UNDER RISK

EXPECTED VALUE APPROACH

- **EVPI Calculation**
 - **Step 1:** Determine the optimal return corresponding to each state of nature.
 - **Step 2:** Compute the expected value of these optimal returns.
 - **Step 3:** Subtract the EV of the optimal decision from the amount determined in step (2).
 - **Expected Value of Perfect Information**
 - Calculate the expected value for the optimum payoff for each state of nature and subtract the EV of the optimal decision.
 - $EVPI = .4(10,000) + .2(18,000) + .4(21,000) - 14,000 = \$2,000$

TASK

		States of Nature			
		Growth	Medium G	No Change	Low
		G	MG	NC	L
Actions	Bonds	12	8	7	3
	Stocks	15	9	5	-2
	Deposit	7	7	7	7

A- Determine which alternative decision maker will choose according to optimistic, pessimistic, and regret approaches

B- If the decision maker know the probabilities for each state of nature as following: 0.3,0.5,0.2, which decision will choose according to the expected value approach and compute the EVPI?



Thanks
Any questions



Mansoura University
Faculty of Computers and Information
Fall 2023-2024



DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

Ibrahim El-Hasnony

QUIZ-I

1. The decision-making process can be defined as the activity that leads to the solution of a decision-making problem involving at leastAlternatives
(one, two, three, four)
2. Data availability is more in which of the following?
(Strategic, management, operational, tactical)
3.supports decision making by analyzing given time-series of data and returning new information gained by those analysis
(data-driven DSS, organizational DSS, document-driven DSS, Knowledge-driven DSS)
4.improving communication and coordination among people with organizations to improve organization performance and solve problems
(web-based DSS, GDSS, KBDSS, ODSS)
5.is a technology-driven process for analyzing data and delivering actionable information that helps executives, managers and workers make informed business decisions.
(BI, DSS, MIS, MSS)

QUIZ-2

1.phase of DSS consists of discovering, identifying, and understanding the problems occurring in the organization
(intelligence, design, implementation, control)
2. The output of the DSS design phase is
(Problem statement, solutions, alternatives, criteria)
3. The decision-maker is viewed as a perfect predictor of the future inenvironment.
(uncertainty, risk, certainty, making decision)
4. Decision-making under uncertainty is more difficult because of insufficient information (T/F)
5. The state of nature are the different possible strategies the decision maker can employ. (T/F)

QUIZ-3

- A. According to the following profit payoff table, using the optimistic, pessimistic and regret approaches, which solution decision maker choose form A,B, and C
- B. If decision maker know that the probabilities of $s_1=0.2$, $s_2=0.3$, and $s_3=0.5$, which decision according to expected value approach, and the value of EVPI?

	S1	S2	S3
A	4	-4	9
B	5	8	3
C	6	12	16

COURSE DETAILS

Course Syllabus

Chapter 1: Introduction to DSS

Chapter 2: Decision Making Concepts

Chapter 3: Decision Making Technologies and Environment

Chapter 4: Model Management

Chapter 5: Data Management System and Dialogue Management

Chapter 6: Designing and Developing Decision Support Systems

Chapter 7: Executive Support Systems

Chapter 8: Automated Decision Systems and Expert Systems

Chapter 9: Knowledge Management and Collaborative Systems

Chapter 10: DSS Intelligent Systems & Tools and Techniques

Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

CHAPTER 4 CONTENTS

- Building Model-Driven Decision Support System (MDSS)
- Model Types
- AHP
- Automating Knowledge Sharing
- Data Mining and Creating Knowledge
- Designing and Developing Web-Based DSS

CHAPTER 4 CONTENTS

- **Building Model-Driven Decision Support System (MDSS)**
- **Model Types**
- **AHP**
- Automating Knowledge Sharing
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BUILDING MODEL-DRIVEN DECISION SUPPORT SYSTEM (MDSS)

- **Model-driven decision support system (MDDSS)** utilizes a model to solve problems or aid in decision making.
- A model can be **statistical, financial, mathematical, analytical, simulation** or **optimization**.
- A model driven DSS may employ a **single model or a combination** of two or more models, depending upon the specific needs of its users.
- Simple models provide basic functionality while combination of two or more models lets users analyze complex data.
- Model-driven DSS are generally **not data intensive**. Rather they use parameters entered by decision makers and help them analyze a situation.

MODELLING DECISION SITUATIONS

- Model-driven DSS can be used to aid decision making in a variety of situations. **It can assist managers in making:**
 - ✓ Credit and lending decisions
 - ✓ Product demand forecasting
 - ✓ Budgeting decisions
 - ✓ Marketing decisions
 - ✓ Production forecasting decisions
 - ✓ Resource allocation decisions
 - ✓ Project planning
 - ✓ Investment decisions

HOW TO BUILD A MODEL-DRIVEN DSS? (I)

- The selection of a model is the most crucial step in building an MDSS
- **Modeling** is the process of identifying an appropriate model for a prospective model- driven decision support system
- Once modeling is done, it's vital to validate the selected model, to ensure it works well and generates appropriate results.
- **Model validation** is done by comparing model's output and the actual behavior of the event.

HOW TO BUILD A MODEL-DRIVEN DSS? (2)

- **Assumptions & Forecasts**

- Assumptions are predictions or best guesses.
- Each model has certain assumptions about the **time** and **risk** involved in a particular situation.
- **These results are tested through sensitivity or what if analysis.**
- Assumptions play an important role in defining a problem and identifying and dealing with uncertainty.
- Decision makers form a hypothesis and attempt to predict results. Basis the outcome, a hypothesis is either accepted or rejected

HOW TO BUILD A MODEL-DRIVEN DSS? (3)

- **Model-driven DSS** are designed assuming any of the analyses – static and dynamic.
- **Static Analysis:** This type of analysis doesn't take into consideration the long-term response of a system.
- It takes a single snapshot of a situation and assumes that it will remain stable all through and won't change.
- **Dynamic Analysis:** Dynamic analysis is testing a program or a software system in real-time.
- This method considers that the situation changes over time, due to any reason, such as cost, rules and regulations, time, etc.

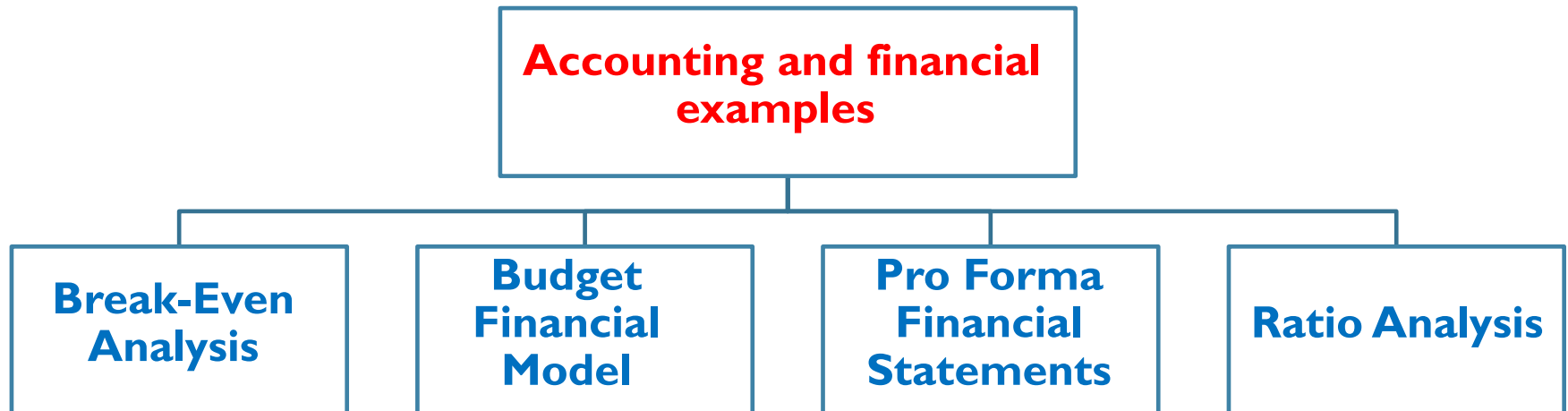
MODEL TYPES (I)

- **A DSS deploys one or combination of below models:**
- **Explanatory/Descriptive Model:** Describes and explains why something is the way it is and why and how it works.
- **Contemplative Model:** Forecasts results or outcomes that may be produced from a specific set of parameters.
- **Algebraic Model:** A high-level modeling system for solving complex equations. It is employed to *optimize* a variable or equation. The best part is that it can handle several *simultaneous* equations.

MODEL TYPES (2)

- **DSS with multiple models is a complete system to perform all three tasks, including:**
 - Identifying relationships between variables
 - Forecasting results based on changes or parameters.
 - Deciding to what extent a variable can be manipulated.

I.ACCOUNTING AND FINANCIAL MODELS (EXAMPLES)



BREAK-EVEN ANALYSIS

- **Break-even analysis** is a financial tool that helps businesses determine the sales volume or revenue needed to cover all costs associated with producing and selling a product or service.
- It is a useful tool for managers to make informed decisions about pricing, production, and marketing.
- **A break-even analysis model typically includes the following components:**
 - I. **Fixed costs:** These are costs that remain the same regardless of the level of production, such as rent, salaries, and insurance.
 - II. **Variable costs:** These are costs that change with the level of production, such as raw materials and direct labor.
 - III. **Selling price:** The price at which the product or service is sold.
- The break-even point is calculated using the following formula:
Break-even point = Fixed costs / (Selling price - Variable cost per unit)

BREAK-EVEN ANALYSIS EXAMPLE I

- For example, if a company has fixed costs of \$10,000, variable costs of \$5 per unit, and a selling price of \$10 per unit, the break-even point would be calculated as follows:

$$\text{Break-even point} = \$10,000 / (\$10 - \$5) = 2,000 \text{ units}$$

- This means that the company must sell 2,000 units to cover all of its costs. Any sales above this level would result in a profit, and any sales below this level would result in a loss.

BUDGET FINANCIAL MODEL

- A DSS with budgeting model can help managers to:
 - Set financial goals and objectives.
 - Develop and allocate resources.
 - Track and analyze financial performance.
 - Forecast future financial performance.
- Some examples of DSS with budgeting model software include:
 - Oracle Hyperion Planning and Budgeting
 - SAP Integrated Business Planning
 - IBM Planning Analytics Workspace
 - Anaplan
 - Workday Adaptive Planning

PRO FORMA FINANCIAL STATEMENTS

- DSS with a pro forma financial statements model can summarize the anticipated financial results for a specific period in the future.
- It does this by estimating costs based on past data and predicting gross sales.
- Pro forma financial statements are useful for a variety of purposes, such as:
 - Evaluating the impact of a new product or service launch.
 - Assessing the feasibility of a merger or acquisition.
 - Developing a financial plan for a new venture.
 - Obtaining financing from investors or lenders.

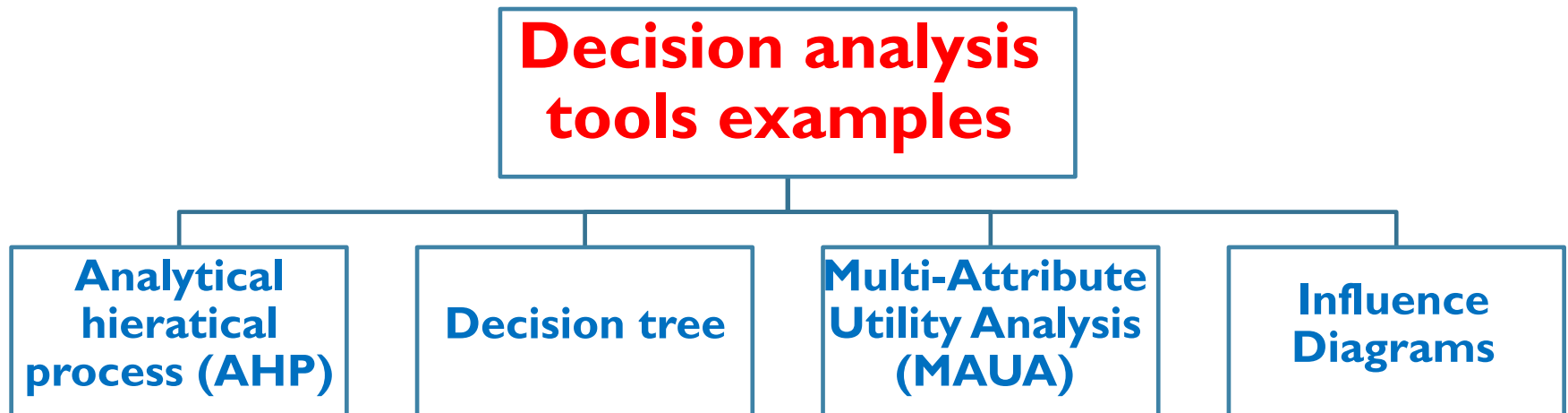
RATIO ANALYSIS

- Here are some of the benefits of using ratio analysis:
 - It can help you to identify trends and patterns in your financial data.
 - It can help you to *compare* your company's performance to that of other companies in your industry.
 - It can help you to identify potential areas of concern or improvement.
 - It can help you to make better decisions about your business

2. DECISION ANALYSIS MODELS (I)

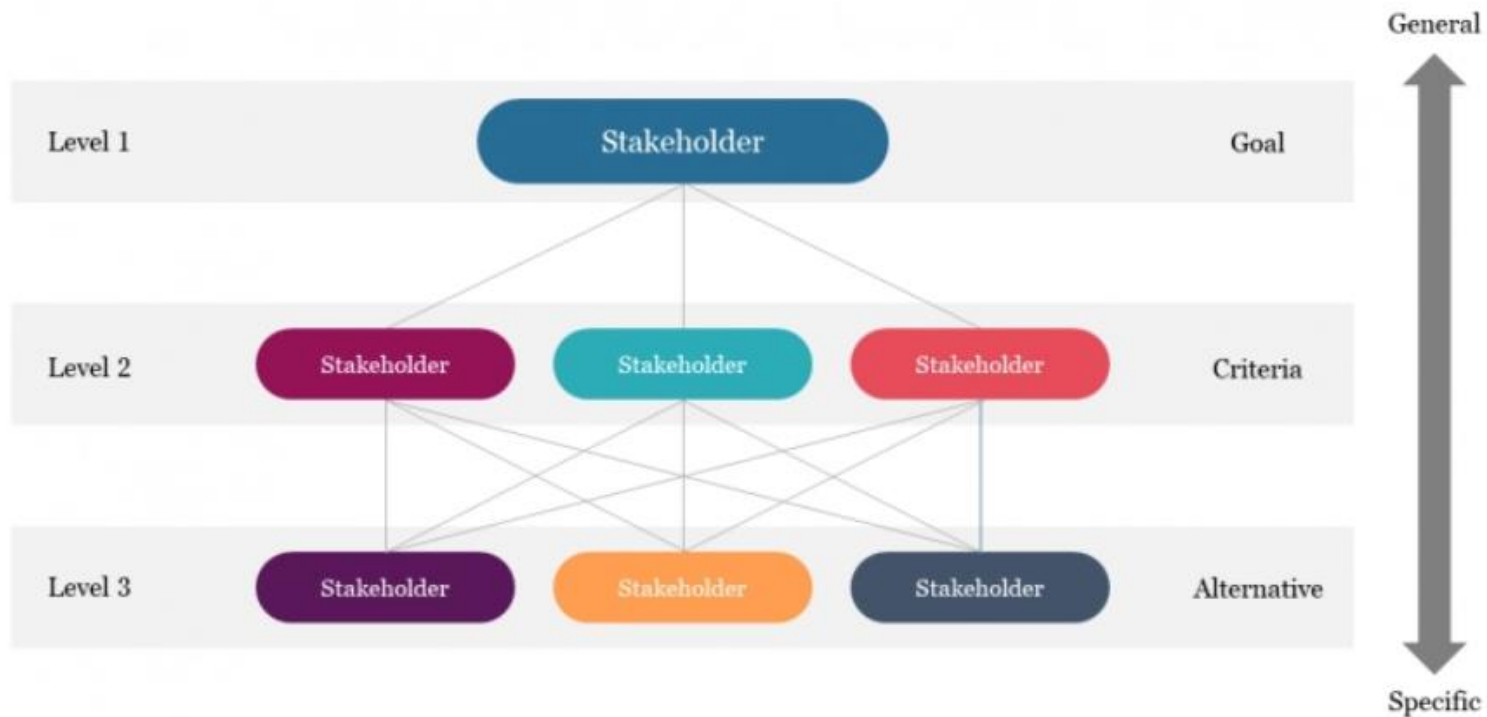
- The main job of decision analysis models is to identify and evaluate alternatives with their respective pros and cons.
- The decision makers then evaluate all the alternatives and pick the one that they think is the best.
- The aim of decision analysis techniques is to:
 - Decompose and restructure the problems.
 - Help decision makers gain in-depth understanding of the problem.
 - Separate facts and figures from preferences and priorities
 - Help users study the performance of decision alternatives.
 - Avoid citing priorities that don't help in decision making.

2. DECISION ANALYSIS MODELS (2)



ANALYTICAL HIERARCHICAL PROCESS (AHP)

- **Analytical Hierarchy Process (AHP):** It's a multi-criteria decision technique (**MCDM**) that combines quantitative and qualitative factors when evaluating alternatives.



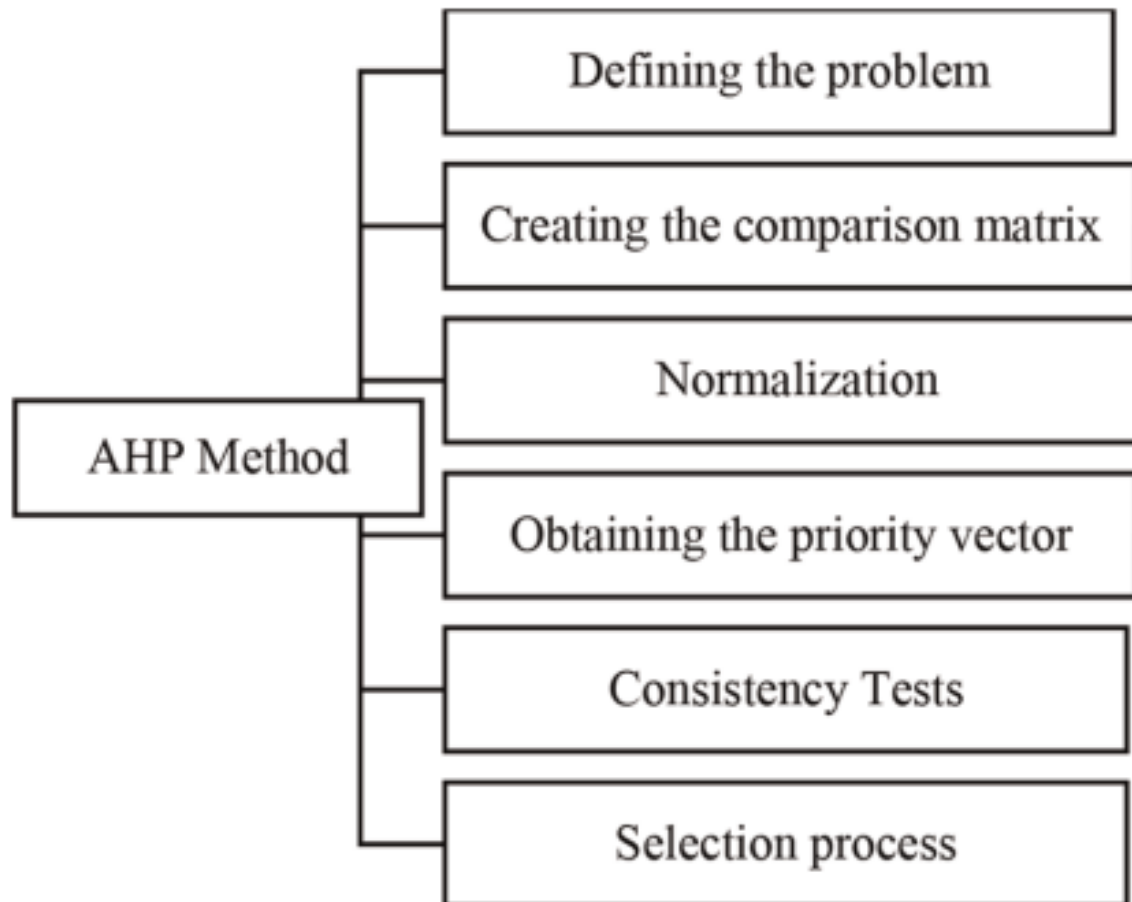
THE ANALYTIC HIERARCHY PROCESS (AHP)

- Founded by Saaty in 1980.
- It is a popular and widely used method for multi-criteria decision-making.
- Allows the use of qualitative, as well as quantitative criteria in evaluation.
- A wide range of applications exists such as:
 - Selecting a car for purchasing
 - Deciding upon a place to visit for vacation
 - Deciding upon an MBA program after graduation.

AHP-GENERAL IDEA

- Develop an **hierarchy of decision criteria** and define the **alternative courses of actions**.
- AHP algorithm is basically composed of two steps:
 1. Determine the relative weights of the decision criteria
 2. Determine the relative rankings (priorities) of alternatives
- ! Both **qualitative and quantitative** information can be compared by using informed judgments to derive weights and priorities.

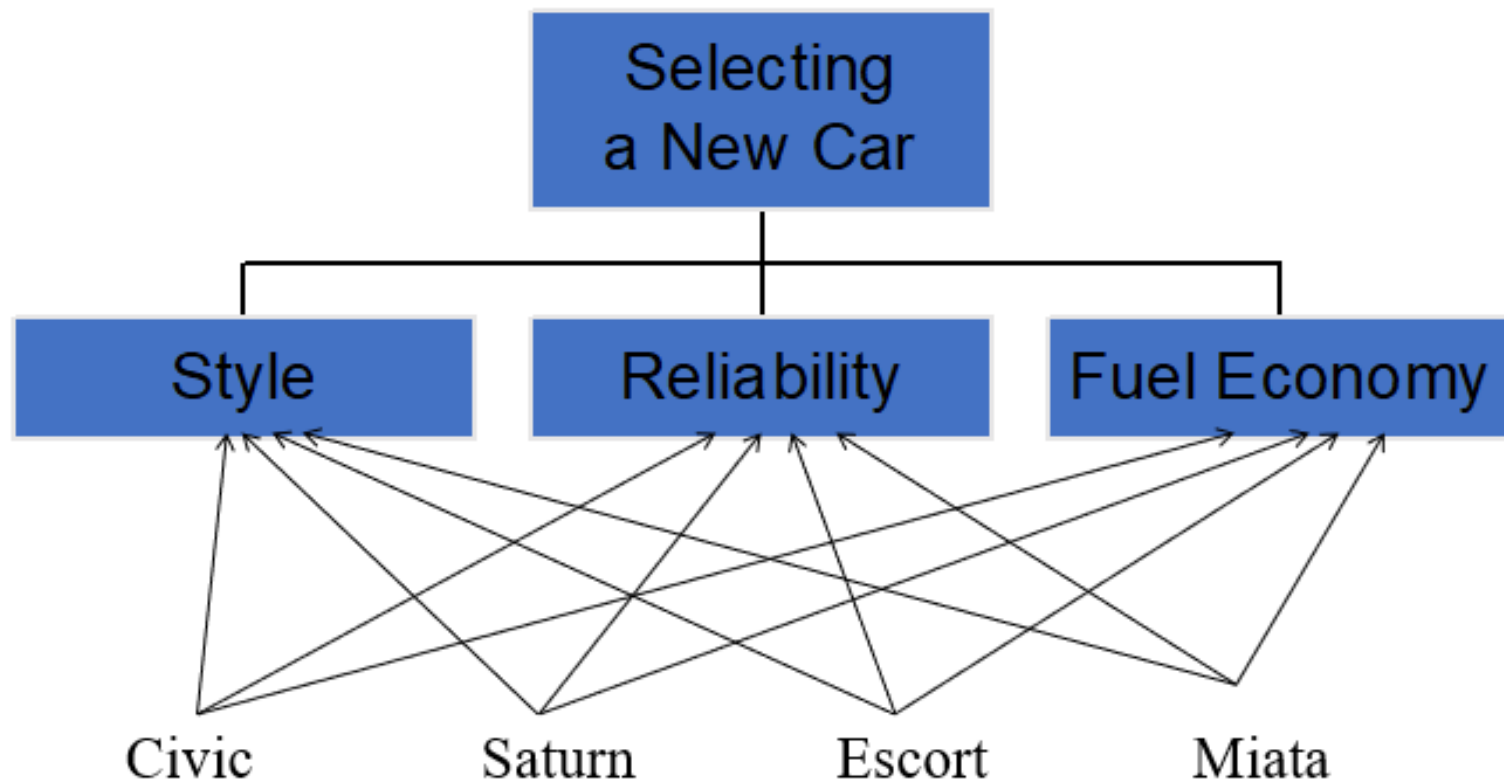
AHP METHOD



EXAMPLE: CAR SELECTION

- Objective
 - Selecting a car
- Criteria
 - Style, Reliability, Fuel-economy?
- Alternatives
 - Civic Coupe, Saturn Coupe, Ford Escort, Mazda Miata

HIERARCHY TREE



Alternative courses of action

RANKING OF CRITERIA AND ALTERNATIVES

- Pairwise comparisons are made with the grades ranging from 1-9.
- A basic, but very reasonable assumption for comparing alternatives:

If attribute A is absolutely more important than attribute B and is rated at 9, then B must be absolutely less important than A and is graded as 1/9.

RANKING SCALE FOR CRITERIA AND ALTERNATIVES

Ranking of importance	Definition	Reciprocal ranking of importance	Explanation
1	Equal importance	1	Two criteria/alternatives contribute equally to the goal/objective
3	Somewhat more important	1/3	Experience and judgement slightly favour one over the other
5	Much more important	1/5	Experience and judgement strongly favour one over the other
7	Very much more important	1/7	Experience and judgement very strongly favour one over the other
Ranking of importance	Definition	Reciprocal ranking of importance	Explanation
9	Absolutely more important	1/9	Experience and judgement assign the highest possible validity of favouring one over the other
2, 4, 6, 8	Intermediate values	1/2, 1/4, 1/6, 1/8	When compromise is needed

RANKING OF CRITERIA

	Style	Reliability	Fuel Economy
Style	1	1/2	3
Reliability	2	1	4
Fuel Economy	1/3	1/4	1

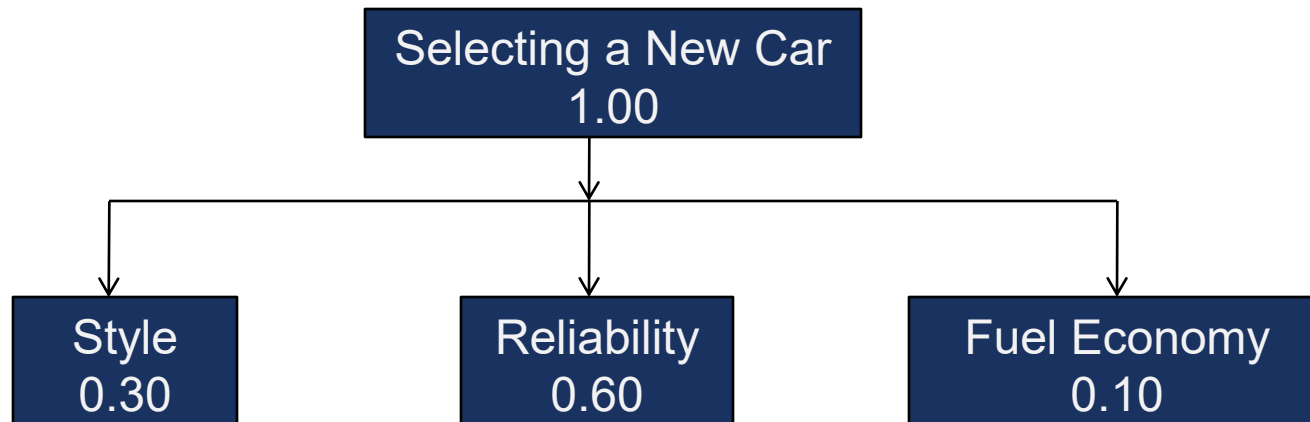
RANKING OF PRIORITIES

- Consider $[Ax = \lambda_{\max}x]$ where
 - A is the comparison matrix of size $n \times n$, for n criteria, also called the priority matrix.
 - x is the Eigenvector of size $n \times 1$, also called the priority vector.
 - $\lambda_{\max}$ is the Eigenvalue, $\lambda_{\max} \in \mathbb{R} > n$.
- To find the ranking of priorities, namely the Eigen Vector X :
 - 1) Normalize the column entries by dividing each entry by the sum of the column.
 - 2) Take the overall row averages.

$$\begin{array}{c}
 A = \begin{bmatrix} 1 & 0.5 & 3 \\ 2 & 1 & 4 \\ 0.33 & 0.25 & 1.0 \end{bmatrix} \xrightarrow{\text{Normalized Column Sums}} \begin{bmatrix} 0.30 & 0.29 & 0.38 \\ 0.60 & 0.57 & 0.50 \\ 0.10 & 0.14 & 0.13 \end{bmatrix} \xrightarrow{\text{Row averages}} X = \begin{bmatrix} 0.30 \\ 0.60 \\ 0.10 \end{bmatrix} \\
 \text{Column sums } 3.33 \quad 1.75 \quad 8.00 \qquad \qquad \qquad 1.00 \quad 1.00 \quad 1.00 \qquad \qquad \qquad \text{Priority vector}
 \end{array}$$

CRITERIA WEIGHTS

- Style .30
- Reliability .60
- Fuel Economy .10



CHECKING FOR CONSISTENCY

- The data obtained in the priority vector is also used as weights in the AHP matrix. After this stage, it is checked whether the answers are consistent.
- The next stage is to calculate a **Consistency Ratio (CR)** to measure how consistent the judgments have been relative to large samples of purely random judgments.
- AHP evaluations are based on the assumption that the decision maker is rational, i.e., if A is preferred to B and B is preferred to C, then A is preferred to C.
- **If the CR is greater than 0.1** the judgments are untrustworthy because they are too close for comfort to randomness and the exercise is valueless or must be repeated.

CALCULATION OF CONSISTENCY RATIO

- The next stage is to calculate $\lambda_{\max}$ so as to lead to the Consistency Index and the Consistency Ratio.
- Consider $[Ax = \lambda_{\max} x]$ where x is the Eigenvector.

$$\begin{array}{c} A \\ \left[\begin{array}{ccc} 1 & 0.5 & 3 \\ 2 & 1 & 4 \\ 0.333 & 0.25 & 1.0 \end{array} \right] \end{array} \begin{array}{c} x \\ \left[\begin{array}{c} 0.30 \\ 0.60 \\ 0.10 \end{array} \right] \end{array} = \begin{array}{c} Ax \\ \left[\begin{array}{c} 0.90 \\ 1.60 \\ 0.35 \end{array} \right] \end{array} = \lambda_{\max} \begin{array}{c} x \\ \left[\begin{array}{c} 0.30 \\ 0.60 \\ 0.10 \end{array} \right] \end{array}$$

$$\lambda_{\max} = \text{average}\{0.90/0.30, 1.60/0.6, 0.35/0.10\} = 3.06$$

- Consistency index , CI is found by

$$CI = (\lambda_{\max} - n) / (n - 1) = (3.06 - 3) / (3 - 1) = 0.03$$

CONSISTENCY RATIO

- The final step is to calculate the Consistency Ratio, CR by using the table below, derived from Saaty's book. The upper row is the order of the random matrix, and the lower row is the corresponding index of consistency for random judgments.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49	1.51	1.48	1.56	1.57	1.59

Each of the numbers in this table is the average of CI's derived from a sample of randomly selected reciprocal matrices of AHP method.

An inconsistency of 10% or less implies that the adjustment is small as compared to the actual values of the eigenvector entries.

A CR as high as, say, 90% would mean that the pairwise judgments are just about random and are completely untrustworthy! In this case, comparisons should be repeated.

In the above example: $CR = CI / 0.58 = 0.03 / 0.58 = 0.05$

$0.05 < 0.1$, so the evaluations are consistent!

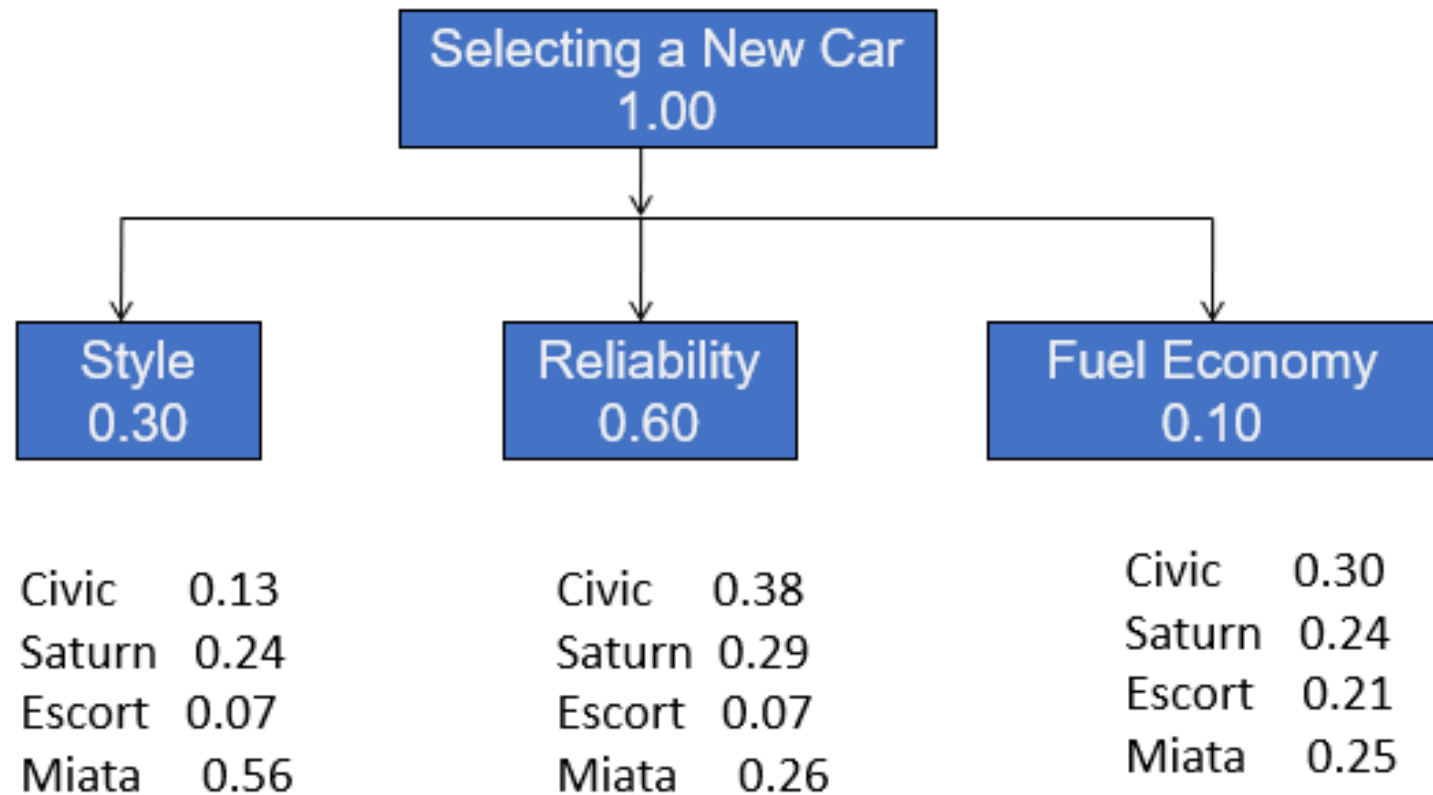
RANKING ALTERNATIVES

<u>Style</u>	Civic	Saturn	Escort	Miata	<u>Priority vector</u>
Civic	1	1/4	4	1/6	$\begin{bmatrix} 0.13 \\ 0.24 \\ 0.07 \\ 0.56 \end{bmatrix}$
Saturn	4	1	4	1/4	
Escort	1/4	1/4	1	1/5	
Miata	6	4	5	1	
<u>Reliability</u>	Civic	Saturn	Escort	Miata	
Civic	1	2	5	1	$\begin{bmatrix} 0.38 \\ 0.29 \\ 0.07 \\ 0.26 \end{bmatrix}$
Saturn	1/2	1	3	2	
Escort	1/5	1/3	1	1/4	
Miata	1	1/2	4	1	

RANKING ALTERNATIVES

		<u>Miles/gallon</u>	<u>Normalized</u>
<u>Fuel Economy</u>	Civic	34	.30
	Saturn	27	.24
	Escort	24	.21
	Miata	<u>28</u>	<u>.25</u>
		113	1.0

! Since fuel economy is a quantitative measure, fuel consumption ratios can be used to determine the relative ranking of alternatives; however this is not obligatory. Pairwise comparisons may still be used in some cases.



RANKING OF ALTERNATIVES

	Style	Reliability	Fuel Economy			
Civic	.13	.38	.30	x	$\begin{bmatrix} .30 \\ .60 \\ .10 \end{bmatrix}$	$= \begin{bmatrix} .30 \\ .27 \\ .08 \\ .35 \end{bmatrix}$
Saturn	.24	.29	.24			
Escort	.07	.07	.21			
Miata	.56	.26	.25			

↓

Priority matrix

↓

Criteria Weights

MORE ABOUT AHP: PROS AND CONS

Pros

- It allows **multi criteria decision making**.
- It is applicable when it is difficult to formulate criteria evaluations, i.e., it allows **qualitative evaluation** as well as quantitative evaluation.
- It is applicable for **group decision making** environments

Cons

- There are hidden assumptions like **consistency**. Repeating evaluations is cumbersome.
- Difficult to use when the **number of criteria or alternatives is high, i.e., more than 7**.
- Difficult to add a **new criterion or alternative**
- Difficult to **take out an existing criterion or alternative**, since the best alternative might differ if the worst one is excluded.

Users should be trained to use AHP methodology.

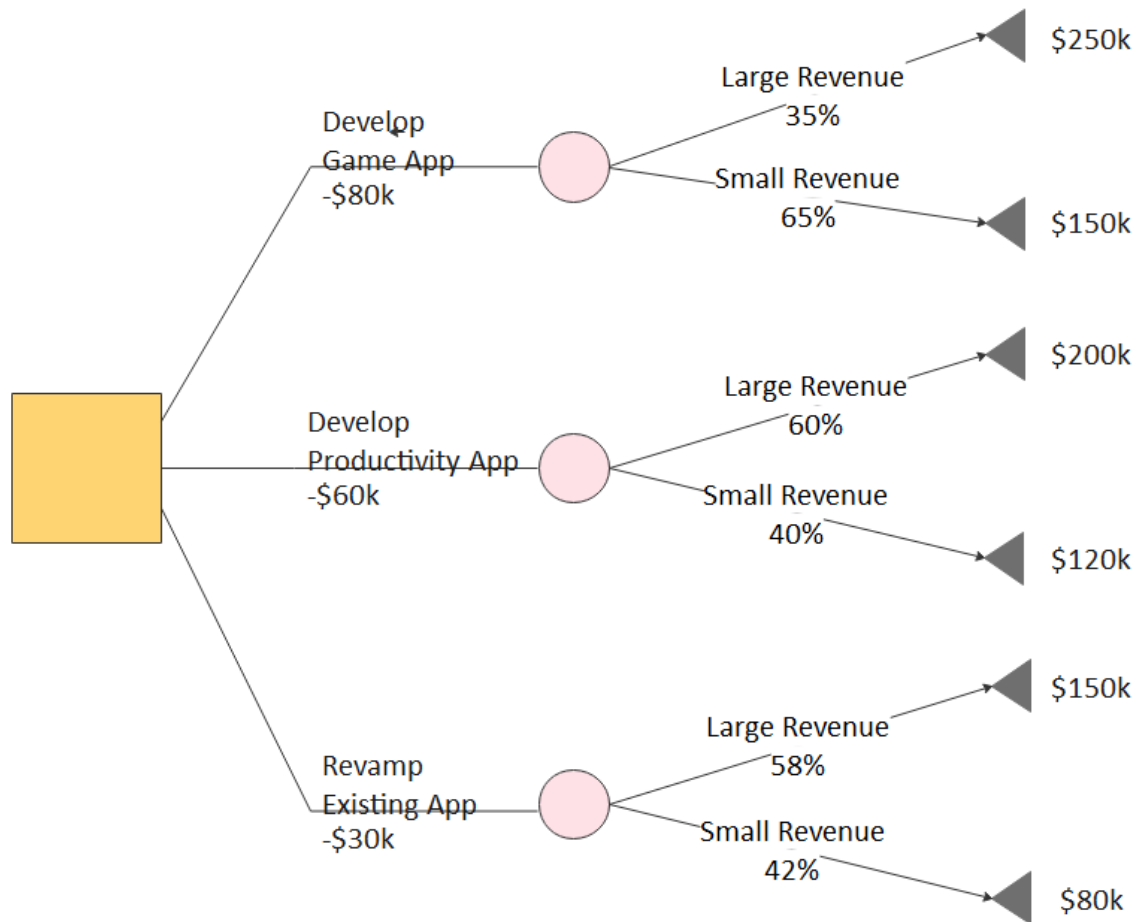
Use **GDSS**
Use **constraints** to eliminate some alternatives

Use **cost/benefit** ratio if applicable

DECISION TREES

- A decision tree uses a tree-like flowchart of decisions, drawn from left to right, with further branches explaining their consequences, cost involved, event outcomes and utility.
- The aim is to identify the most appropriate strategy to reach a goal.
- A decision tree has three types of nodes.
 - **Choice node**: represented by a square
 - **Chance node**: represented by a circle.
 - **End node**: represented by a triangle.

DT EXAMPLE



Expected Value = $(\$250k * 35\%) + (\$150k * 65\%) - \$80k = \$105k$

Expected Value = $(\$200k * 60\%) + (\$120k * 40\%) - \$60k = \$108k$

Expected Value = $(\$150k * 58\%) + (\$80k * 42\%) - \$30k = \$90.6k$

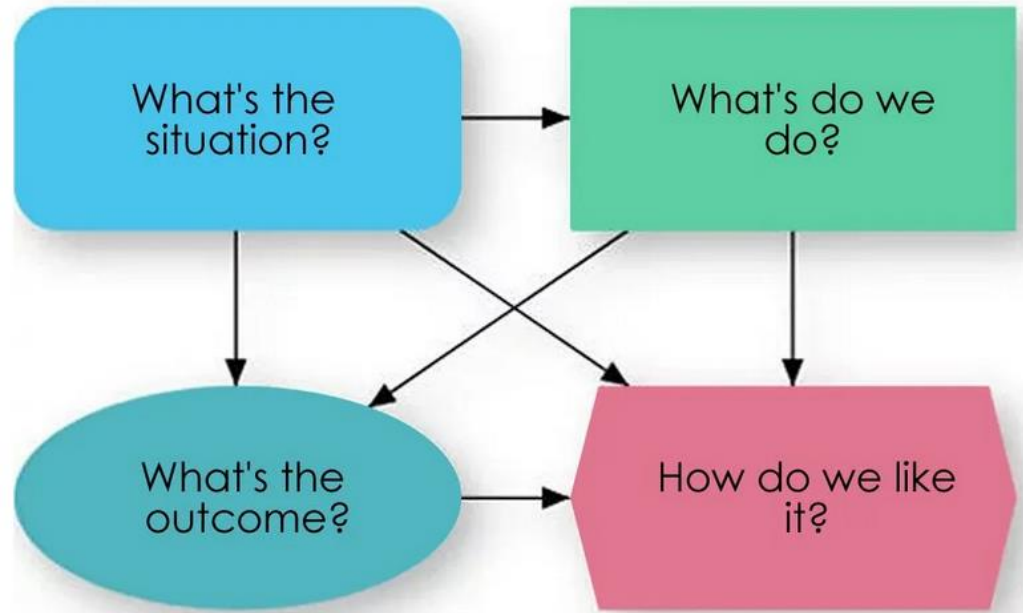
MULTI-ATTRIBUTE UTILITY ANALYSIS (MAUA) (SECTION)

- Multi-attribute utility analysis gives much importance to attribute weights.
 - The information is provided about each decision choice on each attribute.
 - A decision maker then perceives the utility of usefulness of a decision alternative in terms of its attributes.
 - This method is generally used when the attributes of an alternative are certain.
 - For example, if a decision-maker is choosing a new car, they might assign a higher weight to safety than to fuel efficiency.
- MAUA emphasizes the quantification of individual preferences using utility functions, while AHP relies on pairwise comparisons and the determination of relative importance among criteria and alternatives.

INFLUENCE DIAGRAMS (SECTION)

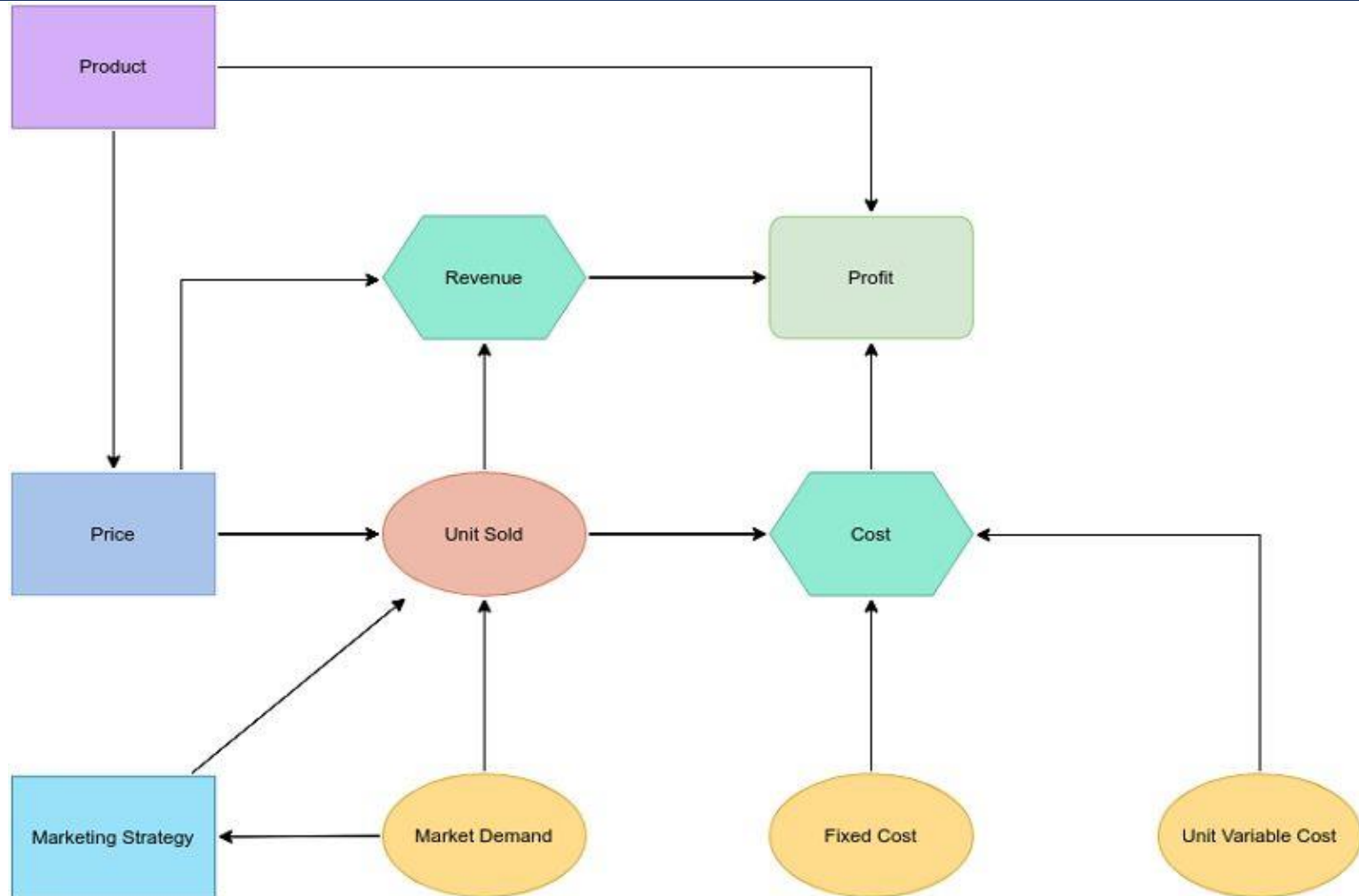
- It's a diagrammatic representation of a decision situation, to express the precise nature of relationships between variables and the influence one variable has on others. . It uses geometric shapes to represent various elements.
- **The elements of an influence diagram are:**
 - **Decision nodes:** These are represented by squares and represent the decisions that need to be made.
 - **Chance nodes:** These are represented by circles and represent the uncertainties in the problem.
 - **Value nodes:** These are represented by hexagons and represent the objectives of the decision-maker.
 - **Arcs:** These represent the relationships between the different elements of the problem.

INFLUENCE DIAGRAMS EXAMPLE (SECTION)



- a standard variable – What's the situation?
- a decision – What do we do?
- a chance variable – What's the outcome?
- a final valuation – How do we like it?

INFLUENCE DIAGRAMS EXAMPLE (SECTION)



3. FORECASTING MODELS

- Forecasting models form an integral part of many decision support systems.
- Their main job is to predict the value of interrelated variables at some point of time in future.
- The two main types of forecasts are:
 - ✓ Short run forecasts: where the prediction will be used anytime soon mainly in deterministic models.
 - ✓ Long run forecasts: where the prediction is used for long term investment/planning decisions.

VARIOUS TYPES OF FORECASTING MODELS

Naïve Exploration:

- simple forecasting that provides limited accuracy.

Judgment Methods

- based on the perceptions and opinions of experts instead on hard data, long term forecasting

Moving Average

- short-run forecasts, the predictions are based upon the historical values

Exponential Smoothing

- short-term forecasts, it alters the historical data mathematically to better reflect the assumptions of a decision maker

Time Series Extrapolation

- considers the economic variables that are measured at consecutive intervals of time. It is believed that the knowledge of past behavior of the variable at successive intervals of time will help understand the behavior of the variables in future better

Regression and Econometric Models

- use of linear and multiple regressions to establish cause and effect relationships

4. NETWORK AND OPTIMIZATION MODELS

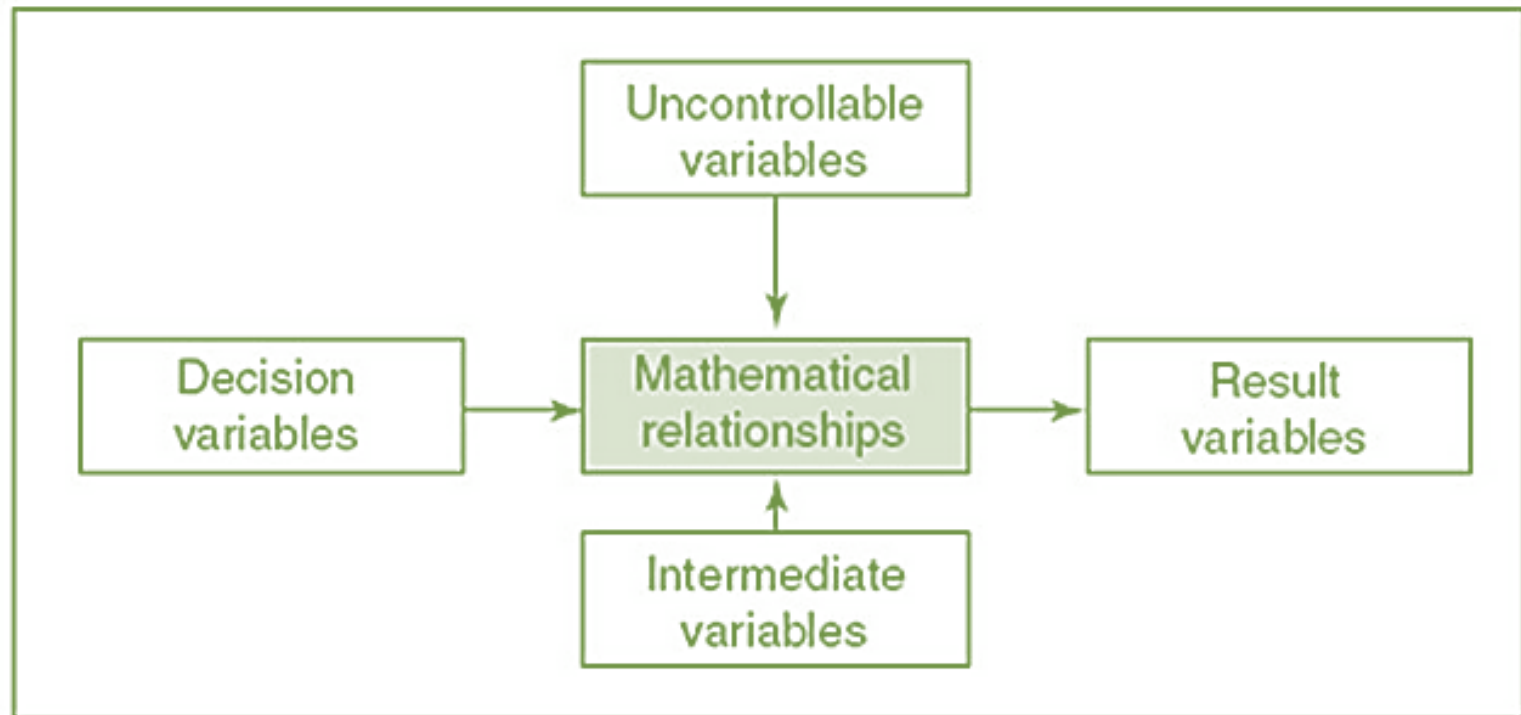
- Network and optimization models are integrated into a DSS when decisions regarding resource allocation, project control, location, scheduling, transportation, distribution, size, shortages, multinational cash flow management, inventory management and distribution and network need to be made. For example:
 - The best location for an operation or manufacturing
 - The resources needed to carry out the operations.
 - Most suitable aircraft route to transport products

5. SIMULATION MODELS

- DSS with simulation models conduct experiments to identify conditions or situations that approximate the actual conditions.
- These models are utilized to solve several problems, including.
 - Manpower planning and assignment
 - Inventory control
 - Reliability and replacement
 - Sequencing and scheduling
 - Stock-in and stock-out
 - Queuing and congestion

6-MATHEMATICAL MODELS

- All models are made up of three basic components: decision variables, uncontrollable variables (and/or parameters), and result (outcome) variables.



RESULT VARIABLES

- Reflect the level of **effectiveness** of the system; that is, they indicate how well the system performs or attains its goal(s).
- These variables are **outputs**.
- Result variables are considered dependent variables. Intermediate result variables are sometimes used in modeling to identify intermediate outcomes.
- In the case of a dependent variable, another event must occur first before the event described by the variable can occur.

DECISION VARIABLES

- Describe alternative courses of action.
- The decision-maker controls the decision variables.
- For example, for an investment problem, the amount to invest in bonds is a decision variable.
- In a scheduling problem, the decision variables are people, times, and schedules.

UNCONTROLLED VARIABLES OR PARAMETERS

- In any decision-making situation, there are factors that affect the result variables but are not under the control of the decision-maker.
- Either these factors can be fixed, in which case they are called parameters, or they can vary (variables).
- Examples are the prime interest rate, a city's building code, tax regulations, and utilities costs.
- Most of these factors are uncontrollable because they are in and determined by elements of the system environment in which the decision-maker works.
- Some of these variables limit the decision-maker and therefore form what are called the constraints of the problem.

INTERMEDIATE RESULT VARIABLES

- Reflect intermediate outcomes.
- For example, in determining machine scheduling, spoilage is an intermediate result variable, and total profit is the result variable (spoilage is one determinant of total profit).

TASK

Ex: Peter is offered 4 jobs from Acme Manufacturing (A), Bankers Bank (B), Creative Consulting (C), and Dynamic Decision Making (D). He bases his evaluation on the criteria such as location, salary, job content, and long-term prospects.

Step 1: Decide upon the relative importance of the selection criteria:

	Location	Salary	Content	Long-term
Location	1	1/5	1/3	1/2
Salary	5	1	2	4
Content	3	1/2	1	3
Long-term	2	1/2	1/3	1

TASK

Relative Scores for Each Criteria

Location Salary Content Long-Term

A	0.174	0.050	0.210	0.510
B	0.293	0.444	0.038	0.012
C	0.489	0.312	0.354	0.290
D	0.044	0.194	0.398	0.188



Thanks
Any questions



Mansoura University
Faculty of Computers and Information
Fall 2023-2024



DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

Ibrahim El-Hasnony

QUIZ-I

- 1) A model driven DSS may employ a single model or a combination of two or more models. (T / F)
- 2) Model-driven DSS are generally data intensive. (T / F)
- 3)is the process of identifying an appropriate model for a prospective model- driven decision support system (Validation. Assumptions, Modelling, Decision making).
- 4)Analysis takes a single snapshot of a situation and assumes that it will remain stable all through and won't change. (Ratio, break even, Dynamic, Static)
- 5)Analysis is a financial tool that helps businesses determine the sales volume or revenue needed to cover all costs associated with producing and selling a product or service. (Break even, ratio, static, dynamic)

QUIZ-2

- 1) If a company has fixed costs of \$100,000, variable costs of \$10 per unit, and a selling price of \$20 per unit, This means that the company must sell units to cover all of its costs. (1000, 2000, 5000, 10000)
- 2)analysis can summarize the anticipated financial results for a specific period in the future. (pro forma, ratio, break even, AHP)
- 3)analysis can help you to compare your company's performance to that of other companies in your industry. (pro forma, ratio, break even, AHP)
- 4) MAUA is considered.....tool. (Financial, budget analysis, decision analysis, optimization)
- 5) AHP is suitable for qualitative and quantitative information. (T, F)

QUIZ-3

Let's say we are making a decision on selecting a location for a new office, and we have three criteria to consider:

- Cost
- Proximity to Clients
- Quality of Life for Employees

We also have three potential locations (alternatives) to choose from:

- A. Downtown
- B. Suburb
- C. Suburb

Pairwise Comparison Matrix for Criteria - Relative Weights

	Cost	Proximity	Quality
Cost	1	3	5
Proximity	1/3	1	3
Quality	1/5	1/3	1

	A	B	C
Cost	1	3	5
	B	1	3
	C	1/5	1/3

Proximity

	A	B	C
A	1	1/3	4
B	3	1	5
C	1/3	1/5	1

Quality

	A	B	C
A	1	1/3	1/5
B	3	1	1/3
C	5	3	1

QUIZ-3-SOL.

- Alternative A: 0.1904
- Alternative B: 0.1881
- Alternative C: 0.6852
- Alternative C has the highest final weight, making it the most preferred option in your decision.

COURSE DETAILS

Course Syllabus

Chapter 1: Introduction to DSS

Chapter 2: Decision Making Concepts

Chapter 3: Decision Making Technologies and Environment

Chapter 4: Model Management (Part 2)

Chapter 5: Data Management System and Dialogue Management

Chapter 6: Designing and Developing Decision Support Systems

Chapter 7: Executive Support Systems

Chapter 8: Automated Decision Systems and Expert Systems

Chapter 9: Knowledge Management and Collaborative Systems

Chapter 10: DSS Intelligent Systems & Tools and Techniques

Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

PART 2 OF CHAPTER 4

- Knowledge driven DSS
- Web-based DSS

KNOWLEDGE-DRIVEN DSS (I)

■ What is a Knowledge-Driven DSS?

- is a computer-based reasoning system
- that provides information, comprehension and suggestions to users
- to support them in decision-making.

Expertise: A knowledge-drive DSS comes with a specific problem-solving expertise. This expertise is based upon three components:

- Knowledge in a particular domain and associated symptoms and signs
- Understanding of the relationships between varied symptoms of a problems
- Skills, ways or methods of solving the problem

KNOWLEDGE-DRIVEN DSS (2)

Expert System: A computer system that imitates the decision-making capability of a human expert is called an expert system or an artificial intelligence system. It is designed to solve problems by

- Using if-then rules
- Reasoning about knowledge
- Drawing inferences from facts and rules

Knowledge Discovery and Data Mining: These are interrelated terms used for the process of extracting valuable knowledge and discovering patterns, in order to transform the knowledge into easily comprehensible structure for further use

CHARACTERISTICS OF KNOWLEDGE-DRIVEN DECISION SUPPORT SYSTEMS

- These systems aid managers in solving complex problems.
- These systems allow users to interact with them during the process of decision making.
- The recommendations made by these systems are based on human knowledge.
- These systems use knowledge base that's engineered keeping in mind the nature of problems they will solve.
- These systems aid in performing limited tasks.
- These systems use heuristic technique of problem solving.

DATA MINING AND CREATING KNOWLEDGE

There are two types of data mining models that can be deployed:

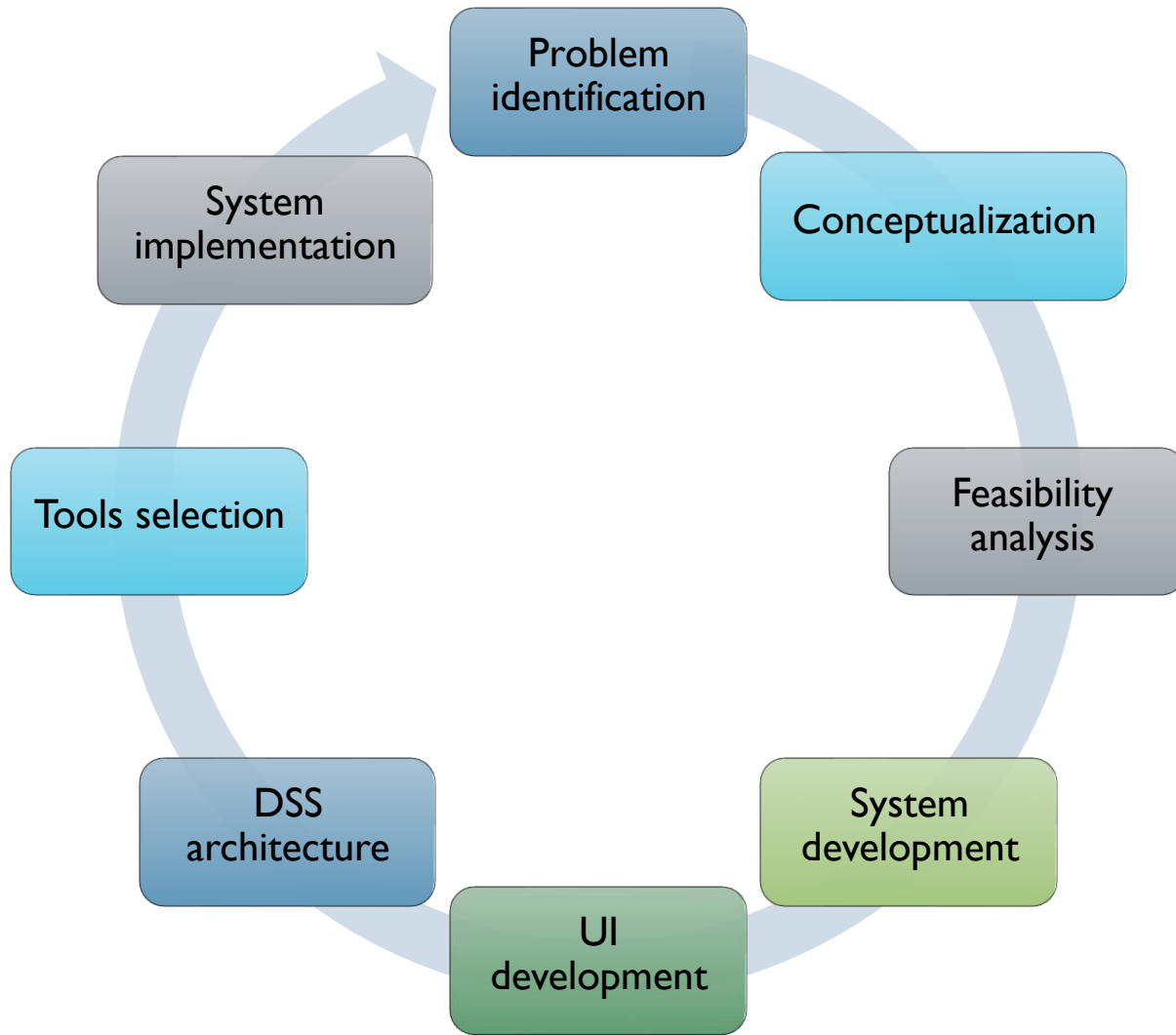
Predictive Model: This data mining model predicts which prospects are likeliest to respond to a particular stimulus. It forecasts clear results based on patterns identified from known results.

Descriptive Model: This model describes patterns in existing data to create significant demographics subgroups, which can then be used for target marketing.

WEB-BASED AND INTER-ORGANIZATIONAL DECISION SUPPORT SYSTEMS

- With the arrival of the **internet and modern communication** technologies, the organizational decision support systems **deploy web-based technologies**.
- The new science of gathering and distributing information adds efficiency to DSS, ultimately helping managers make more appropriate decisions.
- **The best part is that the web-based decision support systems can be knowledge- driven, communications-driven, model-driven, document-driven or data-driven or hybrid**

DESIGNING AND DEVELOPING WEB-BASED DSS



EXAMPLES OF WEB-BASED DSS

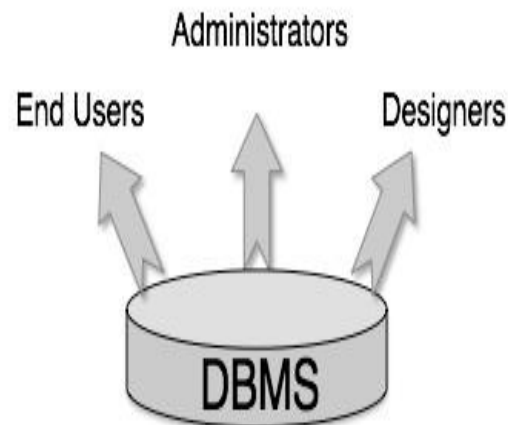
- **Microsoft Carpoint:** a web-based system that allows users to make pair-wise comparisons of car models.
- **TCB Works:** A communication-driven web-based DSS, TCB Works enables people to interact, discuss and make both structured and multi-criteria decisions.
- **Fidelity 'Retirement Planning Calculator':** It is a model-driven DSS to help a person decide how much he or she needs to invest each month for retirement.
- **Netscape Decision Guides:** These are model and knowledge-driven web-based decision support systems, offering 25 different decision guides on diverse topics, including choosing pets, bikes and business schools.
- **Stockfinder:** It's a data-driven DSS to help investors identify stocks based on various criteria, including industry type, price and earnings.

CHAPTER 5 CONTENTS

- Data Management System
- Data dictionary
- Designing a Decision Support System User Interface
- User Interface
- Limitations of dialogue DSS

DATA MANAGEMENT SYSTEM (I)

- A **database** is a collection of related data and data is a collection of facts and figures that can be processed to produce information.
- A **database management system** stores data in such a way that it becomes easier to retrieve, manipulate, and produce information



DATA MANAGEMENT SYSTEM (2)

- Real-world entity
- Relation-based tables
- Isolation of data and application
- Less redundancy
- Consistency
- Query Language
- ACID Properties
- Multiuser and Concurrent Access
- Multiple views
- Security

DATA DICTIONARY

- A **data dictionary** (also called the metadata) is the data about the data.
- It is the self-describing nature of the database that provides program-data independence. It is also called the System **Catalog**.
- It holds the following information about each data element in the databases, it normally includes:
 - + Name
 - + Type
 - + Range of values
 - + Source
 - + Access authorization
 - + Indicates which application programs use the data so that, when a change in a data structure is contemplated, a list of the affected programs can be generated.

ACTIVE AND PASSIVE DATA DICTIONARIES

- **An active data dictionary** (also called integrated data dictionary) is managed automatically by the **database management software**. **Consistent** with the current structure and definition of the database.
- **The passive data dictionary** (also called non-integrated data dictionary) is the one used only for **documentation purposes**. It is managed by the **users of the system** and is modified whenever the structure of the database is changed. Since this modification must be performed manually by the user, it is possible that the data dictionary will **not be current with the current structure of the database**

IMPORTANCE OF DATA DICTIONARY

- Data dictionary provides the name of a data element, its description, and data structure in which it may be found.
- Data dictionary provides great assistance in producing a report of where a data element is used in all programs that mention it.
- It is also possible to search for a data name, given keywords that describe the name.

DESIGNING A DECISION SUPPORT SYSTEM USER INTERFACE

- The effectiveness of a computerized system or a software platform depends on its user **interface design**.
- Be it a routine software program **or** a high-end decision support system, user's interaction with the system should be as **simple, effective** and **hassle-free** as possible.
- Designing a user interface involves attaining complete understanding of user needs and the way they make decisions

USER INTERFACE

- **Interface** means that part of a computerized system that allows its users to interact with it. It is what users see and use.
- User interface is important, yet it facilitates interaction between the user and the software system without drawing much attention to itself. **User interface may include:**
 - ✓ Commands
 - ✓ Set of menus
 - ✓ Graphics
 - ✓ Icons
 - ✓ Guidelines
 - ✓ Acoustics
 - ✓ Tactile
 - ✓ Hardware
 - ✓ Any other presentation

FEATURES OF GOOD USER INTERFACE

- ✓ Influences how users interact with the decision support system
- ✓ Balances the technicality (functionality of the system) and mentality (user's state of mind)
- ✓ Provides users a picture-oriented (visual) way to interact with the system.
- ✓ Creates flawless communication and interaction between the user and the decision support system.
- ✓ Reduces errors, increases speed, supports „good decision making“
- ✓ Logical and intuitive at the same time

USER INTERFACE STYLES

- Command-Line Interface
- Menu Interface
- Graphical Interface (multimedia rather than text)
- Question-and-Answer Interface
- Voice User Interface
- Touch User Interface

DESIGNING USER INTERFACE

- ROMC Design Approach (Representation - Operation - Memory Aid - Control Aid)
- **ROMC** is an organized approach to design specialized decision support systems, more precisely their user interfaces.
- It's a systematic move towards designing the representation, operation, memory and control mechanisms of a large decision support system

REPRESENTATION

- **Representation** is about presenting information or results for that matter, in a structured way.
- All decision-making activities in an organization take place in a certain environment or context.
- **The representation** provides a base to the decision makers backed by concrete information, helping them **interpret DSS outputs**.
- The representation can be in the form of a **table, graph, map, chart** or a **text document** and each value on a map or a table communicates decision making context.

OPERATION

- This stage in user interface design focuses on **specific tasks** performed by/with a decision support system.
- The operation can involve one activity or many, depending upon the specific needs of the decision maker.
- A DSS may be used to **process data, track market trends, carry analytics or suggest alternatives** or perform all the functions.

MEMORY AID

- How will a DSS operate?
- On what basis will it produce outputs?
- What is it going to represent?
- It must have access to data to synthesize and analyze.
- **A data warehouse is its memory aid and so for decision makers.** So, it must give users a link to data warehouse aids their memory.
- In addition, it may provide them with links and command shortcuts or sequences to help users control a decision support system.

CONTROL AID

- Control aid is provided to users, so that they can effectively use representations, operations and memory aids.
- It allows them to edit, redo, undo, save, view, delete any information or representation or the task performed by/with a DSS.
- They make it easier for the user to interact with and control a decision support system.
- The commands, however, depends upon the type of user interface.

FACTORS INFLUENCING USER INTERFACE DESIGN SUCCESS

- Execution Time
- Versatility
- Adaptability
- Learning Time
- Uniformity of Command
- Quality of Help
- Memory Load
- Ease of Recall
- Fatigue
- Errors



Thanks
Any questions



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DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

Ibrahim El-Hasnony

QUIZ-I

- 1)is an umbrella term that combines architectures, tools, databases, analytical tools, applications, and methodologies (DSS, BI, Data mining, MSS).
- 2) The process of is based on the transformation of data to information, then to decisions, and finally to actions. (DSS, BI, Data mining, MSS).
- 3) Managers and specialists use the concept of a system frequently and yet it is hard for most of us to define and understand the concept.(EIS, TPS, MIS, DSS).
- 4) Systems also have mechanisms to provide a means of controlling the operation of the system.(input, output, feedback, results).
- 5) Feedback comes from the(DSS, Decision maker, Customer, System).
- 6) In a Decision Support System, the primary focus is often on the computerized components of the system (T/F)

QUIZ-2

- 1) Information systems capture data from the organization (internal data) and its environment (external data). (T/F).
- 2) Depending on the type of, the information output may take the form a query response, decision outcome, expert-system advice, transaction document, or a report. (DSS, MIS, IS, ES).
- 3) Most of the data captured by information systems relates to the operations of the organization itself. (T/F).
- 4) The objectives of a system are realized in its(input, output, feedback, testing).
- 5) The fewer resources a system consumes in producing given outputs, the more effective it is.(T/F).
- 6) The objective of management reporting systems is to provide routine information to managers (T/F).

QUIZ-3

- 1)support top managers with conveniently displayed summarized information, customized for them. (MIS,DSS,TPS,EIS)
- 2)measuring performance against the planned objectives and initiating corrective action, if needed.(planning, leadership, controlling, organizing)
- 3)clarifies the computer's role in aiding rather than replacing the decisionmaker. (Decision, support, systems, DSS).
- 4) The decision support system is under user control, from early beginning to final putting into use and daily use (T/F).
- 5) DSS refers to a class of systems, which supports the process of decision-making and does not always give a decision itself (T/F).
- 6)consists of various mathematical and analytical models that are used to analyze complex data, thereby producing the required information. (DSS DB, DSS SW, DSS KMS, DSS Dialogue)
- 7) To increase the production level by 40 percent using the backward sensitivity analysis, first, the target value for the production level can be set. (T/F)

QUIZ-4

- 1) Its purpose is to help conduct a meeting, or for users to collaborate (office, data , communication, document) driven DSS.
- 2) An Inter-Organizational DSS provides stakeholders with access to a company's intranet and authority or privileges to use specific DSS capabilities. (T/F).
- 3) ERP is NOT a Decision Support System (T/F).
- 4) Decisions are those in which the decision maker must provide judgment, evaluation, and insights into the problem definition (Structured, semi-structured, unstructured).
- 5)involves identifying and exploring various solutions to the problem.(intelligence, design, choice, implement).
- 6) The set of possible alternatives is the output of thephase. (intelligence, design, choice, implement).

QUIZ-5

- 1) Every time you park downtown, you get a parking ticket because you exceed the time limit on the parking meter although once it did not happen, this is an example for decision making under.....(**certainty, risk, uncertainty**).
- 2) In the payoff table, the rows correspond to the possible decision alternatives.(**T/F**).
- 3)is the increase in the expected profit that would result if one knew with certainty which state of nature would occur.(**EV, EVPI, Regret, Optimistic**).
- 4) In pro forma financial model Costs are estimated based on past data, gross sales are predicted, and profit or loss is then calculated on these relationships (**T/F**).
- 5)models alters the historical data mathematically to better reflect the assumptions of a decision maker.(**exponential smoothing, time series, regression, moving average**)
- 6) DSS with simulation models conduct experiments to identify conditions or situations that approximate the actual conditions (**T/F**).

QUIZ-6

- 1)is an approach to discovery and problem solving by employing practical methods (**Heuristics, inference engine, Knowledge acquisition, case reasoning**).
- 2)used to determine the distance between or relationship among various components. (**Heuristics, inference engine, Knowledge acquisition, case reasoning**).
- 3) “What techniques will it use?” used infor designing web-based DSS.(**conceptualization, problem identification, feasibility analysis, SW development**).
- 4) The passive data dictionary (also called integrated data dictionary) is the one used only for documentation purposes. (**T/F**).
- 5)is an organized approach to design specialized decision support systems, more precisely their user interfaces.(**ROMC, ACID, EVPI, RANI**).
- 6)means that a decision support system must be resourceful enough to perform the entire range of tasks that a decision maker needs to perform when making a decision using DSS. (**Adaptability, Versatility, Quality, Fatigue**).

COURSE DETAILS

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Chapter 4: Model Management (MDSS)

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CHAPTER 6 CONTENTS

- Designing Issues
- Decision oriented analysis
- DSS audit plan

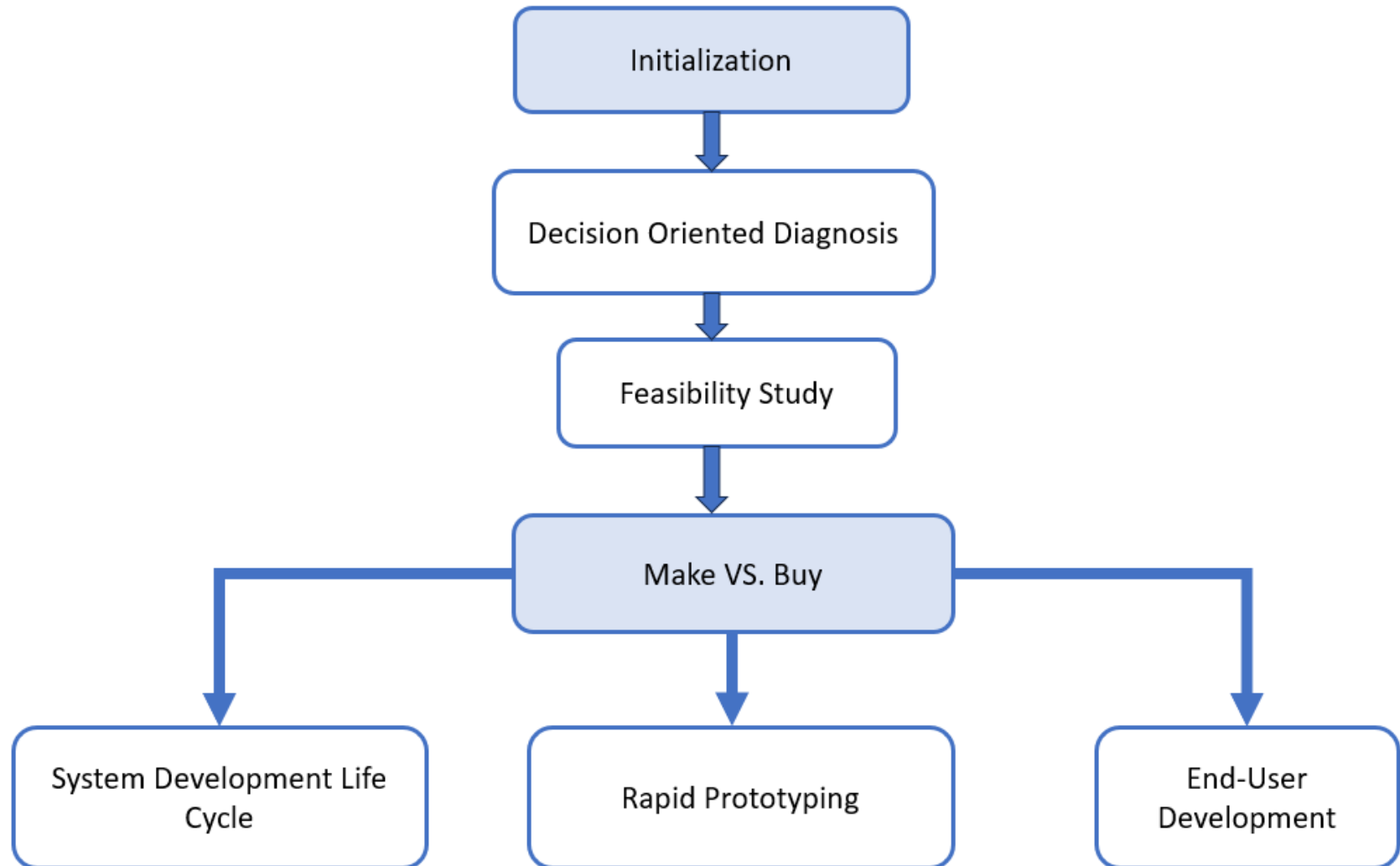
DESIGNING ISSUES

- **Designing and development of a decision support system is a complex process, you may face several issues:**
 - 1) Mismatch between the perspectives of the programmer and decision maker (communication, priorities, risk).
 - 2) Identification of specific requirements
 - 3) Technology selection
 - 4) Approach to software design and development

ISSUES POSSIBLE SOLUTIONS

- **Most of these issues can be resolved by:**
 - 1) Identifying clear agendas
 - 2) Brainstorming with team members
 - 3) Communicating the expectations clearly to the programmers
 - 4) Educating yourself about various technologies used in DSS development.
 - 5) Trying to reach a common platform with the developers.

DSS DESIGN PROCESS



INITIALIZATION

- The initialization phase involves **defining the purpose and objectives** of the DSS.
- This is where you identify the specific decision-making **problems** or **opportunities** that the DSS will address.

DECISION-ORIENTED DIAGNOSIS (I)

- During the diagnosis phase, you conduct a **thorough analysis** of the **current** decision-making processes within the organization.
- This includes understanding **how decisions are made**, the data and information **required**, and the **pain points** or **inefficiencies** in the existing system.
- This step often involves conducting **interviews**, **surveys**, and **data analysis** to pinpoint areas where the DSS can make a significant impact.

DECISION-ORIENTED DIAGNOSIS (2)

- **Diagnosis** is the identification of problems or opportunities for improvement in current decision behavior.
- Diagnosis involves determining **how** decisions are currently made, specifying **how** decisions should be made, and understanding **why** decisions are not made as they should be.
- These statements of improvements provide the objectives for the DSS development.

DECISION-ORIENTED DIAGNOSIS (3)

- **Diagnosis of a decision process involves completing the following three activities:**

- 1) Collecting data on current decision-making using techniques such as interviews, observations, questionnaires and historical records;
- 2) Establishing a coherent description of the current decision process.
- 3) Specifying a norm for how decisions should be made.

DECISION-ORIENTED DIAGNOSIS (4)

- A related diagnostic activity is conducting a **DSS Audit**.
- In general, it can be very useful to **audit operational and managerial decision processes**.

- ✓ In the context of a "DSS audit plan," the term "audit" refers to a systematic and comprehensive examination or review of the Decision Support System (DSS) and its various components to assess its performance, effectiveness, security, compliance with regulations, and overall health.
- ✓ The primary purpose of an audit is to identify strengths and weaknesses in the DSS and to provide recommendations for improvements.

DSS AUDIT PLAN (I)

- **A Decision Support System (DSS) audit plan** is a structured approach for evaluating and assessing the effectiveness, efficiency, and security of a Decision Support System within an organization.
- DSS audit plans are essential for ensuring that the DSS is functioning as intended, providing accurate and reliable information, and complying with relevant regulations and standards.

DSS AUDIT PLAN (2)

	DSS Audit Plan
Step1	Define the decisions, decision processes and related business processes that will be audited. Define the authority of the auditor, purpose of the audit, scope of the audit, timing of the audit, and resources required to perform the audit. Identify a primary contact.
Step 2	Examine the formal design of the process. Diagram the process and specify criteria, etc. Is the design effective and efficient?
Step 3	Examine the actual use of the decision process. Observe the process. Interview decision makers and collect data. Is the process implemented and used as intended?
Step 4	Assess performance of the actual decision process. What works? Can cycle time be reduced? Are decisions appropriate? Timely? Cost effective? Is the process producing value in meeting business objectives? If not, why?
Step 5	Reporting and recommendations. Summarize steps 1-4 in a written report. Discuss what is working well and what needs to be improved. Develop recommendations for improving the process. Hold an exit meeting with decision makers.

PREPARE A FEASIBILITY STUDY (I)

- This study should occur prior to committing resources to developing a proposed DSS.
- Involves assessing the technical, economic, operational, and scheduling feasibility of the project.
- You need to determine if the organization has the resources, budget, and technical capabilities to develop and maintain the DSS. A cost-benefit analysis is often conducted to evaluate the potential return on investment.

PREPARE A FEASIBILITY STUDY (2)

- What should be included in a DSS Feasibility Study? like DSS Scope and Target Users, Anticipated DSS Impacts, Benefits, Risks and Mitigating Factors.
- At this point a decision is made between purchasing an application package and in- house development.

MAKE OR BUY

- In this step, you decide whether to build the DSS in-house or purchase an existing DSS solution from a vendor.
- This decision **depends** on factors like the organization's **expertise**, **budget**, and **time** constraints.
- Buying a DSS may save time and resources, while building one in-house allows for more customization and control.
- Packaged DSS applications are often quite versatile and are usually less expensive to implement than in-house development.
- Packaged solutions are also often faster to implement.

CHOOSE A DEVELOPMENT APPROACH

- **There are three main approaches to system design and development:**
 - 1) The traditional system development life cycle (SDLC) approach;
 - 2) Rapid prototyping and
 - 3) End-user approach.
- You may want to choose a methodology that ensures that the final product accomplishes your objectives

SYSTEMS DEVELOPMENT LIFE CYCLE APPROACH

■ The systems development life cycle (**SDLC**) approach is based on a series of formal steps, including the following seven steps:

- 1) Confirm user requirements.
- 2) Systems analysis.
- 3) System design.
- 4) Programming.
- 5) Testing.
- 6) Implementation.
- 7) Use and Evaluation.

RAPID PROTOTYPING (I)

- Creating a rapid prototype for a Decision Support System (DSS) design involves **quickly building a simplified version** of the system to demonstrate its key features and functionality.
- Creating a rapid prototype for a DSS allows you to visualize and test the system's design early in the development process, helping to ensure that the final DSS will meet user needs and expectations.
- It can save time and resources by identifying issues and improvements before extensive development efforts are undertaken.

RAPID PROTOTYPING (2)

- **A typical prototyping methodology usually includes five steps:**
 1. Identify user requirements.
 2. Develop a first iteration DSS prototype.
 3. Evolve and modify the next iteration DSS prototype.
 4. Test DSS and return to step 3 if needed.
 5. Full-scale implementation.
- Rapid Application Development (RAD) specifies incremental development with constant feedback from potential users.

END-USER DSS DEVELOPMENT (I)

- The end-user approach emphasizes **active involvement of end-users** in the system development process.
- It's often used in **situations where the end-users have a deep understanding of their needs and can contribute directly to system design.**
- This approach can help ensure that the final system **closely aligns with user expectations.**
- End-users may be involved in requirements gathering, testing, and even some aspects of system design.

END-USER DSS DEVELOPMENT (2)

- The major advantage of encouraging end-user DSS development is that the person who wants computer support will be involved in creating it.
- The manager/builder controls the situation and the solution that is developed.
- End-user DSS development can also sometimes result in faster development and cost savings.

END-USER DSS DEVELOPMENT (4)

- End-user DSS development is a **controversial topic**. Information systems staffs have many concerns including:
 - 1) End-users may select an inappropriate software product as a development environment.
 - 2) The end-user may have limited expertise in the use of the product and the IT group may have limited resources to support end-user development.
 - 3) Errors during end-user DSS development are frequent.
 - 4) Unnecessary databases are sometimes developed by end-users for their DSS.
 - 5) A major quality issue involves testing and limited documentation. End-users often perform only limited testing of DSS they develop; and they have limited experience- documenting applications.
 - 6) End-user databases may be poorly constructed and difficult to maintain.
 - 7) End-users rarely follow a systematic development process.

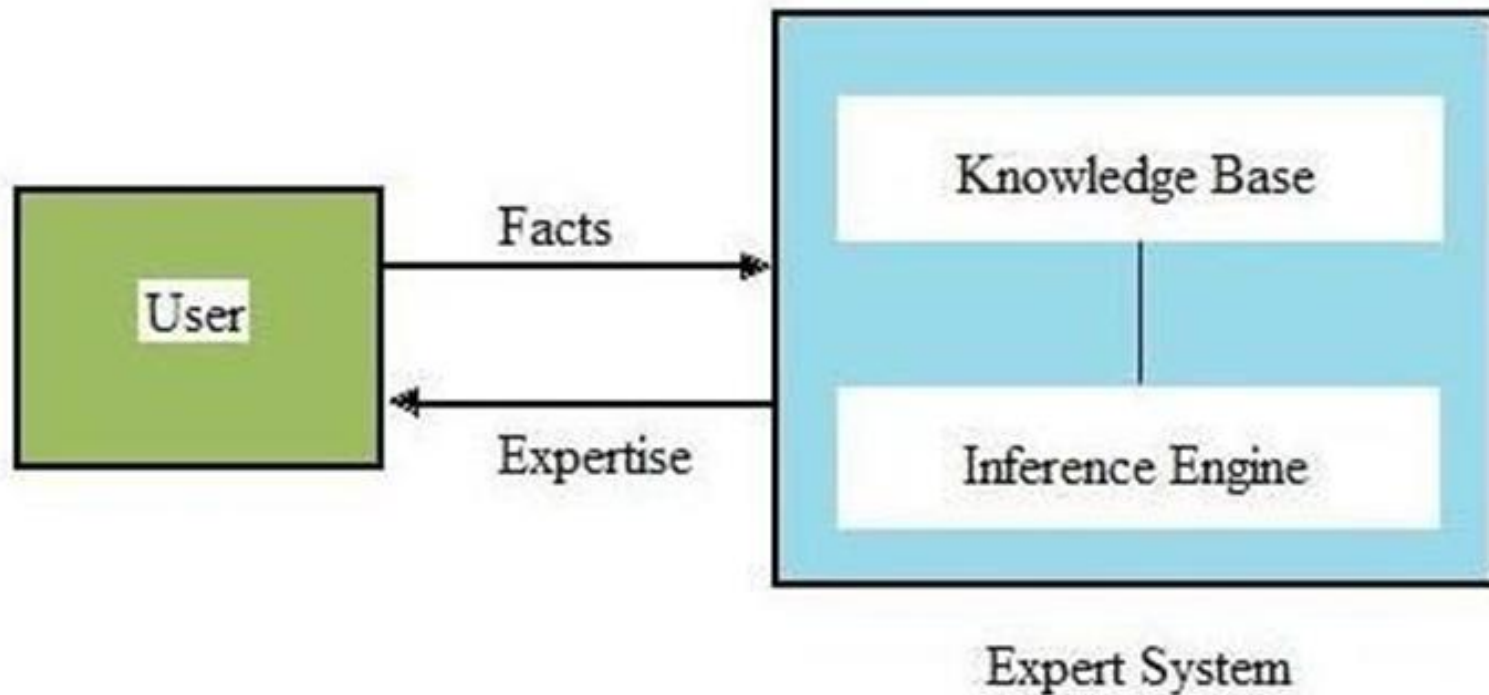
CHAPTER 7 CONTENTS

- Components of executive support systems
- Role of ESS in firms
- Business value of ESS
- ESS for competitive intelligence

ESS

- The ESS help managers with unstructured and semi structured problems by focusing on the information needs of **senior management**.
- Combining data from internal and external sources, **ESS create a generalized computing and communications environment that can be focused and applied to a changing array of problems.**
- ESS help senior executives monitor organizational performance, track activities of competitors, spot problems, identify opportunities, and forecast trends.
- Designed for **top-level executives and senior managers.**
- Its primary purpose is to provide **high-level, summarized information to aid in strategic decision-making.**

ESS COMPONENTS (I)



ESS COMPONENTS (2)

- **The knowledge base:** This is where the information is stored in the expert system in form of facts and rules. Basically, a series of IF statements. Here the programmer writes the code for the expert system.
- **The user interface:** This is where the user interacts with the expert system. This is where questions are asked and advised is produced. The user interface can also output justification features of the expert system.
- **The inference engine:** This applies the facts to the rules and determines the questions to be asked of the user in the user interface and which order to ask them. This is an invisible part of an ESS when the program is run.

THE ROLE OF EXECUTIVE SUPPORT SYSTEMS IN THE FIRM (I)

- Bring together data from all parts of the firm and enable managers to select, access, and tailor them as needed using easy-to-use desktop analytical tools and online data displays
- ESS can drill down, moving from a piece of summary data to lower and lower levels of detail.
- The ability to drill down is useful not only to senior executives but also to employees at lower levels of the firm who need to analyze data
- Many managers have access to news services, financial market databases, economic information, and whatever other public data they may require.

THE ROLE OF EXECUTIVE SUPPORT SYSTEMS IN THE FIRM (2)

- Include tools for modeling and analysis.
- With only a minimum of experience, most managers find they can use these tools to create graphic comparisons of data by time, region, product, price range, and so on

BUSINESS VALUE OF EXECUTIVE SUPPORT SYSTEMS

- Much of the value of ESS is found in their **flexibility**
- Ability to **analyze**, **compare**, and **highlight** trends
- Executives are using ESS to **monitor performance** more successfully in their own areas of responsibility
- Improve **management performance** and increase upper management's span of control

ESS FOR COMPETITIVE INTELLIGENCE

- ESS can help managers to :
 - Identify changing market conditions,
 - Formulate responses,
 - Track implementation efforts,
 - Learn from feedback.

IDENTIFY CHANGING MARKET CONDITIONS,

- ESS can provide managers with [access to a wealth of data and information](#).
- By analyzing this data, ESS can help managers identify changing market conditions, trends, and emerging opportunities or threats.
- This information can be critical for making informed decisions and adjusting strategies in response to market dynamics.

FORMULATE RESPONSES

- Once managers are aware of changing market conditions, ESS can assist in formulating appropriate responses.
- ESS can provide tools for scenario analysis, modeling, and decision support, helping managers evaluate different courses of action and select the most effective strategies to address the identified challenges or opportunities.

TRACK IMPLEMENTATION EFFORTS

- ESS often includes project management and tracking features.
- Managers can use ESS to monitor the progress of various initiatives, ensuring that strategies and plans are implemented as intended.
- This tracking capability is essential for staying on course and ensuring that organizational resources are allocated effectively.

LEARN FROM FEEDBACK

- ESS can capture and analyze feedback and performance data, enabling managers to learn from the outcomes of their decisions and actions.
- This learning process is vital for continuous improvement and adaptability.
- By using ESS, managers can gain insights into what works and what doesn't, allowing them to refine their strategies over time.



Thanks
Any questions



Mansoura University
Faculty of Computers and Information
Fall 2023-2024



DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

Ibrahim El-Hasnony

COURSE DETAILS

Course Syllabus

Chapter 1: Introduction to DSS

Chapter 2: Decision Making Concepts

Chapter 3: Decision Making Technologies and Environment

Chapter 4: Model Management

Chapter 5: Data Management System and Dialogue Management

Chapter 6: Designing and Developing Decision Support Systems

Chapter 7: Executive Support Systems

Chapter 8: Automated Decision Systems and Expert Systems

Chapter 9: Knowledge Management and Collaborative Systems

Chapter 10: DSS Intelligent Systems & Tools and Techniques

Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

CHAPTER 8 CONTENTS

- Automated DSS
- AI field and DSS
- Expert systems concepts
- Expert systems structure
- Knowledge engineering
- Forward and backward chaining

INTRODUCTION

- In many cases, results of **predictive models or even optimization** models get **simplified as rules** that are then implemented in other applications.
- We call these **automated decision systems**.
- In the other hand, It is necessary to find effective ways to incorporate qualitative information and knowledge resides in the **minds of human experts** into decision support systems (DSS).
- A system that integrates knowledge from experts is commonly called a knowledge-based decision support system **(KBDSS) or an intelligent decision support system (IDSS)**

AUTOMATED DECISION SYSTEMS (I)

An ADS is a rule-based system that provides a solution, usually in one functional area (e.g., finance, manufacturing), to a specific repetitive managerial problem, usually in one industry (e.g., to approve or not to approve a request for a loan, to determine the price of an item in a store).

Examples:

“If only 70 percent of the seats on a flight from Los Angeles to New York are sold 3 days prior to departure, offer a discount of x to nonbusiness travelers,”

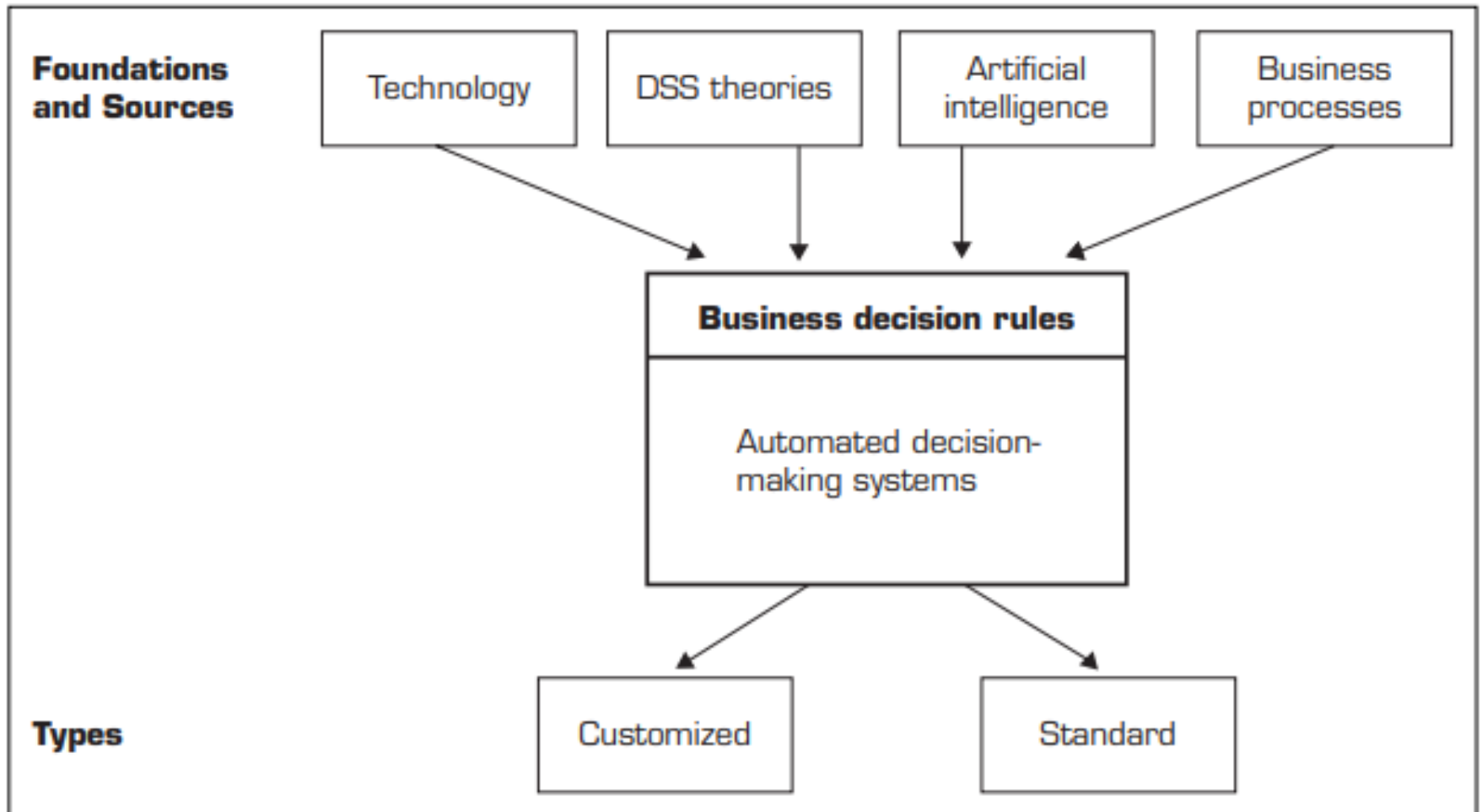
“If an applicant owns a house and makes over \$100,000 a year, offer a \$10,000 credit line,”

“If an item costs more than \$2,000, and if your company buys it only once a year, the purchasing agent does not need special approval.”

AUTOMATED DECISION SYSTEMS (2)

- Such rules in previous slide, which are based on experience or derived through data mining, can be combined with mathematical models to form solutions that can be automatically and instantly applied to problems, or they can be provided to a human, who will make the final decision.
- ADS attempt to automate highly repetitive decisions (in order to justify the computerization cost), based on business rules.
- ADS are mostly suitable for frontline employees who can see the customer information online and frequently must make quick decisions

AUTOMATED DECISION SYSTEMS GENERAL ARCHITECTURE (I)



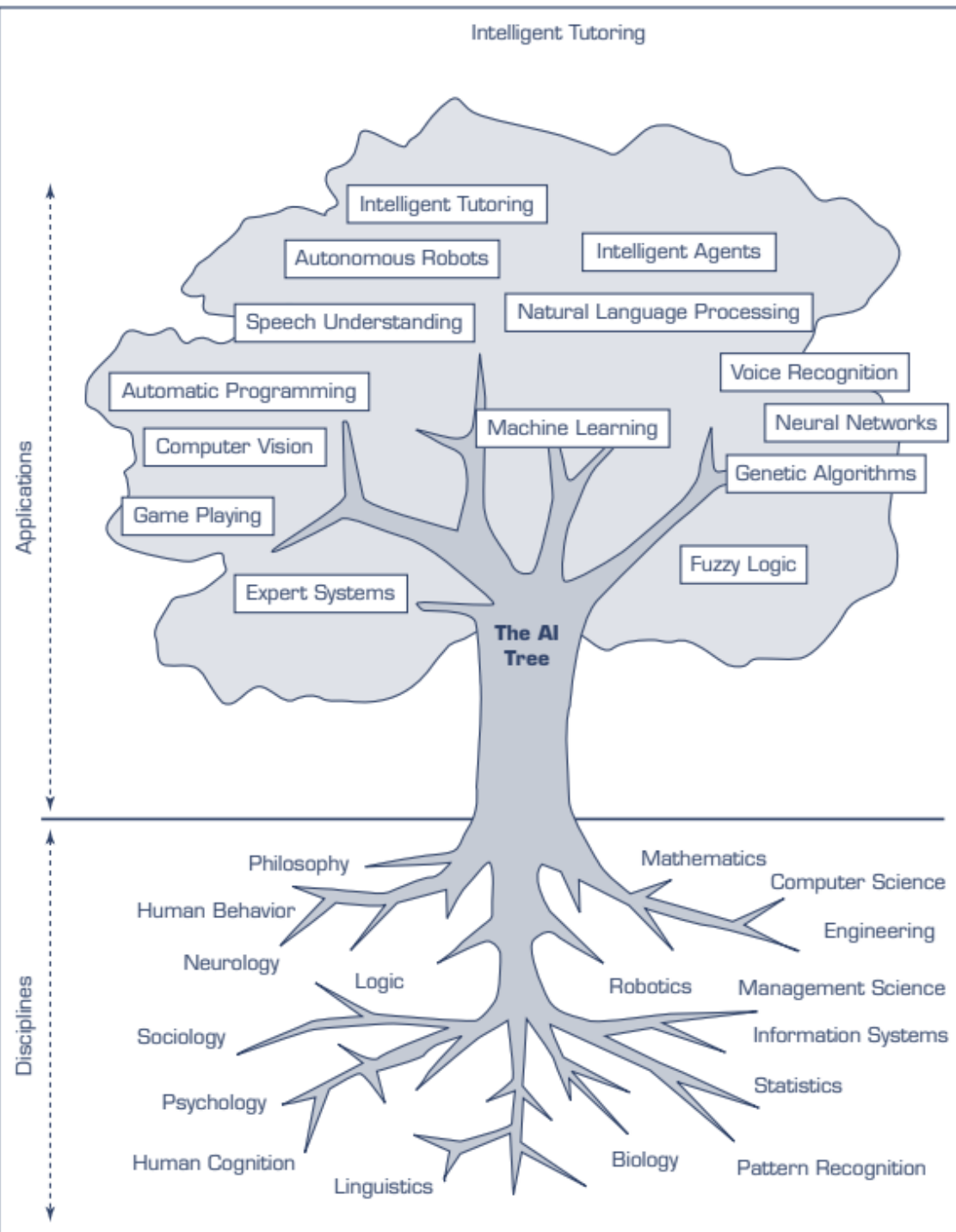
AUTOMATED DECISION SYSTEMS GENERAL ARCHITECTURE (2)

- The building blocks of such intelligent systems are **business rules**.
- A business rule can be as simple as saying—“Offer a discount if average sales drop by 10 percent.”
- To put this in perspective, almost all airlines have automated decision systems to assign dynamic prices based on demand

THE ARTIFICIAL INTELLIGENCE FIELD

- **Artificial intelligence (AI)** is a collection of concepts and ideas that are related to the development of intelligent systems.
- These concepts and ideas may be developed in different areas and be applied to different domains.
- To understand the scope of AI, therefore, we need to see a group of areas that may be called the **AI family**.

AI FAMILY



AI DEFINITION

AI is concerned with two basic ideas:

- 1) The study of human thought processes (to understand what intelligence is)
- 2) The representation and duplication of those thought processes in machines (e.g., computers, robots).

“Artificial intelligence is the study of how to make computers do things at which, at the moment, people are better.”

SIGNS OF INTELLIGENCE ABILITIES

- Learning or understanding from experience
- Making sense out of ambiguous or contradictory messages
- Responding quickly and successfully to a new situation (i.e., different responses, flexibility)
- Using reasoning in solving problems and directing conduct effectively
- Dealing with perplexing situations
- Understanding and inferring in a rational way
- Applying knowledge to manipulate the environment
- Thinking and reasoning
- Recognizing and judging the relative importance of different elements in a situation

BASIC CONCEPTS OF EXPERT SYSTEMS

Expert systems (ES) are computer-based information systems that use expert knowledge to attain high-level decision performance in a narrowly defined problem domain

MYCIN, developed at Stanford University in the early 1980s for medical diagnosis, is the most well-known ES application

The basic concepts of ES include how to determine who **experts** are, the definition of **expertise**, how expertise can be **extracted** and **transferred** from a person to a computer, and how the expert system should **mimic** the reasoning process of human experts

EXPERTS (I)

- **An expert** is a person who has the **special knowledge, judgment, experience,** and **skills** to put his or her knowledge in action to provide sound advice and to solve complex problems in a narrowly defined area
- **Provide knowledge** about how he or she performs a task that a KBS will perform.
- An expert **knows which facts are important** and also understands and explains the dependency relationships among those facts

EXPERTS (2)

- Decision performance and the level of knowledge a person has are typical criteria used to determine whether a particular person is an expert.
- Typically, experts must be able to solve a problem and achieve a performance level that is significantly better than average.
- Experts are relative (and not absolute).
- An expert at a time or in a region may not be an expert in another time or region

EXPERTS (3)

Human experts can do the following:

- Recognizing and formulating a problem
- Solving a problem quickly and correctly
- Explaining a solution
- Learning from experience
- Restructuring knowledge
- Breaking rules (i.e., going outside the general norms), if necessary
- Determining relevance and associations

EXPERTISE (I)

- **Expertise** is the extensive, task-specific knowledge that experts possess.
- The level of expertise determines the performance of a decision.
- Expertise is often acquired through training, reading, and experience in practice.
- It includes **explicit** knowledge, such as theories learned from a textbook or in a classroom, and **implicit** knowledge, gained from experience

EXPERTISE (2)

- **The following is a list of possible knowledge types:**
- Theories about the problem domain
- Rules and procedures regarding the general problem domain
- Heuristics about what to do in a given problem situation
- Global strategies for solving these types of problems
- Metaknowledge (i.e., knowledge about knowledge)
- Facts about the problem area

These types of knowledge enable experts to make better and faster decisions than nonexperts when solving complex problems.

EXPERTISE (3)

- **Expertise often includes the following characteristics:**
- Expertise is usually associated with a high degree of intelligence, but it is not always associated with the smartest person.
- Expertise is usually associated with a vast quantity of knowledge.
- Expertise is based on learning from past successes and mistakes.
- Expertise is based on knowledge that is well stored, organized, and quickly retrievable from an expert who has excellent recall of patterns from previous experiences.

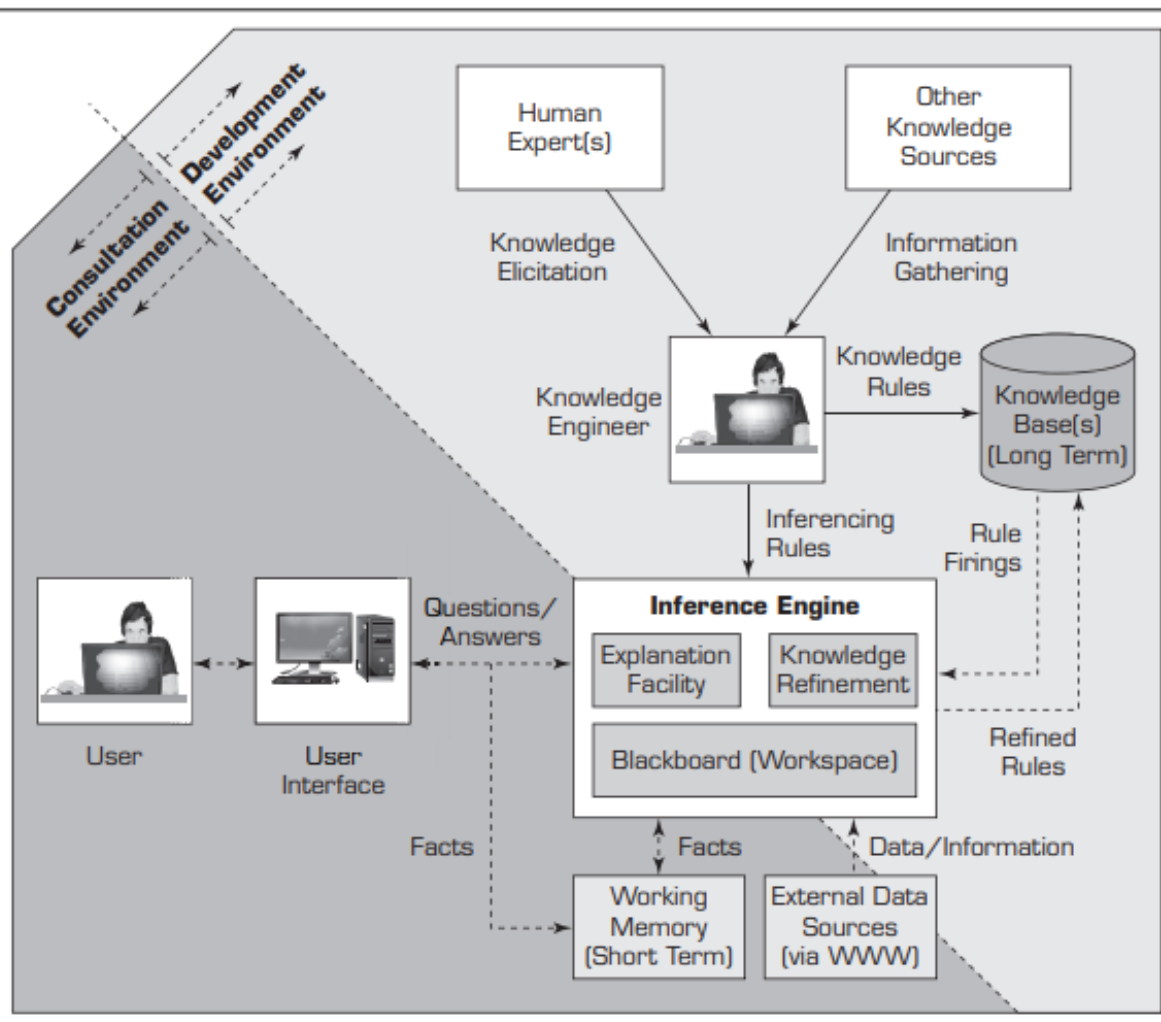
FEATURES OF ES

- **Expertise.** An ES must possess expertise that enables it to make expert-level decisions..
- **Symbolic reasoning.** The basic rationale of artificial intelligence is to use symbolic reasoning rather than mathematical calculation.
- **Deep knowledge.** The knowledge base must contain complex knowledge not easily found among nonexperts.
- **Self-knowledge.** Most experts have very strong learning capabilities to update their knowledge constantly.

APPLICATIONS OF EXPERT SYSTEMS

Expert System	Organization	Application Domain
<i>Classical Applications</i>		
MYCIN	Stanford University	Medical diagnosis
XCON	DEC	System configuration
Expert Tax	Coopers & Lybrand	Tax planning
Loan Probe	Peat Marwick	Loan evaluation
La-Courtier	Cognitive Systems	Financial planning
LMOS	Pacific Bell	Network management
PROSPECTOR	Stanford Research Institute	Discovery of new mineral deposits
<i>Reported Applications</i>		
Fish-Expert	North China	Disease diagnosis in fish
HelpDeskIQ	BMC Remedy	Help desk management
Authorete	Haley	Business rule automation
eCare	CIGNA	Insurance claims
SONAR	NSAD	Stock market monitoring

STRUCTURE OF EXPERT SYSTEMS (I)



- **ES can be viewed as having two environments:**
- An ES builder uses the **development environment** to build the necessary components of the ES and to populate the knowledge base with appropriate representation of the expert knowledge.
- A nonexpert uses the **consultation environment** to obtain advice and to solve problems using the expert knowledge embedded into the system.

STRUCTURE OF EXPERT SYSTEMS (2)

- The three **major components** that appear in virtually every ES are the **knowledge base**, the **inference engine**, and the **user interface**. In general, though, an ES that interacts with the user can contain the following **additional components**:
 - Knowledge acquisition subsystem
 - Blackboard (workplace)
 - Explanation subsystem (justifier)
 - Knowledge-refining system
- Currently, most ES do not contain the knowledge refinement component. A brief description of each of these components in next slides.

KNOWLEDGE ACQUISITION SUBSYSTEM

- **Knowledge acquisition** is the accumulation, transfer, and transformation of problem-solving expertise from experts or documented knowledge sources to a computer program for constructing or expanding the knowledge base.
- Potential **sources of knowledge** include human experts, textbooks, multimedia documents, databases (public and private), special research reports, and information available on the Web.

KNOWLEDGE BASE

- **The knowledge base is the foundation of an ES.**
- It contains the relevant knowledge necessary for understanding, formulating, and solving problems. **A typical knowledge base may include two basic elements:**
- (1) facts that describe the characteristics of a specific problem situation (or fact base) and the theory of the problem area
- (2) special heuristics or rules (or knowledge nuggets) that represent the deep expert knowledge to solve specific problems in a particular domain.
- The inference engine can include general-purpose problem-solving and decision-making rules (or meta-rules—rules about how to process production rules).

INFERENCE ENGINE

- The “brain” of an ES is the inference engine, also known as the control structure or the rule interpreter (in rule-based ES).
- This component is essentially a computer program that provides a methodology for reasoning about information in the knowledge base and on the blackboard to formulate appropriate conclusions.
- The inference engine provides directions about how to use the system’s knowledge by developing the agenda that organizes and controls the steps taken to solve problems whenever a consultation takes place.

BLACKBOARD (WORKPLACE)

- **The blackboard** is an area of working memory set aside as a database for description of the current problem, as characterized by the input data.
- It is also used for recording intermediate results, hypotheses, and decisions.
- Three types of decisions can be recorded on the blackboard: a **plan** (i.e., how to attack the problem), an **agenda** (i.e., potential actions awaiting execution), and a **solution** (i.e., candidate hypotheses and alternative courses of action that the system has generated thus far).

EXPLANATION SUBSYSTEM (JUSTIFIER)

- The ability to trace responsibility for conclusions to their sources is crucial both in the transfer of expertise and in problem solving.
- The explanation subsystem can trace such responsibility and explain the ES behavior by interactively answering questions such as these:
 - Why was a certain question asked by the ES?
 - How was a certain conclusion reached?
 - Why was a certain alternative rejected?
 - What is the complete plan of decisions to be made in reaching the conclusion? For example, what remains to be known before a final diagnosis can be determined?

KNOWLEDGE-REFINING SYSTEM

- **Evaluation** is necessary in expert systems so that a program can analyze the reasons for its **success or failure**, which could lead to **improvements** resulting in a more accurate knowledge base and more effective reasoning.
- Such an intelligent component is not yet mature enough to appear in many commercial ES tools.

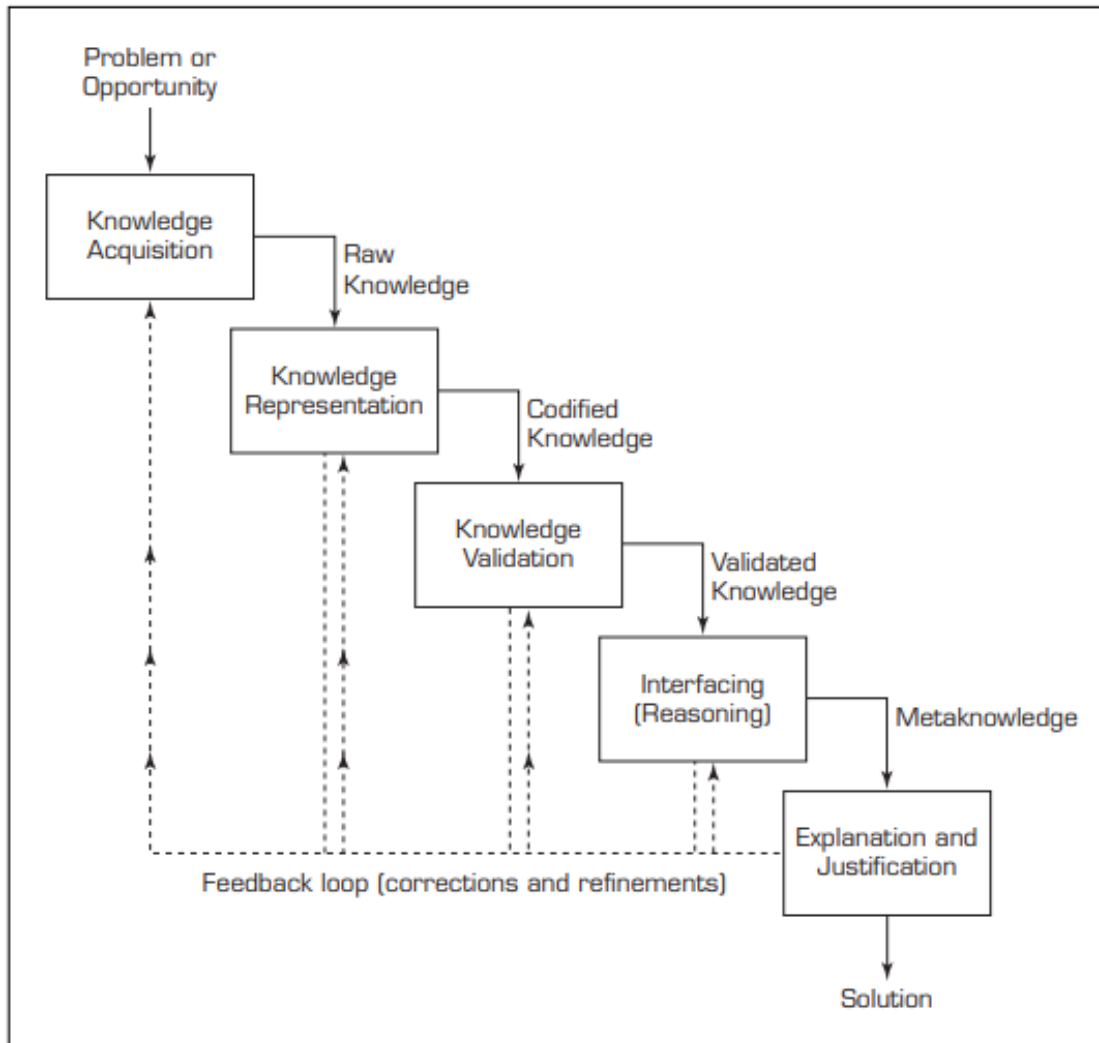
KNOWLEDGE ENGINEERING (I)

- The collection of intensive activities encompassing the acquisition of knowledge from human experts (and other information sources) and conversion of this knowledge into a repository (commonly called a knowledge base) are called **knowledge engineering**.
- Help experts articulate how they do what they do and to document this knowledge in a reusable form.

KNOWLEDGE ENGINEERING (2)

- Knowledge engineering can be viewed from two perspectives: narrow and broad.
- According to the **narrow perspective**, knowledge engineering deals with the steps necessary to build expert systems (i.e., knowledge acquisition, knowledge representation, knowledge validation, inferencing, and explanation/justification).
- According to the **broad perspective**, the term describes the entire process of developing and maintaining any intelligent systems

KNOWLEDGE ENGINEERING PROCESS



KNOWLEDGE ACQUISITION

- **Knowledge acquisition** involves the acquisition of knowledge from human experts, books, documents, sensors, or computer files.
- The knowledge may be specific to the problem domain or to the problem-solving procedures, it may be general knowledge (e.g., knowledge about business), or it may be metaknowledge (knowledge about knowledge).
- By **metaknowledge**, we mean information about how experts use their knowledge to solve problems and about problem-solving procedures in general.

KNOWLEDGE REPRESENTATION

- Knowledge acquired from experts or induced from a set of data must be represented in a format that is both **understandable by humans** and **executable on computers**.
- A variety of knowledge representation methods is available: **production rules**, **semantic networks**, **frames**, **objects**, **decision tables**, **decision trees**, and **predicate logic**.

PRODUCTION RULES

- The most popular form of knowledge representation for expert systems.
- Knowledge is represented in the form of condition/ action pairs: IF this condition (or premise or antecedent) occurs, THEN some action (or result or conclusion or consequence) will (or should) occur.
- Consider the following two examples:

“If the stop light is red AND you have stopped, THEN a right turn is okay”

KNOWLEDGE VERIFICATION AND VALIDATION

- Knowledge acquired from experts needs to be evaluated for quality, including evaluation, validation, and verification. These terms are often used interchangeably.
- **Evaluation** is a broad concept. Its objective is to assess an **ES's overall value**. In addition to assessing acceptable performance levels, it analyzes whether the system would be usable, efficient, and cost-effective.
- **Validation** is the part of evaluation that deals with the performance of the system (e.g., as it compares to the expert's). Simply stated, validation is building the right system (i.e., substantiating that a system performs with an acceptable level of accuracy).
- **Verification** is building the system right or substantiating that the system is correctly implemented to its specifications.

KNOWLEDGE AND INFERENCE (I)

- Knowledge And Inference rules Two types of rules are common in artificial intelligence: **knowledge and inference**.
- **Knowledge rules**, or declarative rules, state all the facts and relationships about a problem.
- **Inference rules**, or procedural rules, offer advice on how to solve a problem, given that certain facts are known.
- The knowledge engineer separates the two types of rules: Knowledge rules go to the knowledge base, whereas inference rules become part of the inference engine that was introduced earlier as a component of an expert system.

KNOWLEDGE AND INFERENCE (2)

- For example, assume that you are in the business of buying and selling gold.

The knowledge rules might look like this:

- **Rule 1:** IF an international conflict begins, THEN the price of gold goes up.
- **Rule 2:** IF the inflation rate declines, THEN the price of gold goes down.
- **Rule 3:** IF the international conflict lasts more than 7 days and IF it is in the Middle East, THEN buy gold.

Inference rules contain **rules about rules** and thus are also called meta-rules. They pertain to other rules (or even to themselves). Inference (procedural) rules may look like this:

- **Rule 1:** IF the data needed are not in the system, THEN request them from the user.
- **Rule 2:** IF more than one rule applies, THEN deactivate any rules that add no new data.

INFERENCE (I)

- **Inferencing (or reasoning)** is the process of using the rules in the knowledge base along with the known facts to draw conclusions.
- Inferencing requires some logic embedded in a computer program to access and manipulate the stored knowledge.
- The inference engine directs the search through the collection of rules in the knowledge base, a process commonly called pattern matching.
- In inferencing, when all of the hypotheses (the “IF” parts) of a rule are satisfied, the rule is said to be fired.
- Once a rule is fired, the new knowledge generated by the rule (the conclusion or the validation of the THEN part) is inserted into the memory as a new fact.

INFERENCE (2)

- The most popular inferencing mechanisms for rule-based systems are forward and backward chaining:
- **Backward chaining** is a goal-driven approach in which you start from an expectation of what is going to happen (i.e., hypothesis) and then seek evidence that supports (or contradicts) your expectation. Often, this entails formulating and testing intermediate hypotheses (or sub-hypotheses).
- **Forward chaining** is a data-driven approach. We start from available information as it becomes available or from a basic idea, and then we try to draw conclusions. The ES analyzes the problem by looking for the facts that match the IF part of its IF-THEN rules.

FORWARD AND BACKWARD CHAINING EXAMPLE (I)

- Here we discuss an example involving an investment decision about whether to invest in IBM stock. The following variables are used:

A = Have \$10,000

B = Younger than 30

C = Education at college level

D = Annual income of at least \$40,000

E = Invest in securities

F = Invest in growth stocks

G = Invest in IBM stock (the potential goal)

FORWARD AND BACKWARD CHAINING EXAMPLE (2)

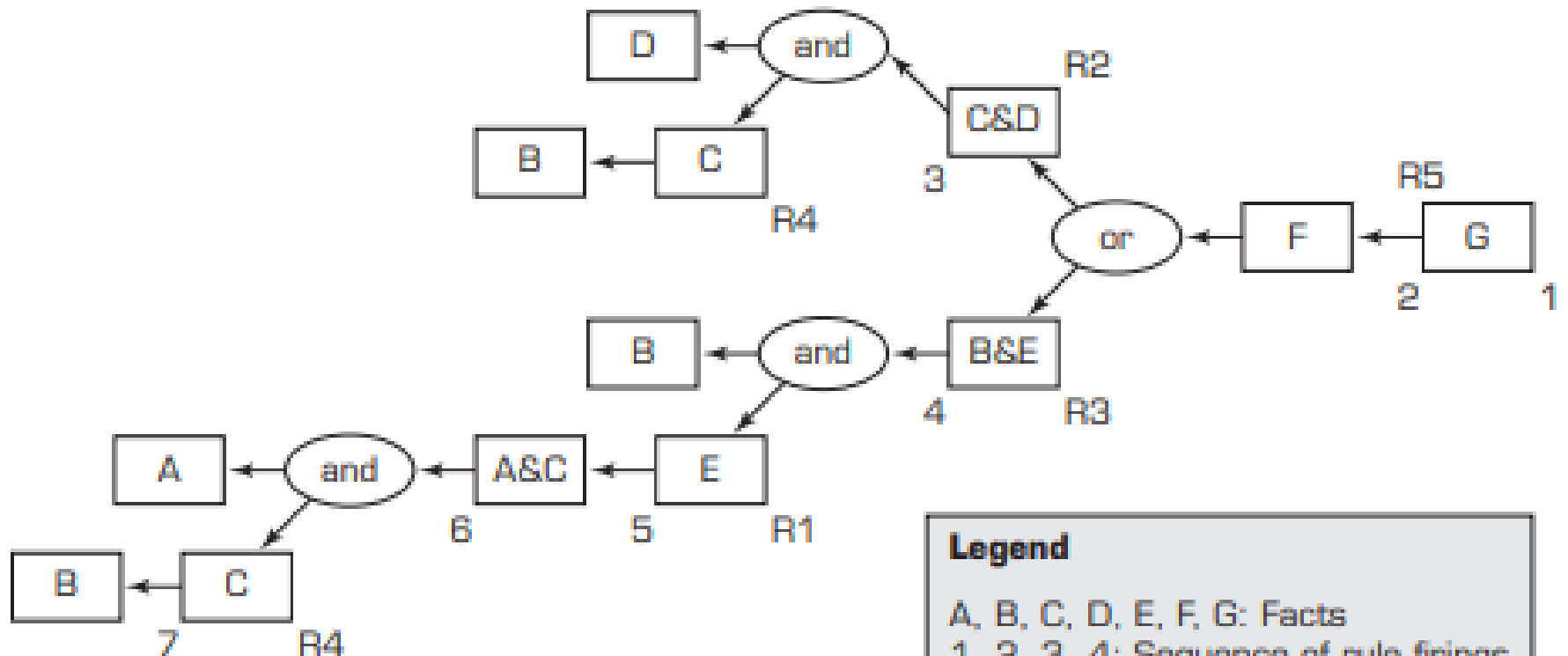
- Each of these variables can be answered as true (yes) or false (no).
- We assume that an investor has \$10,000 (i.e., that A is true) and that she is 25 years old (i.e., that B is true). She would like advice on investing in IBM stock (yes or no for the goal).
- **Our knowledge base includes the following five rules:**
 - R1: IF a person has \$10,000 to invest and she has a college degree, THEN she should invest in securities.
 - R2: IF a person's annual income is at least \$40,000 and she has a college degree, THEN she should invest in growth stocks.
 - R3: IF a person is younger than 30 and she is investing in securities, THEN she should invest in growth stocks.
 - R4: IF a person is younger than 30, THEN she has a college degree.
 - R5: IF a person wants to invest in a growth stock, THEN the stock should be IBM.

FORWARD AND BACKWARD CHAINING EXAMPLE (3)

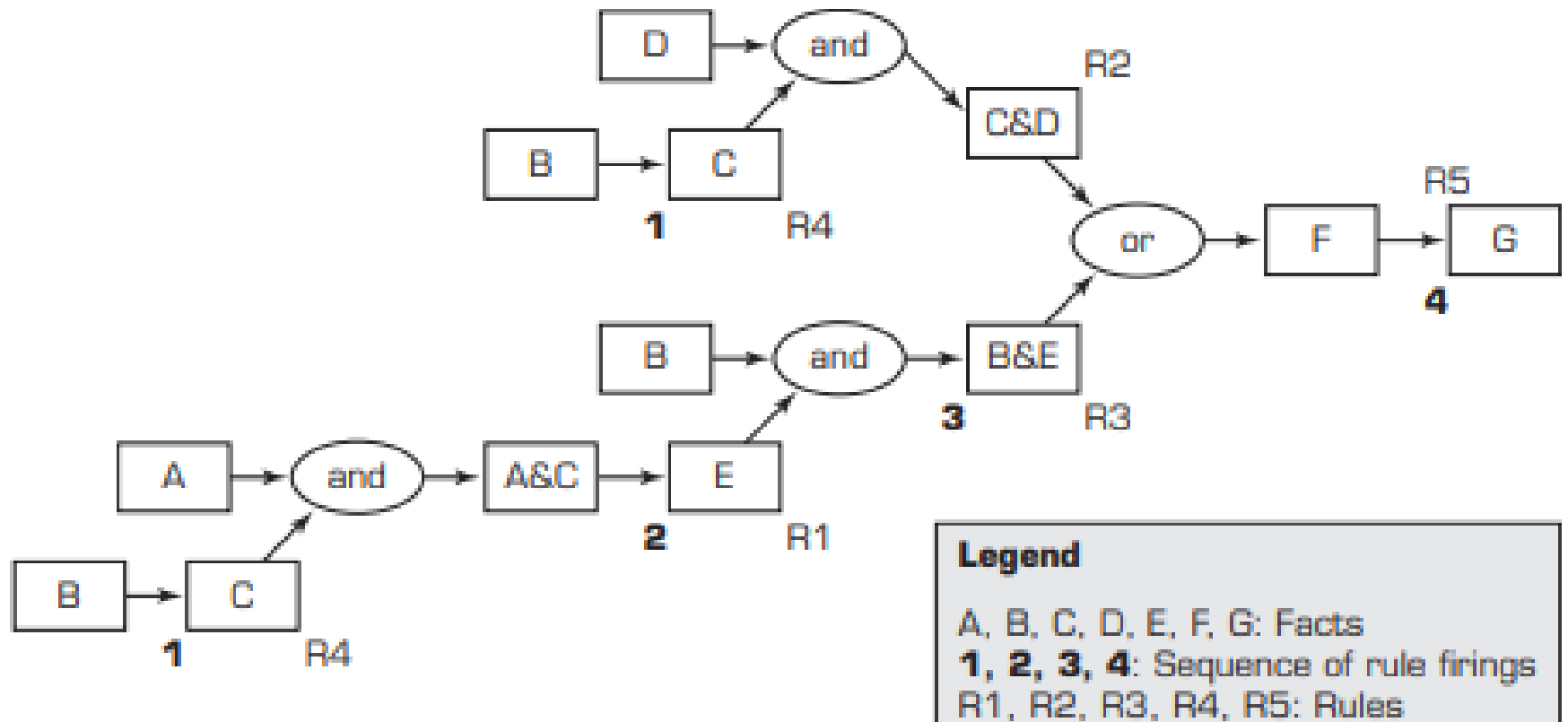
- **These rules can be written as follows:**

- R1: IF A and C, THEN E.
- R2: IF D and C, THEN F.
- R3: IF B and E, THEN F.
- R4: IF B, THEN C.
- R5: IF F, THEN G.

BACKWARD CHAINING



FORWARD CHAINING





Thanks
Any questions



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DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

Ibrahim El-Hasnony

COURSE DETAILS

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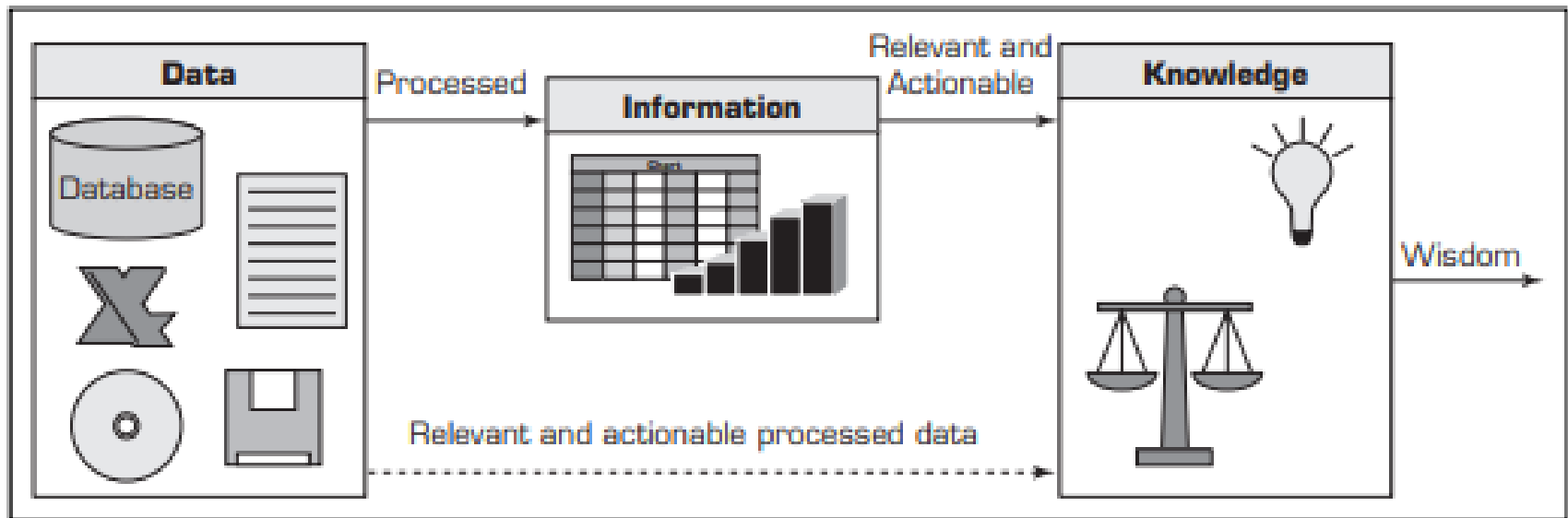
Chapter 10: DSS Intelligent Systems & Tools and Techniques (part-I)

Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

CHAPTER 9 CONTENTS

- How firms use information technology (IT) to implement knowledge management (KM) systems and how these systems are transforming modern organizations.
- Group support is a critical aspect of decision support systems (DSS).

KNOWLEDGE MANAGEMENT (3)



- Having **knowledge** implies that it can be exercised to **solve a problem**, whereas having **information** does **not carry the same connotation**

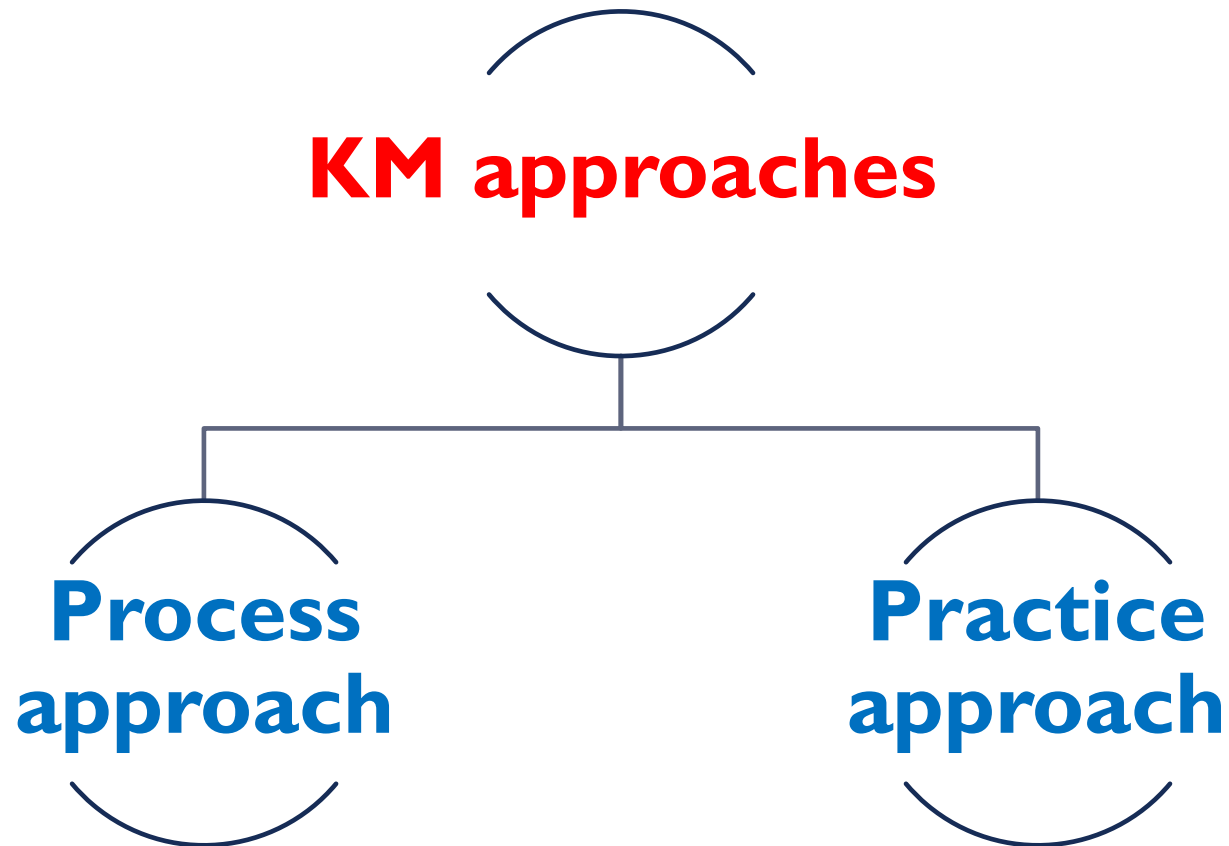
KNOWLEDGE MANAGEMENT (I)

- The idea of **KM** is not new (see Ponzi, 2004; and Schwartz, 2006).
- However, the **application of IT tools to facilitate** the creation, storage, transfer, and application of previously unmodifiable organizational knowledge is a new and major initiative in many organizations.
- **Knowledge management** is a process that helps organizations identify, select, organize, disseminate, and transfer important information and expertise that are part of the organization's memory and that typically reside within the organization in an unstructured manner.

KNOWLEDGE MANAGEMENT (2)

- **Knowledge management (Km)** is the systematic and active management of ideas, information, and knowledge residing in an organization's employees.
- The **structuring** of knowledge enables effective and efficient problem solving, dynamic learning, strategic planning, and decision making.
- The information technologies that make KM available throughout an organization are referred to as **KM systems**.

APPROACHES TO KNOWLEDGE MANAGEMENT



PROCESS APPROACH (I)

- The **process approach** to knowledge management attempts to codify organizational knowledge through formalized controls, processes, and technologies
- Organizations that adopt the process approach may implement **explicit policies** governing how knowledge is to be collected, stored, and disseminated throughout the organization.
- The process approach frequently involves the use of **IT**, such as intranets, data warehousing, knowledge repositories, decision support tools, and groupware to enhance the quality and speed of knowledge creation and distribution in the organization

PROCESS APPROACH (2)

- The main limitations of the process approach are that it **fails to capture much of the tacit knowledge** embedded in firms and it forces individuals into **fixed patterns of thinking**.
- This approach is favored by firms that sell relatively **standardized products** that fill common needs.
- Most of the valuable knowledge in these firms is fairly explicit because of the standardized nature of the products and services
- In these cases, the **knowledge may be typically static** in nature.

PRACTICE APPROACH (I)

- In contrast to the process approach, the practice approach to knowledge management assumes that a great deal of **organizational knowledge is tacit in nature** and that **formal controls, processes, and technologies are not suitable for transmitting this type of understanding.**
- Rather than build formal systems to manage knowledge, the focus of this approach is to **build the social environments or communities of practice** necessary to facilitate the sharing of tacit understanding.

PRACTICE APPROACH (2)

- This approach is typically adopted by companies that provide **highly customized solutions to unique problems.**
- For these firms, **knowledge is shared mostly through person-to-person contact.**
- Collaborative computing methods (e.g., group support systems [GSS], e-mail) help people communicate.
- **The valuable knowledge for these firms is tacit in nature,** which is difficult to express, capture, and manage.
- In this case, the environment and the nature of the problems being encountered are extremely dynamic

HYBRID APPROACHES TO KNOWLEDGE MANAGEMENT

- Many organizations use a hybrid of the process and practice approaches.
- Early in the development process, when it may not be clear how to extract tacit knowledge from its sources, the practice approach is used so that a repository stores only explicit knowledge that is relatively easy to document.
- The tacit knowledge initially stored in the repository is contact information about experts and their areas of expertise.

MAKING DECISIONS IN GROUPS: CHARACTERISTICS, PROCESS, BENEFITS, AND DYSFUNCTIONS

- When people work in groups (i.e., teams), they perform groupwork (i.e., teamwork).
- Groupwork refers to work done by two or more people together.

CHARACTERISTICS OF GROUPWORK

- A group performs a task (sometimes decision making, sometimes not).
- Group members may be located in different places.
- Group members may work at different times.
- Group members may work for the same organization or for different organizations.
- A group can be permanent or temporary.
- A group can be at one managerial level or span several levels.
- It can generate productivity gains and/or losses.
- The task may have to be accomplished very quickly.
- It may be impossible or too expensive for all the team members to meet in one place.

THE GROUP DECISION-MAKING PROCESS

- Even in hierarchical organizations, decision making is usually a shared process.
- A group may be involved in a decision or in a decision-related task, such as creating a short list of acceptable alternatives or choosing criteria for evaluating alternatives and prioritizing them.

DIFFICULTIES ASSOCIATED WITH GROUPWORK

-
- Waiting to speak
 - Dominating the discussion
 - Fear of speaking
 - Fear of being misunderstood
 - Inattention
 - Lack of focus
 - Inadequate criteria
 - Premature decisions
 - Missing information
 - Distractions
 - Digressions
 - Wrong composition of people
 - Groupthink
 - Poor grasp of problem
 - Ignored alternatives
 - Lack of consensus
 - Poor planning
 - Hidden agendas
 - Conflicts of interest
 - Inadequate resources
 - Poorly defined goals
-

BENEFITS OF WORKING IN GROUPS (PROCESS GAINS)

- It provides learning.
- People readily take ownership of problems and their solutions.
- Groups are better than individuals at catching errors.
- A group has more information (i.e., knowledge) than any one member.
- A group may produce synergy during problem solving. The effectiveness and/ or quality of groupwork can be greater than the sum of what is produced by independent individuals.
- Working in a group may stimulate the creativity of the participants and the process.
- A group may have better and more precise communication working together.
- Risk propensity is balanced. Groups moderate high-risk takers and encourage conservatives.

GROUP DECISION SUPPORT SYSTEMS (GDSS)

- An interactive computer-based system that facilitates the solution of semi structured or unstructured problems **by a group of decision makers**.
 - The goal of **GDSS** is to improve the productivity of decision-making meetings by speeding up the decision-making process and/or by improving the quality of the resulting decisions.
-
- **A group support system (GSS)** is any combination of hardware and software that enhances groupwork either in direct or indirect support of decision making.



Chapter 10

DSS Intelligent Systems & Tools and Techniques

INTRODUCTION

- In addition to rule-based systems, a few advanced techniques are available for designing intelligent systems that **can learn by themselves**.
- These include **case-based systems**, **artificial neural networks**, **genetic algorithms**, and **fuzzy reasoning** systems.
- The case-based system contains a case base that maintains unique expert experience.
- Artificial neural networks use artificial neurons executable on computers to mimic the human brain.
- Genetic algorithms modify the natural-selection process to find useful knowledge.
- Fuzzy systems convert between symbolic reasoning and mathematical calculation to improve decision performance in certain problem domains.

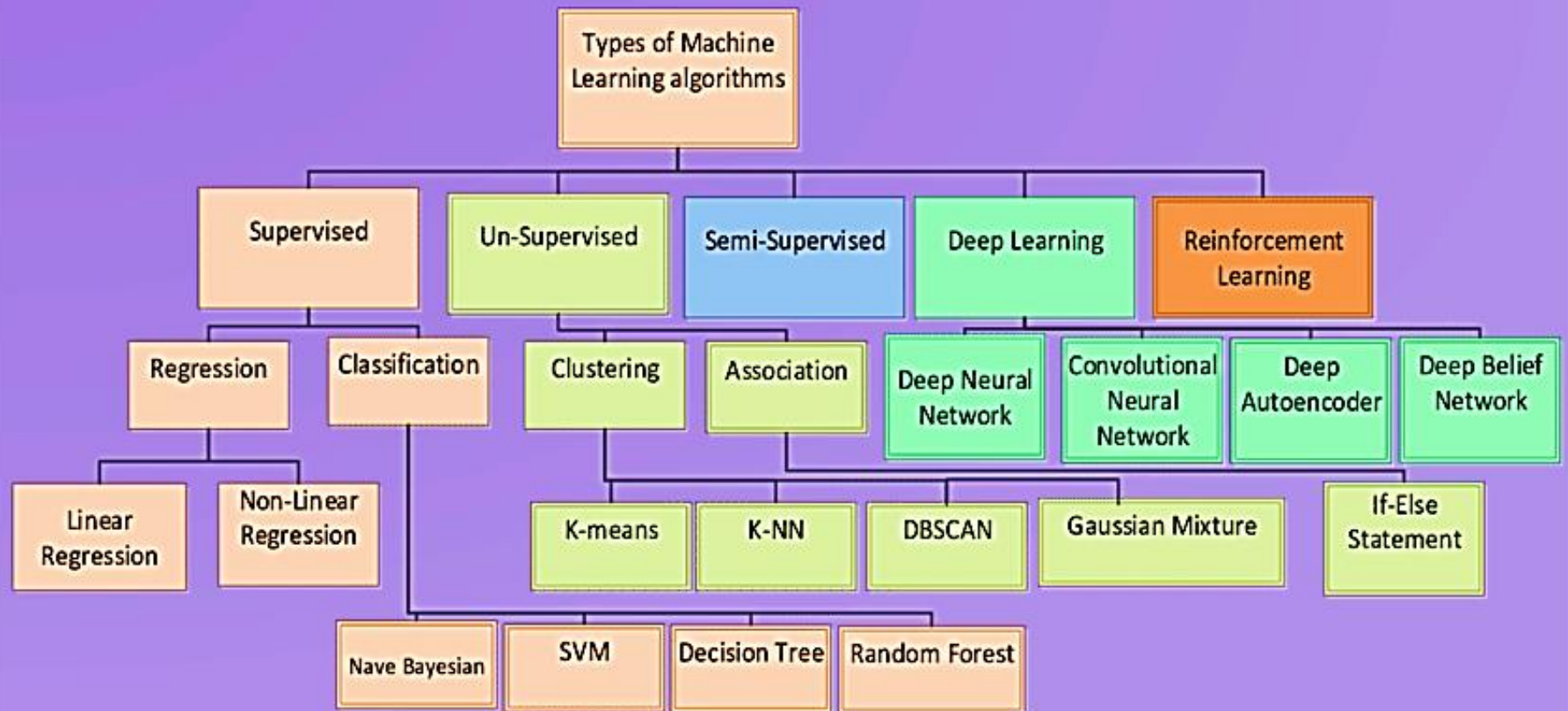
MACHINE-LEARNING

- Many organizations use neural networks to automate complex decision-making.
- Since such networks learn from past experience to improve their own performance, they are members of a technology family called **machine learning**.
- Machine learning is different in several ways from the conventional knowledge acquisition methods.
- Knowledge acquisition from human experts often suffers from the expert's inability to provide accurate knowledge, **whereas machine learning is an attempt to implicitly induce expert knowledge from historical decisions**.
- In other words, instead of asking the experts to articulate their decision knowledge, the learning module of the system can identify patterns and rules from the historical data available in the database or organizational files.

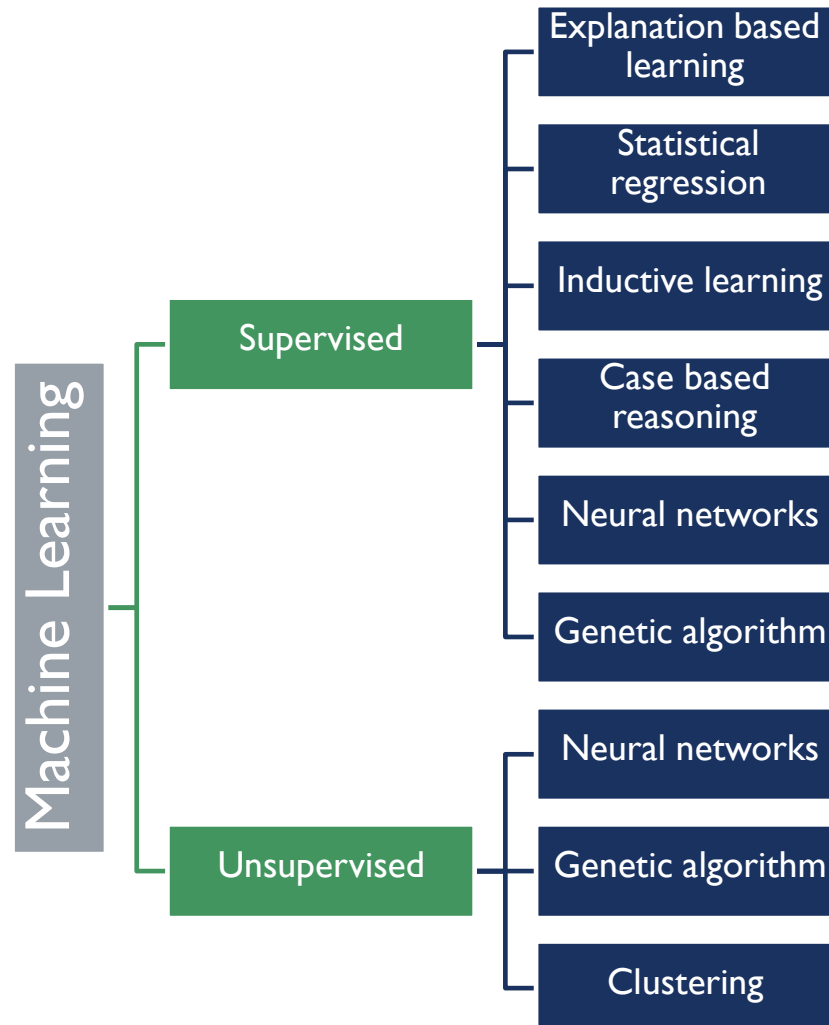
MACHINE-LEARNING

- **Machine learning** is a family of methods that attempt to allow machines to acquire knowledge for problem-solving by showing them historical cases.
- Machine learning is nontrivial.
- One **problem** is that there are **many models of learning**.
- Sometimes it is difficult to **match the learning model with the problem** type (e.g., scheduling).
- Machine learning is considered an artificial intelligence method even though some of its technologies do not formally exhibit intelligence.
- **However, it is useful for designing an intelligent decision-support system.**

MACHINE LEARNING METHODS



MACHINE LEARNING METHODS



MACHINE LEARNING METHODS

- **Supervised machine learning** is used for making predictions from data. To be able to do that, we need to know what to predict, which is also known as the target variable.
- **Reinforcement Learning (RL)** is the science of decision making. It is about learning the optimal behavior in an environment to obtain maximum reward.
- **Unsupervised learning** is self-organized learning. Its main aim is to explore the underlying patterns and predicts the output

MACHINE LEARNING METHODS

Criteria	Supervised ML	Unsupervised ML	Reinforcement ML
Definition	Learns by using labelled data	Trained using unlabelled data without any guidance.	Works on interacting with the environment
Type of data	Labelled data	Unlabelled data	No – predefined data
Type of problems	Regression and classification	Association and Clustering	Exploitation or Exploration
Supervision	Extra supervision	No supervision	No supervision
Algorithms	Linear Regression, Logistic Regression, SVM, KNN etc.	K – Means, C – Means, Apriori	Q – Learning, SARSA
Aim	Calculate outcomes	Discover underlying patterns	Learn a series of action
Application	Risk Evaluation, Forecast Sales	Recommendation System, Anomaly Detection	Self Driving Cars, Gaming, Healthcare



Thanks
Any questions



Mansoura University
Faculty of Computers and Information
Fall 2023-2024



DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

Ibrahim El-Hasnony

COURSE DETAILS

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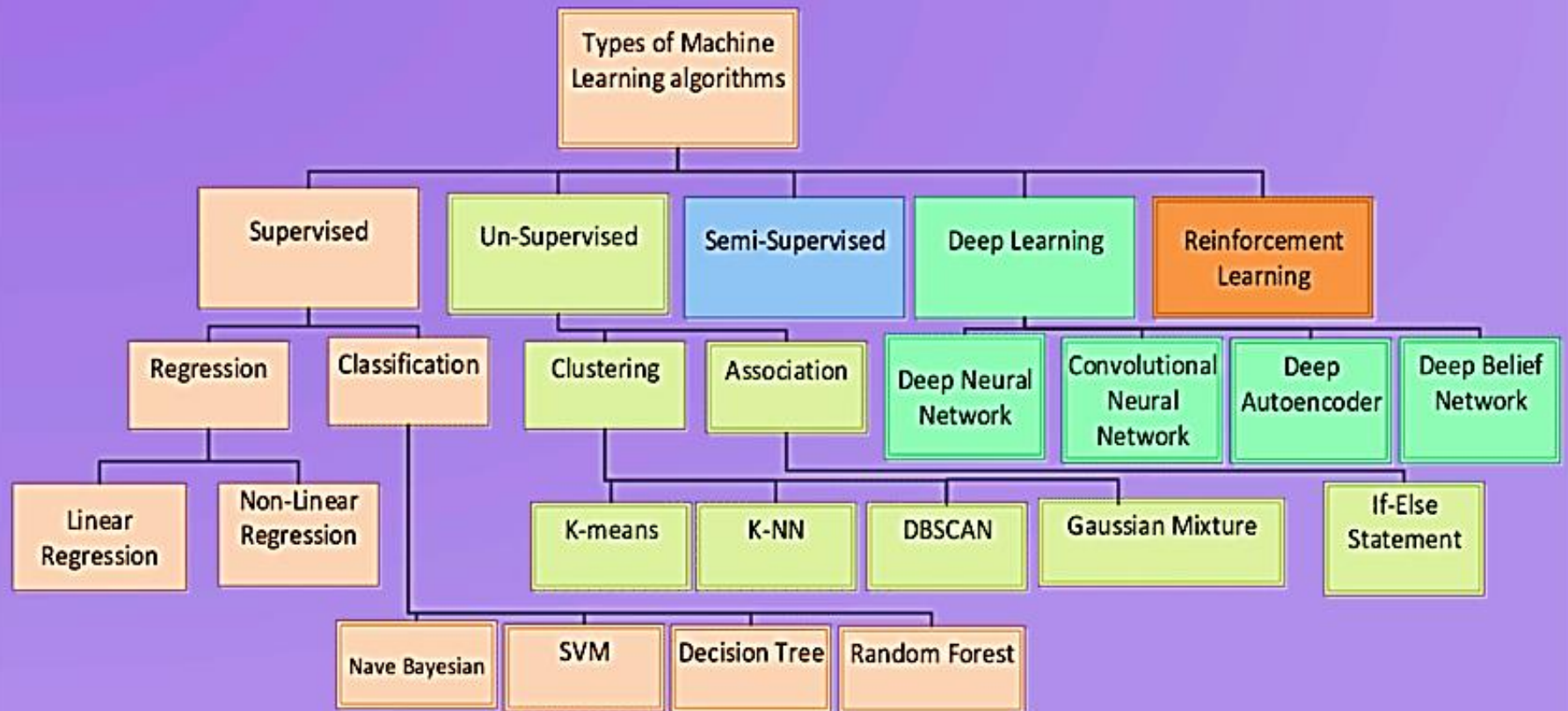
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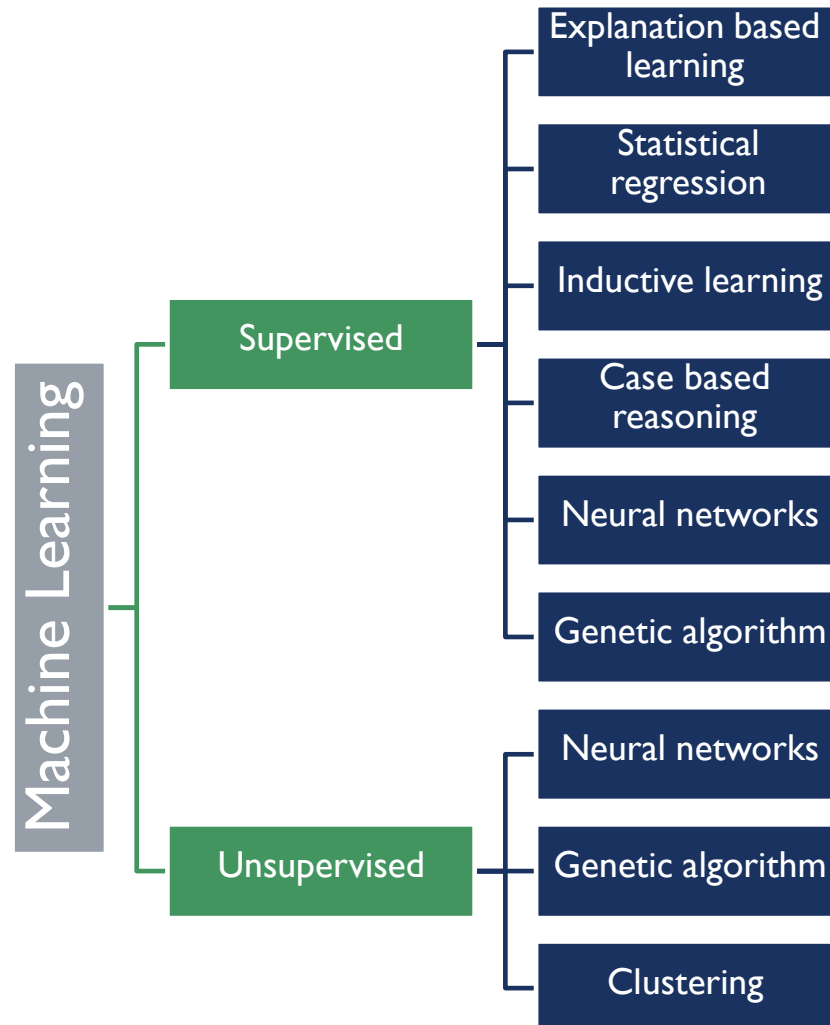
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MACHINE LEARNING METHODS

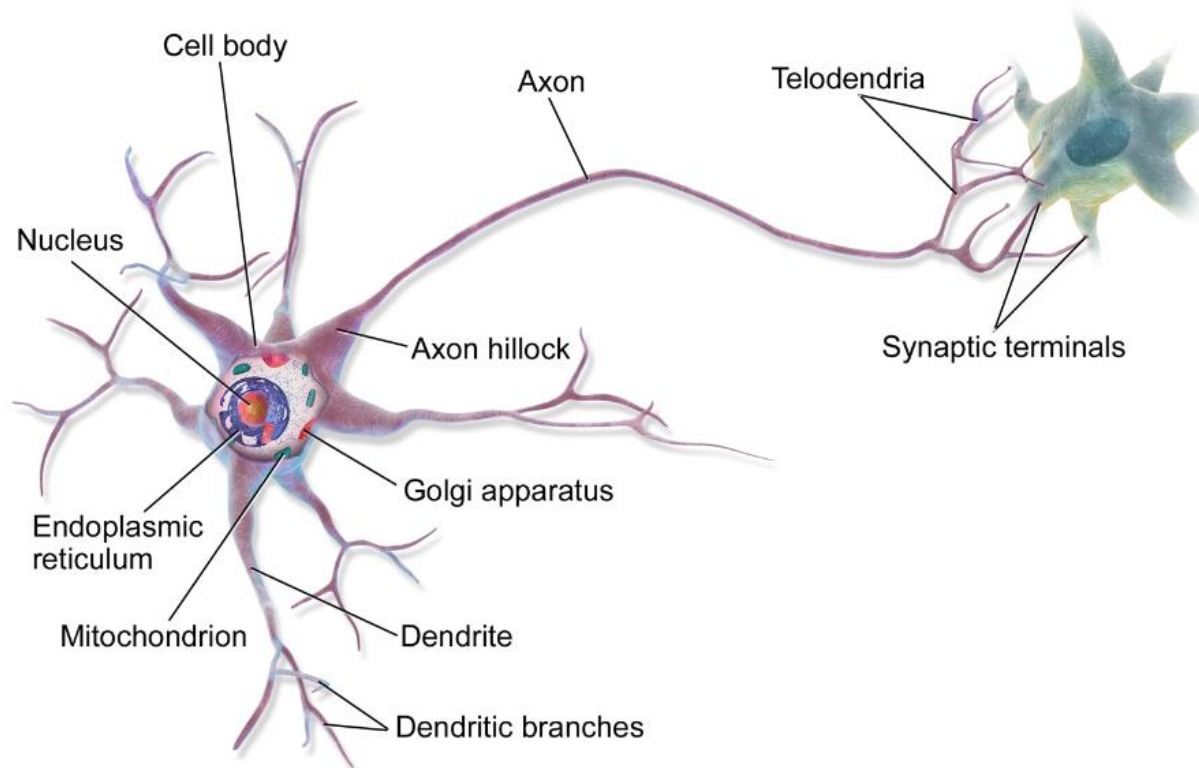
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BASIC CONCEPT OF NEURAL COMPUTING

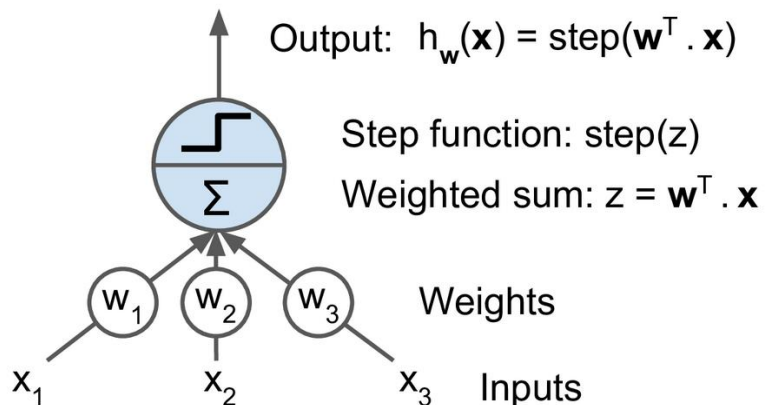
- **Neural computing** is a problem-solving methodology that attempts to mimic how our brains function.
- It is one of several successful approaches to machine learning.
- The resulting model from neural computing is often called an artificial neural network (ANN) or a neural network (NN).
- The study of neural networks began from a **single neuron**, called a **perceptron**, which is capable of converting a set of inputs to its outputs. After some initial success, the perceptron model fell into disfavor because it was unable to solve the simple XOR problem.
- The **multiple-layer** architecture was proposed and proved powerful in solving complex problems.

BIOLOGICAL AND ARTIFICIAL NEURAL NETWORKS

- The human brain is composed of special cells called **neurons**.
- The estimated number of neurons in a human brain is 50 billion to 150 billion, of which there are more than 100 different kinds



AN ARTIFICIAL NEURAL NETWORK



- An ANN model emulates a biological neural network.
- Neural computing uses a very limited set of concepts from biological neural systems
- Neural concepts are usually implemented as software simulations of the massively parallel processes that involve processing elements (also called artificial neurons) interconnected in a network architecture

BIOLOGICAL AND ARTIFICIAL NEURAL NETWORKS

Biological Neuron	Artificial Neuron
Cell Nucleus (Soma)	Node
Dendrites	Input
Synapse	Weights or interconnections
Axon	Output

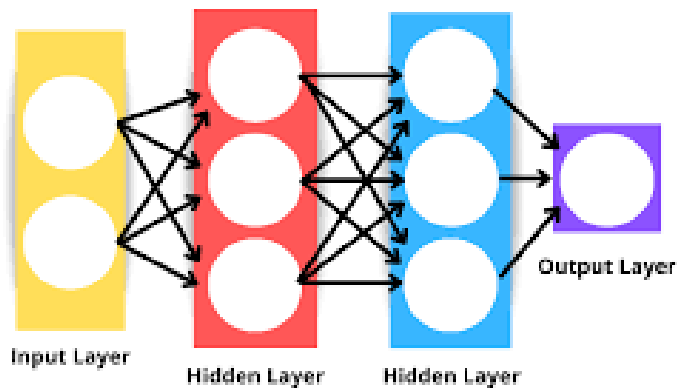
ELEMENTS OF ARTIFICIAL NEURAL NETWORKS

- A neural network is **composed of processing elements** organized in different ways to form the network's structure.
- The basic processing unit is the **neuron**.
- Several neurons are organized into a network.
- There are many ways to organize neurons, and these are referred to as **topologies**.
- One popular approach, known as the feedforward-backpropagation paradigm (or simply **backpropagation**),

PROCESSING ELEMENTS

- The processing elements (PEs) of an ANN are **artificial neurons**.
- Each of the neurons receives inputs, processes them, and delivers a single output
- The input can be raw input data or the output of other processing elements.
- The output can be the result (e.g., 1 means yes, 0 means no), or it can be input to other neurons.

NETWORK STRUCTURE



- Each ANN is composed of a collection of neurons grouped in layers.
- Note the three layers: **input**, **intermediate** (called the **hidden layer**), and **output**.
- Several hidden layers can be placed between the input and output layers.

INPUTS

- Each input corresponds to a single attribute.
- For example, if the problem is to decide on approval or disapproval of a loan, some attributes could be the applicant's income level, age, and home ownership.
- The numeric value, or representation, of an attribute is the input to the network.
- Several types of data, such as text, pictures, and voice, can be used as inputs.
- **Preprocessing** may be needed to convert the data to meaningful inputs from symbolic data or to scale the data.

OUTPUTS

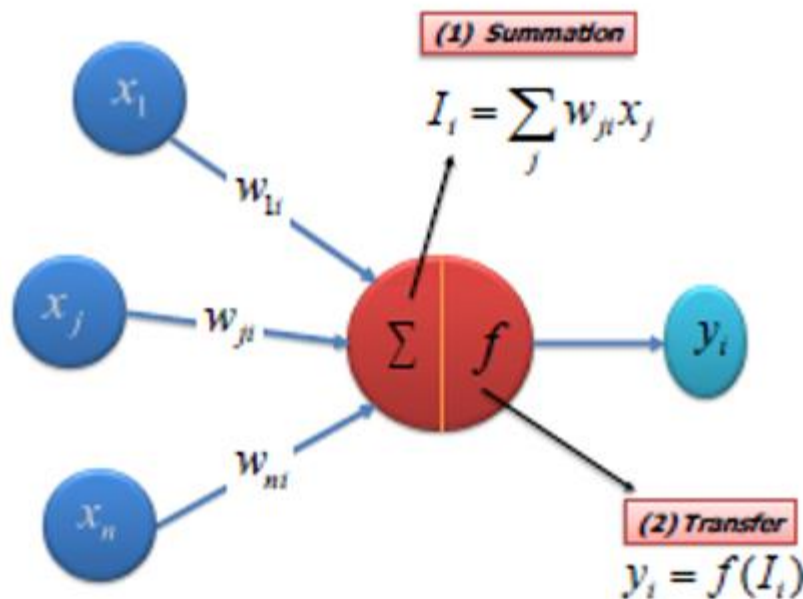
- The outputs of the network contain the solution to a problem.
- For example, in the case of a loan application it can be yes or no.
- The ANN assigns numeric values, like 1 for yes and 0 for no.
- The purpose of the network is to compute the values of the output.
- When the resulting value is not exactly 0 or 1, **postprocessing** of the outputs is required.

CONNECTION WEIGHTS

- Connection weights are the key elements in an ANN.
- They express the **relative strength** (or mathematical value) of the input data or the many connections that transfer data from layer to layer.
- Weights express the **relative importance** of each input to a processing element and ultimately the outputs.
- Weights are crucial in that they store learned patterns of information.
- It is through repeated adjustments of weights that the network learns.

SUMMATION FUNCTION

- The summation function computes the **weighted sums** of all the input elements entering each processing element.
- A summation function multiplies each input value by its weight and totals the values for a weighted sum Y .



TRANSFORMATION (TRANSFER) FUNCTION

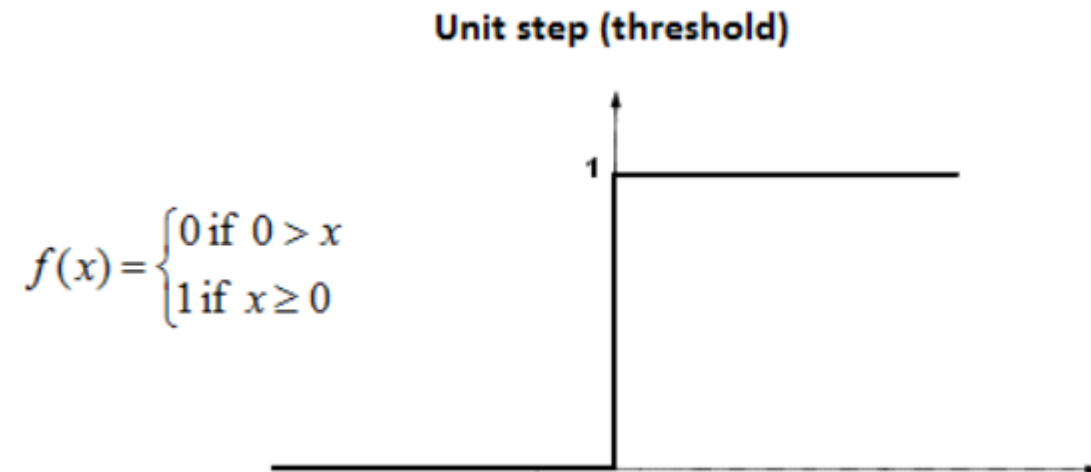
- The summation function computes the internal stimulation, or activation level, of the neuron.
- Based on this level, the neuron may or may not produce an output.
- The relationship is expressed by one of several types of transformation (transfer) functions.
- The selection of the specific function impacts the network's operation.
- The transformation modifies the output levels to reasonable values (typically between 0 and 1). This transformation is performed before the output reaches the next level
- The sigmoid (logical activation) function (or transfer function) is a popular and useful nonlinear transfer function.

TRANSFORMATION (TRANSFER) FUNCTION

- Without such a transformation, the value of the output becomes very large, especially when there are several layers of neurons.
- Sometimes, instead of a transformation function, a threshold value is used.
- For example, any value of 0.5 or less becomes 0, and any value above 0.5 becomes 1.
- A transformation can occur at the output of each post-processing element, or it can be performed only at the final output nodes.

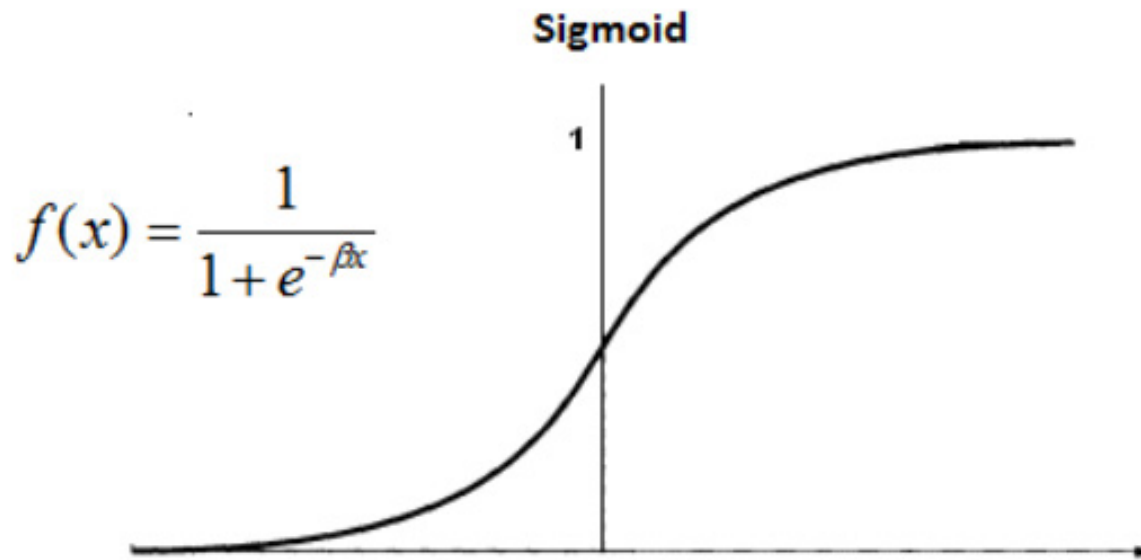
UNIT STEP (THRESHOLD)

- The output is set at one of two levels, depending on whether the total input is greater than or less than some threshold value.

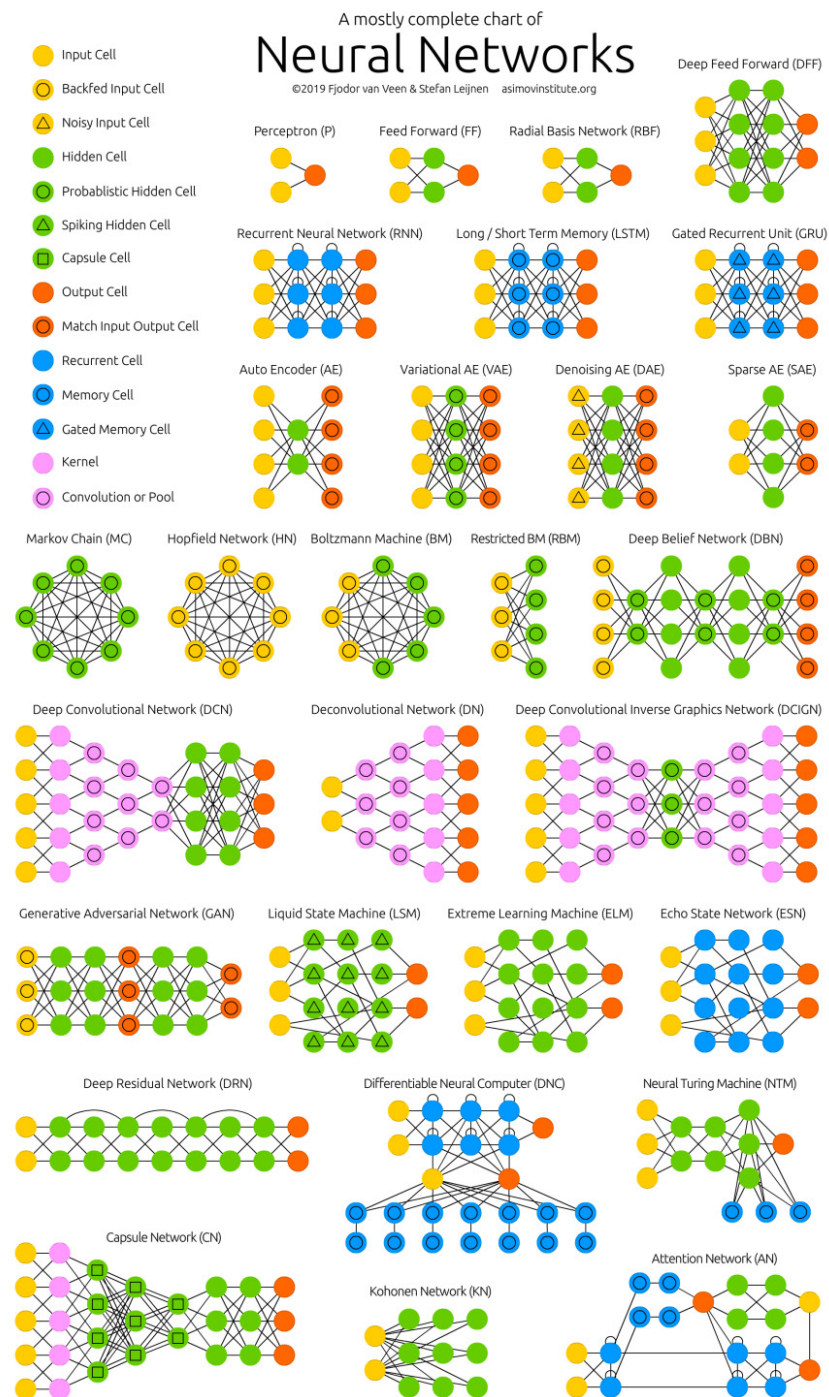


SIGMOID

- The sigmoid function consists of 2 functions, logistic and tangential. The values of the logistic function range from 0 and 1 and -1 to +1 for tangential functions.



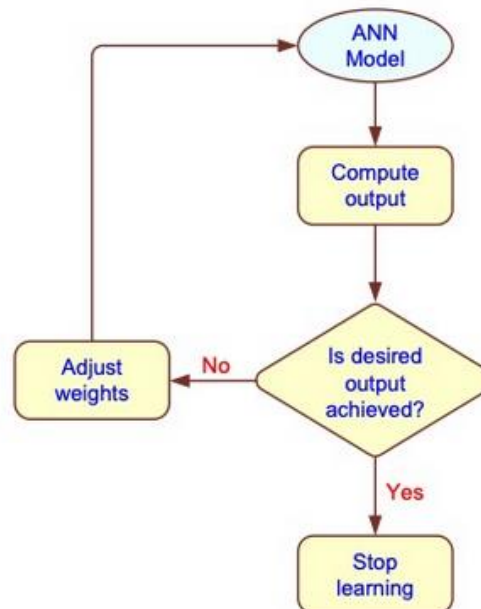
NEURAL NETWORK ARCHITECTURE



GENERAL LEARNING PROCESS

- In supervised learning, the learning process is inductive; that is, connection weights are derived from existing cases. The usual process of learning involves three tasks:

A Supervised Learning Process



Three-step process:

1. Compute temporary outputs
2. Compare outputs with desired targets
3. Adjust the weights and repeat the process

GENERAL LEARNING PROCESS

- When existing outputs are available for comparison, the learning process starts by setting the connection weights, either by rules or randomly.
- The difference between the actual output (For YT) and the desired output (Z) for a given set of inputs is an error called **delta**.
- The objective is to minimize the delta (reduce it to 0 if possible), and this is done by adjusting the network's weights.
- The key is to change the weights in the right direction, making changes that reduce the delta (error).

HOW A NEURAL NETWORK LEARNS (EXAMPLE)

- Consider a single neuron that learns the **inclusive OR operation**

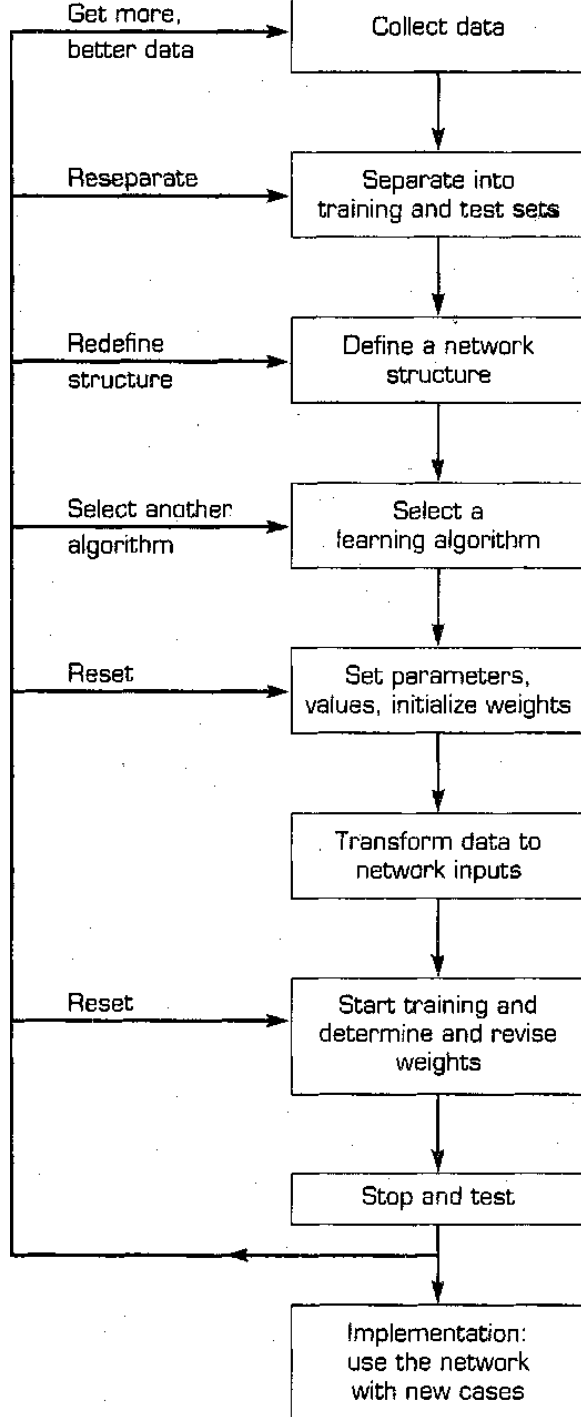
<i>Inputs</i>			
<i>Case</i>			<i>Desired Results</i>
1	0	0	0
2	0	1	1 (positive)
3	1	0	1 (positive)
4	1	1	1 (positive)

HOW A NEURAL NETWORK LEARNS (EXAMPLE)

- **Delta = $Z_j - Y_j$**
- Z and Y are the desired and actual outputs
- **$W(\text{ final}) = W_i(\text{initial}) + \text{alpha} * \text{delta} * X$**
- alpha is a parameter that controls how fast the learning takes place
- The threshold value (0.5) sets the output Y to 1 in the next row if the weighted sum of inputs is greater than 0.5; otherwise, Y is set to 0
- In subsequent steps, the learning algorithm improves the results until it finally produces a set of weights that give the correct results ($W_j = W_2 = 0.7$ in step 4 of Table below).

<i>Step</i>				<i>Initial</i>		<i>Y</i>	<i>Delta</i>	<i>Final</i>	
			<i>z</i>	$\bullet w_1$	w_2			w_1	w_2
1	0	0	0	0.1	0.3	0	0.0	0.1	0.3
	0	1	1	0.1	0.3	0	1.0	0.1	0.3
	1	0	1	0.1	0.5	0	1.0	0.1	0.5
	1	1	1	0.3	0.5	1	0.0	0.3	0.5
2	0	0		0.3	0.5	0	0.0	0.3	0.5
	0	1	1	0.3	0.5	0	0.0	0.3	0.5
	1	0	1	0.3	0.7	0	1.0	0.5	0.7
	1	1	1	0.5	0.7	1	0.0	0.5	0.7
3	0	0		0.5	0.7	0	0.0	0.5	0.7
	0	1	1	0.5	0.7	1	0.0	0.5	0.7
	1	0	1	0.5	0.7	0	1.0	0.7	0.7
	1	1	1	0.7	0.7	1	0.0	0.7	0.7
4	0	0		0.7	0.7	0	0.0	0.7	0.7
	0	1	1	0.7	0.7	1	0.0	0.7	0.7
	1	0	1	0.7	0.7	1	0.0	0.7	0.7
	1	1	1	0.7	0.7	1	0.0	0.7	0.7

Parameters: alpha = 0.2; threshold = 0.5.



DEVELOPING NEURAL NETWORK-BASED SYSTEMS

- Although the development process of ANNs is similar to the structured design methodologies of traditional computer-based information systems, some phases are unique or have some unique aspects.
- In the process described, we assume that the preliminary steps of system development, such as determining information requirements, conducting a feasibility analysis, and gaining a champion in top management for the project, have been completed successfully. Such steps are generic to any information system.



Thanks
Any questions



Mansoura University
Faculty of Computers and Information
Fall 2023-2024



DECISION SUPPORT SYSTEMS

Grade: 4TH YEAR (GENERAL)

Prepared by:

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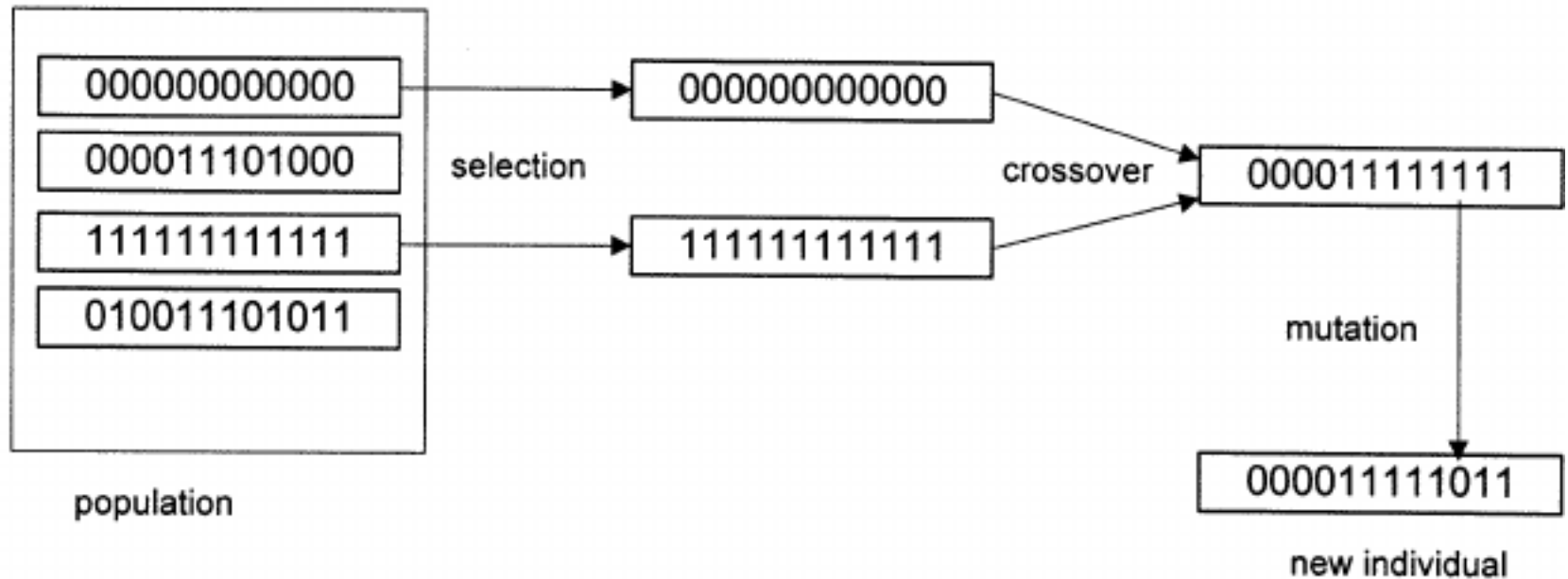
Chapter 10: DSS Intelligent Systems & Tools and Techniques (part-3)

Chapter 11: Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization

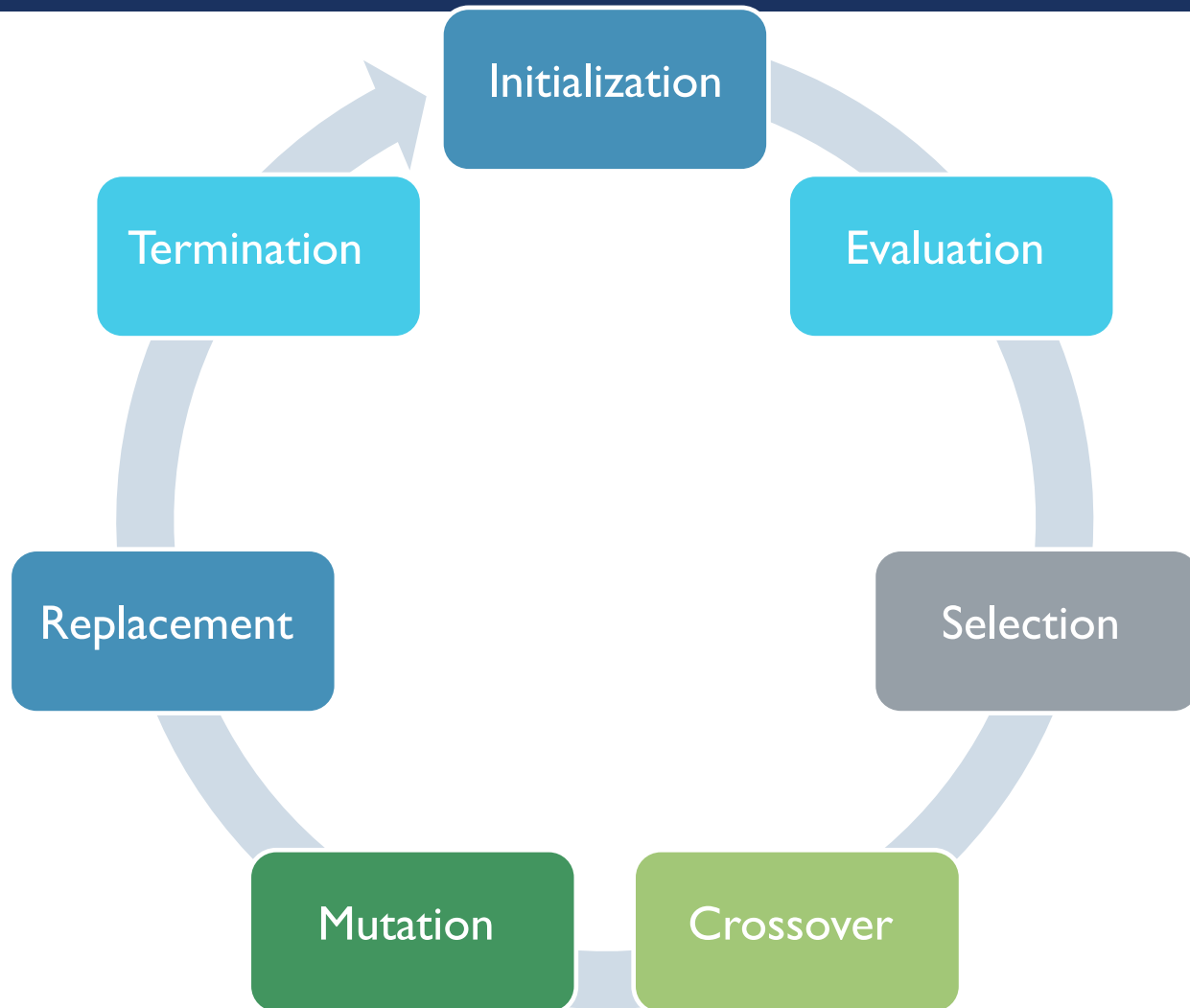
GENETIC ALGORITHM FUNDAMENTALS

- The most popular among all nature-based techniques is the genetic algorithm (GA)
- GA is based on genetic reproduction among animals.
- Genetic reproduction is the process of generating offspring.
- There are three main sub-processes of GA
 1. Selection
 2. Crossover
 3. mutation

GENETIC ALGORITHM FUNDAMENTALS



GENETIC ALGORITHM STEPS (I)



GENETIC ALGORITHM STEPS (2)

- **Initialization:**

- A population of potential solutions (individuals or chromosomes) is randomly generated. Each solution represents a possible solution to the problem at hand.

- **Evaluation:**

- Each individual in the population is evaluated for its fitness, which is a measure of how well it solves the problem. The fitness function is problem-specific and reflects the objective or goal of the optimization.

- **Selection:**

- Individuals are selected from the population based on their fitness. The more fit individuals have a higher chance of being selected..

GENETIC ALGORITHM STEPS (3)

- **Crossover (Recombination):**

- Pairs of selected individuals (parents) exchange genetic information to create new individuals (offspring). This is typically done by combining portions of the parent solutions to create one or more new solutions.

- **Mutation:**

- Random changes are applied to some of the individuals in the population. This introduces new genetic material and helps explore the solution space more broadly.

- **Replacement:**

- The new offspring and possibly some of the existing individuals are used to form the next generation. The less fit individuals may be replaced by the fitter offspring.

- **Termination:**

- The algorithm continues to iterate through the selection, crossover, mutation, and replacement steps for a fixed number of generations or until a termination condition is met (e.g., reaching a satisfactory solution).

EXAMPLE (I)

- Let's consider a simple numerical example of using a genetic algorithm to solve the traveling salesman problem (TSP). The TSP is a classic optimization problem where the goal is to find the shortest possible route that visits a given set of cities and returns to the starting city.
- **Problem Statement:**
- Imagine we have 5 cities labeled A, B, C, D, and E, and we want to find the shortest possible route that visits each city exactly once and returns to the starting city.
- **Representation:**
- A solution in this context is a permutation of the cities, representing the order in which the cities are visited.

EXAMPLE (2)

- **Initialization:**
- Generate a population of potential solutions. For simplicity, let's consider an initial population of 5 individuals:
 1. ABCDE
 2. ACBED
 3. AEDCB
 4. BACDE
 5. CBAED

EXAMPLE (3)

- **Evaluation:**
- Calculate the total distance for each individual in the population. The fitness is inversely proportional to the total distance. So, shorter routes have higher fitness.
- **Example Distances (for illustration):**
- ABCDE: 20 units
- ACBED: 18 units
- AEDCB: 22 units
- BACDE: 21 units
- CBAED: 19 units

EXAMPLE (4)

- **Selection:**
- Select individuals based on their fitness. Higher fitness increases the chance of being selected. For simplicity, let's say we select the top 3 individuals:
 - ACBED
 - CBAED
 - ABCDE

EXAMPLE (5)

- **Crossover:**

- Perform crossover to create new individuals. Let's use a simple one-point crossover:
- ACBED and CBAED crossover at the third position, resulting in two offspring:
- Offspring 1: ACBDE
- Offspring 2: CBADD

- **Mutation:**

- Apply mutation to introduce small changes. Let's say we randomly swap two cities in one of the offspring:
- Mutate Offspring 2 (CBADD): CBADD -> CDBAD

EXAMPLE (6)

- **Replacement:**

- Replace some individuals in the old population with the new offspring. For simplicity, let's replace the two least-fit individuals:

- ACBED

- CBAED

- ACBDE

- CDBAD

- ABCDE

- **Termination:**

- Repeat the process for a certain number of generations or until a termination condition is met.

EXAMPLE-2(I)

- **Problem Statement:**

- Imagine you have a dataset with 10 features ($F_1, F_2, \dots, F_{10}$), and you want to select a subset of features that maximizes the performance of a classification model.

- **Initialization:**

- Create a population of potential solutions, where each solution is a binary string indicating which features are selected (1) and which are not selected (0). For example:

- 1100100010
- 0011001100
- 1010101010
- 0101010101
- 1110001110

EXAMPLE-2(2)

- **Evaluation:**
- Train and evaluate a classification model using each solution in the population. The performance of the model (accuracy, precision, etc.) serves as the fitness score. The goal is to maximize the fitness score.
- **Selection:**
- Select individuals based on their fitness. Higher-performing feature subsets are more likely to be selected.

EXAMPLE-2(3)

- **Crossover:**

- Perform crossover to create new individuals. Let's use a simple one-point crossover:
- Individuals 1 and 2 crossover at the fifth position, resulting in two offspring:
- Offspring 1: 1100111100
- Offspring 2: 0011000010

- **Mutation:**

- Apply mutation to introduce small changes. Randomly flip some bits in one of the offspring:
- Mutate Offspring 1 (1100111100): 1100111100 -> 1100111000

EXAMPLE-2(4)

- **Replacement:**
- Replace some individuals in the old population with the new offspring. For simplicity, replace the two least fit individuals.
- **Termination:**
- Repeat the process for a certain number of generations or until a termination condition is met.

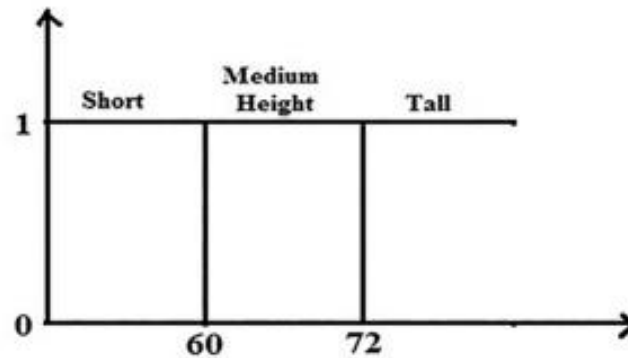
FUZZY THEORY AND FUZZY LOGIC

- Fuzzy theory is a widely applied concept to deal with uncertainty.
- Fuzzy theory and fuzzy logic are concepts that belong to the field of fuzzy set theory, a mathematical framework for dealing with uncertainty and vagueness.
- Developed by Lotfi A. Zadeh in the 1960s, fuzzy set theory provides a way to represent and manipulate uncertainty in information and decision-making.

FUZZY THEORY AND FUZZY LOGIC

- **Fuzzy Sets:**
- In classical set theory, an element either belongs to a set or does not. In contrast, fuzzy set theory allows elements to partially belong to a set.
- Instead of using binary membership (0 or 1), fuzzy sets use membership degrees between 0 and 1, indicating the degree of belongingness of an element to a set.

CRISP SETS



Crisp representation of height example.

Values of Characteristic Function for Height of Four Persons

	Height (inches)	Short	Medium Height	Tall
Mr. X	80	0	0	1
Mr. Y	73	0	0	1
Mr. Z	70	0	1	0
Mr. K	60	1	0	0

FUZZY SETS

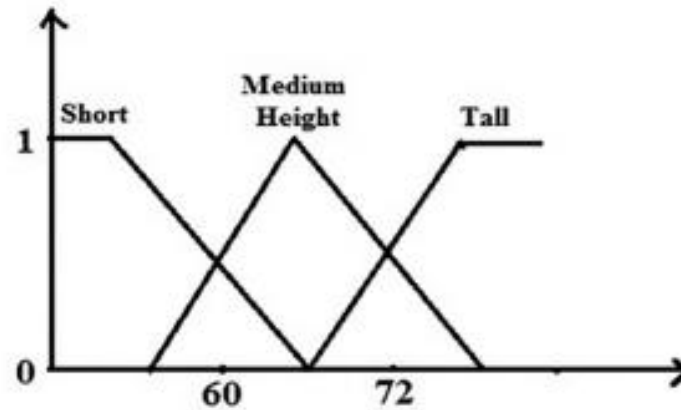
- Thus, the sets can be taken as “tall” and “moderate height”, and instances of various degrees for six different persons are shown in Table below.

Degree of “Tall” and “Moderate Height”

Set	Tom	Harry	Ray	Moni	Bentinc	Suzi
Tall	0.3	0.9	0.7	0.8	0.6	0.1
Moderate height	0.8	0.2	0.6	0.8	0.4	0.9

In the crisp set, the characteristic function for each member has a value of either 0 or 1.

FUZZY SETS

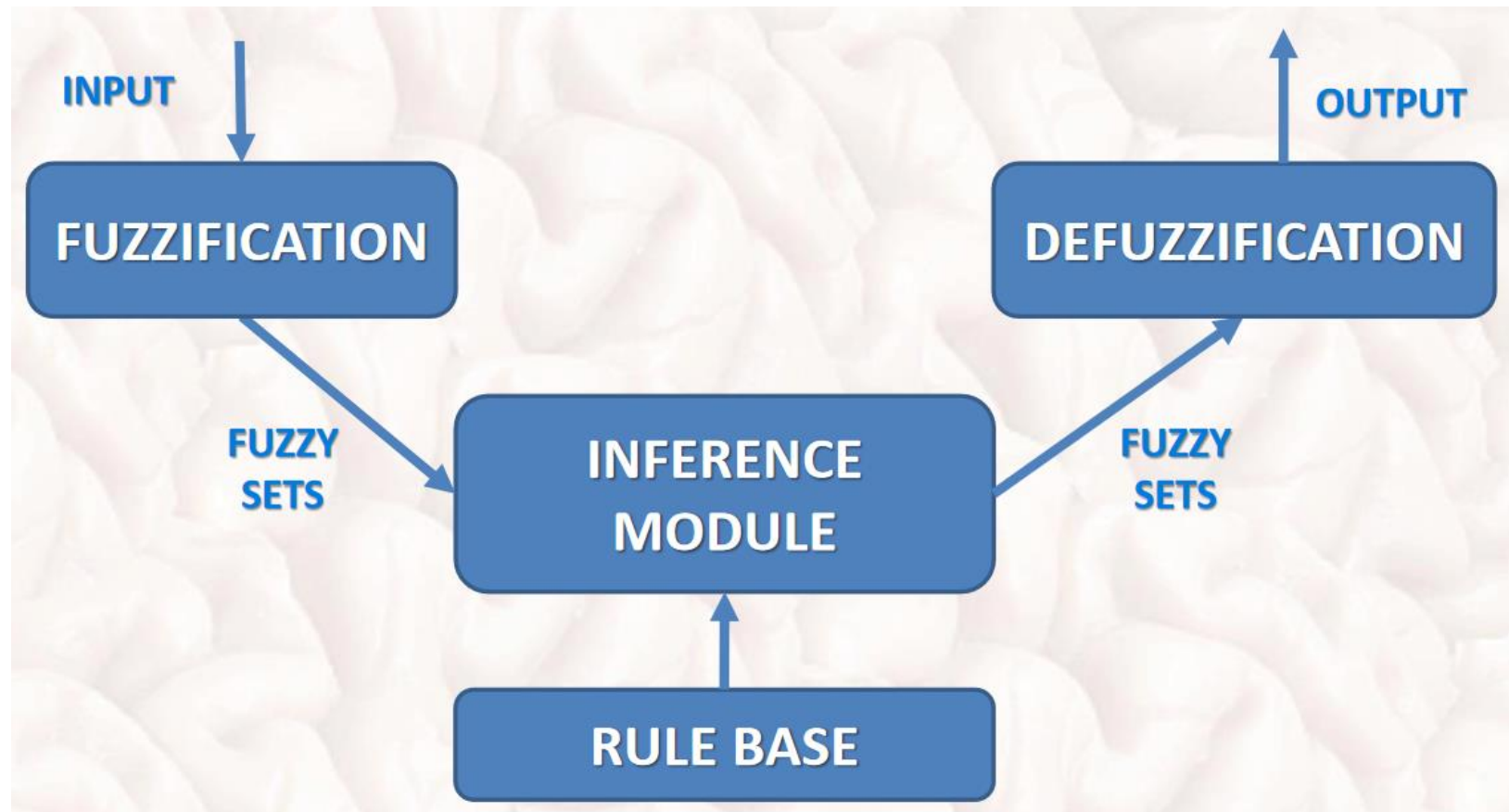


Fuzzy representation of height example.

Values of Membership Function for Height of Four Persons

	Height (inches)	Short	Medium Height	Tall
Mr. X	80	0	0.1	0.9
Mr. Y	73	0	0.3	0.7
Mr. Z	70	0	0.6	0.4
Mr. K	60	0.8	0.2	0

FUZZY SYSTEMS



EXAMPLE

- Define the Fuzzy Sets:

Input 1 (Temperature):

Cold, Warm, Hot

Input 2 (Humidity):

Low, Medium, High

Output (Fan Speed):

Slow, Medium, Fast

EXAMPLE

- Specify Membership Functions:
- Let's use triangular membership functions for simplicity:
- **For Temperature:**
 - Cold: 0,0,20
 - Warm: 10,20,30
 - Hot: 20,30,30
- **For Humidity:**
 - Low: 0,0,40
 - Medium: 20,40,60
 - High: 40,60,60
- **For Fan Speed:**
 - Slow: 0,0,40
 - Medium: 20,40,60
 - Fast: 40,60,60

Let's assume the current temperature is 25°C and humidity is 50% The goal is to determine the fan speed as an output.



Thanks
Any questions